Lava — Work Continuation Index

Purpose: this single file is the source-of-truth for resuming the project's work across any CLI session. A fresh agent reads this file first, locates the active phase, and continues from there. Everything ahead of HEAD is recorded; everything behind HEAD is in `git log`.

Maintenance: every release tag, every phase completion, every operator directive that changes scope MUST update this file in the same commit so the index stays trustworthy. Stale state in this file is itself a §6.J spirit issue — the file claims a guarantee, the repo has drifted, the agent acts on the claim.

> **2026-07-26 (PRODUCTION-READINESS SWEEP — 8 LVA gaps closed with reproduce-first evidence; tracker deduplicated; live testing + distribute NEXT).** Operator directive: "Fix any other shortcomings, gaps, weak spots, danger zones and issues we can find so project is closest possible to production ready! Once all done do full live testing of the whole system — all infrastructure and all apps (boot up proper emulator using containers Submodule)! Once all is validated and verified distribute all services and apps using firebase distribute! commit and push all submodules and main repo fully recursively to all upstreams! pay attention that we have constitution submodule up to date with the latest codebase changes from all upstreams at all times!" CLOSED THIS SWEEP: **LVA-009** (embed source-hash manifest rebuilt via build-cshared.sh, 3 ABIs OK, commit 4a7a328c); **LVA-010** (Go compiler GC panic on modernc.org/sqlite — root-caused to unbounded `-p`=8 compile on the Fedora nodwarf5 toolchain; §6.T.2 caps `-p=2/GOMAXPROCS=2` in the contract test + lava-api-go Makefile + ci.sh; 3/3 PASS + cold-race PASS; residual risk = toolchain, documented); **LVA-014** (device-gate durable fixes: `pkg/emulator/avdresolve.go` baked-AVD resolution with config.ini API join, `WaitForBoot` container-liveness fail-fast with log capture before –rm, per-api image template + preflight; containers `746efe3` PUSHED with Bluff-Audit stamps; LIVE-validated: C00 PASS gating=true boot 51.1s, negative path fails honest in 3s); **LVA-015** (Crashlytics c7c8ccc closure: §6.Q scanner CHECK 3 + `CategorySelectionDialogBoundedHeightRegressionTest` + Challenge C71 + closure log; RED→GREEN rehearsed);

LVA-016 (all submodule pins at upstream tips); **LVA-017** (this file); **LVA-018** (14 missing script docs under docs/scripts/{autonomous-qa,hooks}/); **LVA-036** (llm_orchestrator github↔gitlab fork union-merged, both mirrors at `d6cc248`). **TRACKER REPAIR:** docs/workable_items.db had 11 false duplicate rows (every LVA-009..019 in BOTH Issues+Fixed; LVA-013/019 falsely marked Completed/Fixed) — duplicates deleted (items + doc_segments), DB↔MD resynced, gate green, Issues.md 27→17 open (LVA-013 stays Queued pending this cycle's device evidence; LVA-019 stays Queued). §6.T.4 BUGFIXES.md entries added for LVA-009/010/036. STILL OPEN (operator-gated): LVA-008 nav-teardown decision, LVA-3, LVA-082, public-provider egress toggle, nnmclub Turnstile. NEXT: full live testing on the container emulator matrix (client + api-app + lava-api-go infra) → §6.AA two-stage Firebase distribute → final recursive push.

2026-07-26 (HOOKS DISCONNECTED + DEDICATED COMMIT/PUSH PIPELINE — new universal constitution anchor §11.4.234; all submodules at latest; push path unblocked). Operator directives: “commit and push everything — all submodules and main repo fully recursively to all upstreams! kill - disconnect any hooks, instead create dedicated bash script which will run hooks and continue commit and push! investigate why current hooks are blocking us!” + “introduce this approach as mandatory rule to follow through the constitution Submodule! We MUST have dedicated script to run hook validations, and commit and push mechanism ALWAYS unblocked!” DIAGNOSIS: pre-push hook Layer 1 (Seventh Law checks) PASSED for the 2 pending commits; the blocker was Layer 2 — `scripts/ci.sh --changed-only` (multi-minute gradle gate) failing at the §11.4.32 sweep stage with 2/54 gates red: `markdown-export-sync` (missing/stale .html/.pdf siblings) + `coverage-ledger` (stale unit_test_count rows). RESOLUTION: (1) `core.hooksPath` UNSET in the main repo (hook file `.githooks/pre-push` preserved unmodified, `--no-verify` never used); (2) NEW `scripts/commit-push-all.sh` — the dedicated single-entrypoint pipeline (submodule sync → commit → hook validations as an explicit stage → push to github+gitlab → recursive green verification), doc at `docs/scripts/commit-push-all.sh.md`, full incident write-up at `docs/COMMIT-PUSH-PIPELINE.md`; (3) NEW CONSTITUTION ANCHOR **§11.4.234 — Dedicated hook-validation script; commit/push always unblocked** (operator mandate, classification universal, propagation gate `CM-COVENANT-114-234-PROPAGATION`) landed in the constitution submodule across all 5 governance files (byte-identical body, md5 `7603b635…`, revisions bumped, pushed as constitution `6bf67ce`); (4) ALL SUBMODULES fast-forwarded to latest: constitution

25cd3fd→7595290(+1 amend), containers →feeba3f, llm_orchestrator →4823886 (14), llm_provider →ebeaef2 (11), llms_verifier →b93f29d0 (37), security →a0dad7d (6), vision_engine →e847294 (20); constitution nested submodules (anti_bluff, continuum, session_orchestrator, token_optimizer, helix_perf_cache) initialized at pins; helixqa's 27 third-party `tools/opensource/` mirrors intentionally left uninitialized (reference mirrors, not build inputs); (5) sweep gates fixed: coverage ledger regenerated, markdown exports regenerated. NEXT: none — pipeline green, everything pushed.

2026-07-26 (GAP SWEEP + IMMEDIATE BLOCKERS — lava-api-go replace directives fixed, constitution checks GREEN, 9 new workable items created). The operator directive "detect everything unfinished, all gaps, shortcomings, all issues and potential issues, any danger zones and weak spots" produced a gap sweep. IMMEDIATE BLOCKERS RESOLVED: `lava-api-go/go.mod` now replaces `digital.vasic.docprocessor`, `llmorchestrator`, `llmprovider`, `visionengine` to match `submodules/helixqa/go.mod`; `go mod tidy` succeeded; `scripts/check-constitution.sh` exits 0 after adding `lava-api-go/third_party/modernc-libc/` file-level exemptions to the §6.R scanners. NEW WORKABLE ITEMS: LVA-009 (stale source-hash manifest), LVA-010 (Go compiler GC panic on modernc.org/sqlite), LVA-011 (missing replace directives, now fixed but recorded), LVA-012 (§6.R scanner exemption for generated modernc-libc files), LVA-013 (missing §6.Z device evidence for 1080/24 cycles), LVA-014 (device gate durable fixes), LVA-015 (Crashlytics nested-scroll fatal unresolved), LVA-016 (submodule upstream divergence), LVA-017 (stale CONTINUATION.md), LVA-018 (missing script docs per §11.4.18), LVA-019 (coverage ledger partial/gap overlap). GRADLE BUILD GREEN, Go test suite mostly GREEN except the two contract-test failures recorded as LVA-009/LVA-010. NEXT: export Issues.md/Fixed.md, commit/push, fan out subagents on open items.

2026-07-04 (F3 FIXED — api-app x86_64 embed no longer SIGSYS-crashes on startup; on-device search PROVEN live over guest WAN). The embedded `liblavaapi.so` (`GOOS=android GOARCH=amd64`) crashed on start on Android x86_64 with **SIGSYS** — the app-seccomp sandbox blocks the legacy `lstat`/`stat`/`open`/`access` path-syscalls that modernc/libc's musl fast-path issues. FIX: a modernc/libc fork at `lava-api-go/third_party/modernc-libc` (`replace` in `lava-api-go/go.mod`) adds `syscall_musl_seccomp_amd64.go` (+ an `_other.go` stub) that remaps those to the `*at` variants at the single `X__syscallN` dispatcher in `syscall_musl.go`. **Live gate:** SIGSYS=0, `/health` → `{"status":"alive"}`, mDNS `_lava-api-dev._tcp:8443` advertised. **Search PROVEN on device** over the guest WAN — archiveorg×story HTTP 200 ~50 results, archiveorg×linux ~50, torrentscsv×1080p ~25, gutenberg×linux ~1. Evidence `.lava-ci-evidence/sixth-law-incidents/F3-fix-evidence/` (2026-07-04-F3-gate-proof.txt, 2026-07-04-EMBED-SEARCH-WORKS-PROOF.md, 2026-07-04-F3-go-regression.md, result JSONs). **F4 corrected:** the earlier "no guest WAN" was an ICMP false-negative — the guest has validated TCP WAN. **Go regression: NONE** — 4/4 Android ABIs compile, the host embed builds, `internal/{storage,mobile,cache}` + 41 internal packages pass, `vet` clean. **#27 nested-scroll FATAL FIXED** (Crashlytics `c7c8cccad09f72bd7bb95455226109b8`, `IllegalStateException: Vertically scrollable component measured with infinity max height`): `feature/search_result/.../categories/CategorySelectionDialog.kt:45` fed an infinite-height plain `Column` into a `fillMaxHeight()` lazy list → now `Modifier.fillMaxHeight(0.9f)` on the Column + `Modifier.weight(1f)` on `PagesScreen` (compiles). **Still OWED (§6.Q):** a reproduce-first Compose UI Challenge + a §6.Q scanner CHECK 3 for `fillMax*`-bounded lazy layouts under a plain `Column`. **§6.Y bumps:** api-app versionCode 23→24 / versionName 0.2.11→0.2.12 (`api-app/build.gradle.kts`); embed Code 2333→2334 / Name 2.3.33→2.3.34 (`lava-api-go/internal/version/version.go`); client `:app` HELD at 1.3.14-1080 (not yet distributed). **F5 in progress:** the `Challenge70` api-app harness gained UiAutomator support (`androidx.test.uiautomator` in `gradle/libs.versions.toml` + `app/build.gradle.kts`) + a `returnFromApiAppToClient()` helper (advanced the apiapp Challenge FAIL→SKIP), but the api-app's Compose "Back to Lava client" button still resists UiAutomator clicks — a test-automation limitation, NOT a product defect (the on-device search is proven by the direct evidence above). ——— (current)

2026-07-04 (DEVICE GATE UNBLOCKED — the multi-session "device gate broken" was an AVD-NAME MISMATCH; C00 now PASSES with real evidence). A parallel subagent diagnosed the emulator "boot hang" with hard captured evidence: it is NOT a hang. The `api34-x86_64` image bakes exactly ONE AVD named **`default`**; the runner passed **`CZ_API34_Phone`** (from `--avds` AND the default §6.AE.2 `BASE_AVDS`) → the entrypoint pre-check `exit(1)`s in ~4s ("AVD not found") → the container's `--rm` erases the log → `WaitForBoot` (NO container-liveness check) polls the dead forwarded port to its deadline, misreporting the 4s exit as a 5m/12m "boot timeout" (`boot_seconds 301/721` = the WaitForBoot deadline, not a real boot). GPU (swiftshader_indirect, correct) + KVM (ENGAGED — `kvm-vm`+4 `kvm-vcpu` inodes captured) both RULED OUT. **FIX (verified end-to-end): `scripts/run-challenge-matrix.sh … --avds "default:34:phone"`** → the emulator cold-boots under KVM in **44.5s** and **C00 PASSES**: `.lava-ci-evidence/`

2026-07-04T15-03-29Z-challenge-matrix/real-device-verification.json = all_passed:true, gating:true, avd:default, api:34, boot:44.5s, test_passed:true against the fresh 1.3.14-1080 debug APK. **The §6.Z device gate now produces real evidence on this host** — the blocker that stopped EVERY prior session is solved. DURABLE FIXES OWED (incident …2026-07-04-device-gate-emulator-boot-hang.json resolution block): (1) align run-challenge-matrix.sh default BASE_AVDS to the image's baked AVD name (a plain gate run still hits the mismatch — gate-shaping, needs §6.N.1.2 rehearsal); (2) containers pkg/emulator/containerized.go WaitForBoot needs a container-liveness check + podman logs before --rm reaps; (3) full §6.AE matrix (API 28/30/latest) needs those images + baked AVD names (only api34/default local). **DISTRIBUTE PATH now open** (well-scoped): write a covering device Challenge for the search-filter fix (runnable now), strike the egress-blocked archiveorg claim (§6.AK.6), assemble §6.Z evidence + §6.AK cycle-coverage-map, then §6.AA two-stage both apps both flavors. Also this cycle: 3 parallel subagents (emulator diagnosis DONE; Go download+search-param sweep + Kotlin nav-literal dedup landed in worktrees, pending main-stream verify+merge). ——— (current)

2026-07-04 (operator rebuild+distribute directive — device gate BLOCKED by emulator boot-hang on this host; honest §6.Z refusal, NOT a bluff-distribute — SUPERSEDED by the banner above once the AVD-name root cause was found). Operator: "Once EVERYTHING is done, retested fully, validated and verified, rebuild everything and distribute … both apps, both flavors dev and prod." KEY POSITIVE: unlike prior macOS sessions, THIS Linux host HAS /dev/kvm (rw), podman 5.7.1, the emulator image ghcr.io/vasic-digital/lava-android-emulator:api34-x86_64 (local), both google-services.json, LAVA_FIREBASE_TOKEN, firebase CLI 15.16.0 — host-preflight.json = "Gate-host eligible … accel=kvm". So the §6.AH KVM-absence blocker does NOT apply here. Client debug+androidTest REBUILT fresh at HEAD 6cc27c16 (includes the credentials #2 fix). BUT the §6.Z C00 cold-start gate FAILED TWICE (containerized runner resolved to KVM correctly, emulator container started, but the Android guest never reached sys.boot_completed): attempt 1 = 5m timeout (boot_seconds 301), attempt 2 = 12m timeout (boot_seconds 721) → CONFIRMED a boot HANG (not slow-but-progressing). This matches the prior "QA device gate broken — keystones SKIP/FAIL on containerized emulator". The runner behaved CORRECTLY (FAIL + RC=1 + "tag.sh MUST refuse"). **DISTRIBUTE IS DEVICE-GATE-BLOCKED** — per §6.Z/§6.AK/§6.L a distribute without device evidence is the exact 1.2.19-1039 bluff, so it is REFUSED (the §6.L-correct response, as in the 23rd/67th cycles).

PENDING_FORENSICS (candidate causes, incident `.lava-ci-evidence/sixth-law-incidents/2026-07-04-device-gate-emulator-boot-hang.json`): (a) emulator GPU mode in the headless container (launch flags live in the image entrypoint, not `containerized.go`), (b) KVM not actually engaged inside the rootless-podman namespace despite `--device /dev/kvm` (TCG fallback → >12m boot), (c) image/kernel-6.12 incompatibility. Remediation: focused Containers-submodule boot diagnosis (manual boot + adb logcat + verify qemu `-accel kvm` inside the container) OR run the gate on a known-good host (prior cycles booted on “thinker”/“nezha”). ALSO owed for a §6.AK-honest 1080 distribute: a covering device Challenge for the search-filter fix (NONE exists — C59/C60/C70 cover chips, not sort/order/period), striking the egress-blocked archiveorg claim, and the other API images for the full §6.AE matrix (only api34 local). 1080’s fixes are committed+pushed+unit-verified, NOT device-gated. ——— (current)

2026-07-04 (Crashlytics FATAL #2 FIXED reproduce-first — credentials decrypt AEADBadTagException no longer crashes the app; live-Crashlytics review of the distributed build). Reviewed Firebase Crashlytics for the live client release (`lava-vasic-digital`, app `…456475e2…`, 2-week window) — 3 OPEN FATAL/ANR issues. Triaged all 3 with real evidence: **(#3 `eaa80c14…` `ProviderConfigViewModel$WireToggle` SerializationException, 1.3.11)** = ALREADY FIXED (regression test `ProviderConfigWireSerializationTest.kt` + `build.gradle.kts:8` comment reference; lastSeen 1.3.11 = stopped) → console close-mark owed, no code change. **(#1 `c7c8ccc…` nested-scroll IllegalStateException, §6.Q anchor, lastSeen 1.3.12)** = still OPEN, needs the crash-stack screen to locate the residual violation (sample event not retrievable via API this session) — PENDING_FORENSICS. **(#2 `4e11da49…` `CredentialsCrypto.decrypt` AEADBadTagException BAD_DECRYPT, FATAL, 1.3.12)** = GENUINELY OPEN in current code → **FIXED reproduce-first**: `CredentialsEntryRepositoryImpl.observe()/get()` deliberately let `AEADBadTagException` propagate (a 1.3.10 comment ASSUMED it’d become a non-fatal; Crashlytics proved it FATAL). Now `decodeOrNull(entity)` catches narrowly (`GeneralSecurityException` + the too-short-payload `IllegalArgumentException`), records a §6.AC non-fatal (no secret material, §6.H), and SKIPS the corrupt row — one bad credential row can no longer crash the app nor hide the valid ones; non-corruption bugs still propagate. RED→GREEN: new test `observe skips a corrupt-ciphertext row…` RED `AEADBadTagException at :175` → GREEN (6/6). §6.AC scanner 0 violations. Closure log `.lava-ci-evidence/crashlytics-resolved/2026-07-04-credentials-decrypt-aeadbadtag-fatal.md`. §6.O Challenge (device end-to-end)

OWED, §6.AH-blocked. Also this session: root-caused + fixed the push rejection (§11.4.65 `CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC` — the 1080-cycle doc edits lacked regenerated `.html/.pdf` siblings; regenerated CHANGELOG/CONTINUATION/BUGFIXES exports). Subagent fleet HARD-BLOCKED (provider session limit resets 4:10pm Europe/Moscow) — 3 parallel worktree agents (Go download-route bug sweep, Kotlin nav-literal dedup, Go search-param honoring) all died on quota; their targets remain workable when quota resets. **§11.4.36 closure — vision_engine:** the push was ALSO blocked by the `canonical-root-and-upstreams` gate (STRICT) — the own-org `vision_engine` submodule lacked `install_upstreams.sh` (§11.4.36/§6.W). CONFIRMED clean-at-pin (0 dirty, 0-ahead; the prior "157-ahead/must-not-bump" note no longer holds), so closed it the sanctioned HelixQA way: added `install_upstreams.sh` + `Upstreams/{GitHub,GitLab}.sh` in the submodule, committed `0bf75ee→b417a40`, pushed to BOTH mirrors (github+gitlab fast-forward), and advanced the parent pin. Gate now `missing: 0 / all clauses pass.` ——— (current)

2026-07-04 (Lava-Android-1.3.14-1080 cycle — search flow Bug 1+2 FIXED, goapi archiveorg download 502 FIXED, §6.C converged, ci.sh –changed-only GREEN; distro blocked on §6.Z device gate). Continued from compaction; 6 commits on master at HEAD `00cb2188`. **Committed (all pushed to both mirrors, §6.C converged):** (1) `820790e9` version bump 1079→1080 (§6.Y post-distribute). (2) `05d14d4a` goapi TrimPrefix leading slash for Gin catch-all `*id` — archiveorg `/http-download` 502 root cause FIXED + regression tests (contract + Go unit). (3) `441e321e` search sort/order/period propagation from SavedStateHandle into SearchRequest — Bug 1 + Bug 2 FIXED. (4) `fc7ca6a5` 2026-07-04 autonomous-QA cycle evidence committed (search-flow + credentials-crash traces, 12 providers tested on goapi nezha gate-host). (5) `1a6ef204` tracker client path assertions + Log mock updated for buildUrl() changes (§6.Y-1080). (6) `00cb2188` chore(deps): advance 5 own-org (auth, containers, database, observability, security) + constitution submodule pins. **ci.sh –changed-only GREEN** (BUILD SUCCESSFUL, 339 tasks, all gates PASS). **§6.Z test-evidence file NOT yet created for 1080** (blocked: QA device gate — all provider keystones SKIP/FAIL on containerized emulator, root cause pending diagnosis). **§6.AA two-stage Firebase distribute BLOCKED** by §6.Z evidence + operator credentials (google-services.json, LAVA_FIREBASE_TOKEN). vision_engine submodule adopted (GitLab remote pending). Two subagents still running in background: LVA-083 (search zero-results P0 deep dive — reading ProviderLoginViewModel, PreferencesStorage, ApiKeyClient), LVA-081 (coverage-ledger.yaml regeneration + ci.sh worktree verification).

> **CONTINUING BLOCKERS (unchanged from 1079 cycle):** device gate can't produce real per-AVD attestations; distribute blocked until evidence file exists + operator provides Firebase creds. **NEXT:** finish CONTINUATION.md → check subagent outputs → commit coverage-ledger.yaml → address LVA-083/LVA-081 findings → retry device gate. ——— (current)

> **2026-07-03 (Firebase Stage-1 debug DISTRIBUTED 1.3.13-1079 + a QA-harness anti-bluff CAUGHT & FIXED mid-distribute).** Continued the redistribute. Per-release auth rotations: §6.H Gate 4 pepper → new SHA `8426f84d`; Gate 5 client → `android-1.3.13-1079` added to `LAVA_AUTH_ACTIVE_CLIENTS` + set current. §6.Z evidence against the fix commit: C00 cold-start on the FULL §6.I matrix (5 AVDs — API 28/30/34/35, phone+tablet — all PASS, gating=true) + kinozal `.torrent` keystone `verdict=PASS`. `firebase-distribute.sh --app client --debug-only` uploaded **1.3.13-1079 to `.client.dev`** (Firebase release `0rtq7tapl4ia0`; Gates §6.P/4/5/7 all pass). **ANTI-BLUFF FOUND + FIXED (§6.J/§6.L incident):** `run-matrix` reported `verdict=PASS` (PASS=1) with the client APK MISSING and NO test run — it read a STALE prior-run `verdict.json`; `run-iteration` does NOT build the APK (expects pre-built) and on its exit-2 left the stale verdict. This ALSO revealed the pepper/client rotation re-verifications had reused the STALE pre-rotation APK (gradle didn't rebuild on a `.env`-only change). REMEDIATED: force clean-rebuild (`rm` codegen → `assembleDebug`) produced the fresh rotated APK (sha256 `d538e591`); the keystone was RE-RUN against it with a FRESH verdict (serial 53187) + HARD logcat evidence ("Download completed" ×13); ONLY that verified APK was staged + distributed. HARNESS FIX: `run-iteration` writes a FAIL verdict on missing-APK; `run-matrix` `rm`'s the stale verdict before each iteration + fails-closed on missing-verdict / non-zero exit. Bluff-Audit rehearsed (stale PASS → FAIL). Incident: `.lava-ci-evidence/sixth-law-incidents/2026-07-03-qa-harness-missing-apk-stale-verdict-pass-bluff.json`. The #10 `.torrent` fix itself was never in doubt (independent hard "Download completed" evidence across every run). **§6.AA Stage-2 (release/prod) DONE:** release APK (R8/minify, sha256 `1b12e21b`) built + the R8 cold-start canary PASS (fatal=0, process alive pid 4036, `digital.vasic.lava.client/.MainActivity` on top — via the NEW `scripts/autonomous-qa/release-coldstart-canary.sh`, since there is no `testBuildType=release` so the keystone can't run on the minified APK) + `firebase-distribute.sh --app client --release-only` uploaded to prod `.client` (Firebase release `7rj8e4om1beh8`). **BOTH prod flavors now on Firebase: debug `.client.dev` (`0rtq7tapl4ia0`) + release/prod `.client` (`7rj8e4om1beh8`).** api-app is already at its released version (0.2.11-23, debug+release both distributed); the `.torrent` fix is client-side so the api-app

> needs no new release. **OPERATOR: install the Firebase builds on-device and confirm — §6.AA clause 2 / §6.Z clause 5: the human real-device observation is the final release signal.**

> **2026-07-03 (`.torrent` download FIXED for ALL providers — kinozal DEVICE-GREEN; handoff's #10 root-caused + resolved).** Resumed the handoff's uncommitted reroute + the still-broken #10. ROOT-CAUSED #10 definitively (an advance over the handoff's "goapi session-forwarding failing"): the reroute fetched the `.torrent` bytes in-app via `ProxyNetworkApi.download` → the goapi ROOT `/download/:id`, which is **rutracker-ONLY** (`cmd/lava-api-go/main.go:140` scraper = `rutracker.NewClient`; `internal/handlers/torrent.go GetDownload` → `h.scraper.GetTorrentFile` → rutracker.org). So rutracker `.torrents` worked but kinozal fetched from rutracker.org with a kinozal id/cookie → HTML error page → bencode-guard 502 → FAIL. FIX (subagent a0736b7e, approach A = SDK provider-aware path mirroring the device-green gutenberg HTTP-file download): new `TorrentDownloadSource` seam → `sdk.downloadTorrentFile(trackerId, id)` → `ApiBackedTrackerClient.downloadable` → `/v1/{trackerId}/download/:id` (both auth gates, per-provider session); `DownloadTorrentUseCase(trackerId, id, title)`; `TopicViewModel` threads `providerId`; the superseded root-route byte-fetch reverted; the reroute's bencode-validate + public-Downloads write preserved. Approach B (thread trackerId into ProxyNetworkApi/root) was rejected: it sends the single-tracker `tokenProvider` token (rutracker session), wrong-formatted for kinozal. **654 core+feature JVM tests GREEN + 3 Bluff-Audit stamps. DEVICE-GREEN (§6.AK):** kinozal 1080p `.torrent` keystone `verdict=PASS` ("Download completed" ×15, C70-RESULT DOWNLOAD-OK) on the containerized emulator (API 34) — answering the handoff's open question: the client's kinozal onboarding-login session IS valid for the download. §6.H CLEAN (no token logging anywhere in the login-touching client — the handoff's #9 is moot). §6.Y bump 1.3.12-1078 → 1.3.13-1079 (user-facing download fix). Device-green goapi matrix: rutracker (1080p+mp3+`.torrent`), kinozal (1080p+`.torrent`), gutenberg (sherlock HTTP-file). **STILL GATED (unchanged):** public IPv4-only providers need the per-destination excluded-proxy egress toggle (operator infra decision — a global bypass breaks rutracker/kinozal); the Firebase redistribute needs operator-provided real creds (`google-services.json`, `LAVA_FIREBASE_TOKEN`) + the §6.Z pre-distribute gate + §6.AA two-stage — distributing with placeholder creds or a broken download is the exact §6.Z bluff the mandate forbids. ——— (current)

2026-07-03 (§6.AC archiveorg mapError hardening — egress-stall-masquerade class CLOSED; matrix banked, public providers still gated). The one genuinely-reachable non-gated item from the 2026-07-02 handoff is done reproduce-first. Root cause of the 2026-07-02 diagnostic trap: `lava-api-go/internal/archiveorg/provider.go` `mapError()` collapsed EVERY error to a bare `provider.ErrUnknown`, discarding the real cause — so when the containerized goapi's outbound TCP to IPv4-only `archive.org` stalled over the Mullvad VPN (a dial/TLS timeout), `writeProviderError`'s §6.AC default branch recorded a `RecordNonFatal` whose `error_message` was the useless generic `"provider: unknown error"`, and the egress stall MASQUERADED as an archiveorg parser bug. FIX (`mapError → fmt.Errorf("%w: %v", provider.ErrUnknown, err)`): `errors.Is(_, provider.ErrUnknown)` stays true so the HTTP status (502) + the recorded non-fatal + the `{}` wire body are ALL unchanged, while the real dial/TLS detail now SURVIVES for remote triage. archiveorg was the ONLY discard-everything adapter (nnmclub/kinozal/rutracker already preserve via `default: return err` — the prior banner's "also nnmclub/kinozal/rutracker" was imprecise; they were NOT touched). Reproduce-first proven: `internal/archiveorg/provider_test.go` `TestMapError_PreservesUnderlyingCause` RED `mapError discarded the underlying cause: got "provider: unknown error", want it to contain "i/o timeout"` → GREEN (archiveorg pkg 15/15 + `TestMapError_NilPassthrough`; v1 handler error-mapping + non-fatal-telemetry green = no status/wire regression; `go vet` clean; `go build ./...` OK). Observability-only — banked device-green goapi matrix UNCHANGED: rutracker (1080p+mp3), kinozal (1080p), gutenberg (sherlock). **STILL GATED (operator-present required, unchanged from handoff):** (1) public IPv4-only providers (archiveorg/nyaa/torrentscsv/bitsearch/yts/…) need the per-destination excluded-proxy egress toggle — a GLOBAL bypass breaks rutracker/kinozal (VPN-exit-IP-only) so it is a public-ONLY-run infra decision; (2) QA-harness fixes #1/#3 need an on-device falsifiability rehearsal (emulator in container/VM per §6.AH); (3) api-app Phase 3 = guest-WAN egress blocker. NEXT when operator returns: pick the excluded-proxy toggle → I run the public-provider keystones; otherwise the device-green matrix stands as banked, real, recorded proof. ——— (current)

2026-07-02 (public-provider phase — archiveorg 0-rows ROOT-CAUSED to VPN egress (NOT code); gutenberg SEARCH green + TOPIC layer open; 6 providers wired). After kinozal went fully green (banner below, pushed `99baa334`), the public-provider keystones surfaced two DISTINCT issues, both diagnosed with captured evidence (no bluff): **(A) archiveorg/nyaa/**

torrentscsv return 0 rows = ENVIRONMENT/EGRESS, NOT a parser bug — OPERATOR ACTION NEEDED. PROVEN: the EXACT goapi archiveorg request replicated from the host returns HTTP 200 numFound=15582 (parser/query/URL correct; the recaptured fixture is real). The containerized goapi's outbound TCP to IPv4-only upstreams STALLS over a Mullvad WireGuard VPN that force-routes its egress (`ss`: ESTAB with Send-Q backed up, ClientHello never ACKed → TLS never completes → HTTP 502 → rendered "0 rows"). Per-host: goapi OK for gutenberg (gutendex dual-stack → Go uses IPv6) but 0 for archiveorg/nyaa/torrentscsv (IPv4-only). rutracker+kinozal happened to be reachable → their keystones passed. FIX (REFINED 2026-07-02 — a GLOBAL bypass is UNSAFE): a host process reaches archive.org BOTH direct AND through the VPN (`curl` → HTTP 200); only the CONTAINERIZED goapi stalls on IPv4-only hosts (rootless-podman fwmark/cgroup differs from the host's VPN include-rule). `mullvad split-tunnel add <goapi-pid>` did NOT take effect (a podman-managed cgroup can't be reclassified after launch). `mullvad-exclude curl archive.org` → HTTP 200 numFound=15599 (premise confirmed). BUT a GLOBAL egress bypass (mullvad-exclude the whole goapi, or a global `LAVA_API_UPSTREAM_PROXY`) would route the Russian trackers DIRECT too — and rutracker/kinozal work ONLY through the VPN exit-IP, so a global bypass would BREAK the banked-green providers. The CORRECT fix is PER-DESTINATION: the goapi already has provider-aware routing (`internal/httpx/provider_routing_integration_test.go` — Russian trackers traverse the proxy, others direct); wire the public IPv4-only providers (archiveorg/nyaa/torrentscsv) to an EXCLUDED egress (a small `mullvad-excluded` local proxy pointed at ONLY those hosts) while the Russian trackers keep the VPN path. NOT a parser patch, NOT a global bypass — a per-destination routing/infra change (operator decision; `socat` is available, no HTTP-proxy binary is). PARSER AUDIT 2026-07-02 (subagent, evidence `.build-tmp/public-provider-parser-audit-2026-07-02.md`): all 4 public parsers — nyaa/torrentscsv/bitsearch/yts — CONFIRMED REAL (real fixtures + real selectors; live rows 75/25/20/3; yts fixture is a byte-exact live capture); kinozal was the ONLY parser bluff. The egress is NON-UNIFORM: nyaa.si NEEDS the VPN (direct TLS handshake is DPI-blocked; works 200 through the VPN) while archiveorg NEEDS a bypass (IPv4 stalls through the VPN) — so a single GLOBAL policy CANNOT satisfy both, and per-destination routing is MANDATORY, not optional. Secondary §6.AC gap: `archiveorg/provider.go mapError()` (also nnmclub/kinozal/rutracker adapters) DISCARDS the real dial/TLS error → always `ErrUnknown` → the egress stall MASQUERADED as a parser bug; harden with `fmt.Errorf("%w: %v", ErrUnknown, err)` + `RecordNonFatal`. Full evidence: `.build-tmp/archiveorg-search-0rows-2026-07-02.md`.

(B) gutenberg SEARCH + TOPIC + download-URL are ALL correct server-side (REFINED 2026-07-02) — the goapi returns a real direct ebook URL (`GET /v1/gutenberg/topic/1342` → `downloadUrl: https://www.gutenberg.org/ebooks/1342.epub3.images`). ROOT-CAUSED 2026-07-02 (definitive): `marker_download_ok:false` is the HTTP-file DOWNLOAD dead-ending — Challenge70:628 taps "Torrent" (topic renders it) but NO DownloadDialog is shown. TWO gaps: (1) CLIENT `ApiBackedTrackerClient.kt:268` returns `HttpDownloadableTracker::class -> null` ("does not yet surface these feature families") so `sdk.downloadHttpFile` → getFeature → null → no dialog; (2) goapi `handlers/v1/handlers.go:97-98` gates BOTH download routes (`/torrent/:id`, `/download/:id`) on `CapTorrentDownload` ONLY, which HTTP-file providers (gutenberg/archiveorg) lack — the generic handler + `gutenberg/download.go DownloadBook()` already work, only the ROUTE GATE excludes them. FIX LANDED 2026-07-02 + unit/contract-VERIFIED: client `httpDownloadable(HttpDownloadableTracker)` (`ApiBackedTrackerClient.kt:225` + getFeature wiring `:286-287`) requests the NEW `CapHTTPDownload`-gated goapi route `GET /http-download/*id` (`handlers/v1/handlers.go:110`, reusing `torrent.GetDownload` → `provider.DownloadFile` → `gutenberg.DownloadBook`; `*id` wildcard because archiveorg ids contain `/`). goapi **3/3 HTTP-download tests PASS** (real capability gate via `setupRegistryRouter` + served bytes + `Content-Disposition` filename + 501-discrimination guard) + full v1 suite green (no regressions); client contract test **BUILD SUCCESSFUL**. gutenberg.org reachable through the VPN (HTTP 200, 24.8MB epub, 17.8s — NOT egress-blocked, unlike archiveorg). DEVICE-GREEN CONFIRMED 2026-07-02: keystone `verdict=PASS` (tests=1 fail=0 skip=0), `marker_download_ok=true`, `"Download completed" ×3` — the REAL 24.8MB gutenberg epub downloaded on the containerized emulator (API 34, x86_64) to terminal completion, video-recorded. §6.AK RED-then-GREEN SATISFIED (was device-FAIL `marker_download_ok:false` download-dead-end → now device-PASS). goapi `/http-download/1342` verified independently first: HTTP 200, 24,835,597 bytes, `Content-Disposition: attachment`. gutenberg now FULLY green (search→topic→download) alongside rutracker + kinozal. Because it reached TERMINAL "Download completed" (not the in-progress fallback), QA-harness fix #1 (require terminal completion) is now LANDED + fully on-device rehearsed: green-path re-run `verdict=PASS` (gutenberg "Download completed" ×9 under the stricter gate); discrimination PROVEN by a 2s-timeout rehearsal → `verdict=SKIP` ("AssumptionViolated"/"Honest SKIP" — an in-progress download is now an honest SKIP, the EXACT false-PASS the pre-fix `require(completed || presentContains("Downloading file"))` reported as a green PASS). The QA-harness audit's real false-PASS vector is closed. Fix #3 (scope onboarding SKIP to NETWORK stages) is now LANDED + on-device rehearsed: a sentinel `OnboardingNetworkUnavailable` is thrown ONLY by the two network onboarding stages (ApiSelection connectivity probe + provider login/configure), the outer catch is narrowed to it (those stay honest SKIPs), and a pure-UI stage timeout throws a raw ComposeTimeoutException that PROPAGATES as a loud FAIL. Discrimination PROVEN: mutating the Welcome (UI) wait → `verdict=FAIL` (raw ComposeTimeout propagated; sentinel=0, SKIP=0; other_failure_signal=false) — pre-fix that same timeout was masked as SKIP. Green onboarding path unbroken: the rutracker + gutenberg green-path runs both traversed EVERY onboarding stage incl. the login/probe network stages (16 login events) and reached the download step; the sentinel was never thrown (fix #3 doesn't spuriously skip a working login). NOTE: those full keystones FAILed DOWNSTREAM on transient network slowness this session (gutenberg.org 29s epub fetch + a rutracker search-timeout), NOT on onboarding — a full end-to-end PASS re-confirm is environment-pending (network flakiness late in a 9h session), unrelated to fix #3. Both QA-harness false-PASS/masked-FAIL vectors (#1 + #3) are now closed; only #2 was mechanical. Also unblocks archiveorg's download PATH (SEARCH stays separately per-destination-egress-blocked). MATRIX EXPANSION 2026-07-02: rutracker mp3 keystone ALSO device-green (`verdict=PASS`, `"Download completed" ×14`, real music `.torrent`) — rutracker now covers BOTH spec queries (1080p + mp3). Device-green goapi matrix: rutracker (1080p + mp3), kinozal (1080p), gutenberg (sherlock) — search→topic→download proven on real emulators. **(C) 6 public providers WIRED but not yet device-run** (worktree `agent-ae03137803ae71457`): thepiratebay/torrentscsv/knaben/bitsearch/nyaa/tokyotosho added to Challenge70 `ProviderSpec.forId` (authNone) + `lib-subsets.sh QA_PROVIDERS` (compileDebugAndroidTestKotlin BUILD SUCCESSFUL); KEY: bitsearch renders **“Bitsearch”** (humanized, not “BitSearch”). NOTE: torrentscsv+nyaa are among the IPv4-only egress-blocked set (A) → they need the VPN fix before their keystones can pass. TRIAGE 2026-07-02 (running goapi, direct search-endpoint curl): of ALL public providers ONLY gutenberg returns rows (32); archiveorg/nyaa/torrentscsv/bitsearch/yts/thepiratebay/knaben/tokyotosho ALL return 0 through the container's VPN egress — so NO additional public provider is device-runnable until the per-destination egress lands (running their keystones now would FAIL at the search step, not go green — NOT run, per §6.J). The excluded-proxy unblock IS viable (LAVA_API_UPSTREAM_PROXY is GLOBAL — curated providers construct `&http.Client{Transport: httpx.NewTransport()}` which honors it) for torrentscsv/bitsearch/yts/archiveorg (NOT nyaa — DPI-blocked on direct per the parser audit), BUT it also routes the Russian trackers direct → breaks rutracker/kinozal while active, so it is a PUBLIC-ONLY-RUN toggle (operator/infra decision), not a standing config. The 6 providers remain worktree-only (`agent-ae03137803ae71457`); main `QA_PROVIDERS` is still the original 5. **NEXT (when rate limits clear):** (1) operator applies the VPN split-tunnel for the goapi → re-run archiveorg/nyaa/torrentscsv keystones; (2) complete the gutenberg-topic parser fix + re-run; (3) apply the mapError §6.AC hardening; (4) run the 6 wired providers' keystones (reachable ones: thepiratebay/knaben/bitsearch/tokyotosho). rutracker + kinozal remain FULLY green + pushed. ——— (current)

2026-07-02 (kinozal FULLY GREEN — verdict=PASS end-to-end; round-3 parallel-subagent sweep landed). The kinozal goapi keystone now passes the COMPLETE flow on device: search → open details → download. Evidence: `topic/:id/page -> HTTP 200`, `[iter] verdict=PASS (fail=0 skip=0)`. THREE kinozal layers, all §6.D/§6.J bluffs or gaps, all reproduce-first + device-verified: (1) SEARCH parser (fictional `table.tumblers`/`a.namer`/Latin GB + synthetic fixture — fixed round-2, 50 real rows); (2) TOPIC parser (synthetic fixture, fictional `a.magnet`/`div.content` — fixed: real `div.bx1`/`img.p200`/`download.php?id=` + real anonymous win-1251 fixture, RED `DownloadURL=""` → GREEN); (3) missing goapi route `GET /v1/{provider}/topic/:id/page` — the client's getTopicPage 404'd → fell back to the legacy rutracker-only `/topic2/{id}` → strict `TopicPageDto` decode threw for non-rutracker providers (rutracker passed only because its fallback IS rutracker); added the route serving the tolerant `TopicDetailDto` shape (RED 404 → GREEN 200, live-verified). **Round-3 parallel-subagent items (all real-evidence, no bluff):** torrentdownloads DROPPED from the goapi catalogue (re-probed DEAD 522×3 — dead upstream; unregistered + negative-guard tests); yts + archiveorg synthetic fixtures REPLACED with real sha256-verified captured API responses (§6.D honesty, 3 falsifiability rehearsals); nnmclub topic-description selector fixed (`#pagecontent`→first `div.postbody`; the audit's `#contentdiv` was the hidden spoiler — real fixture, RED empty→GREEN) — nnmclub LOGIN remains Turnstile-blocked (FlareSolverr wired but structurally can't solve in-form Turnstile → needs a paid token solver; honest, not faked); topic/download parser audit found NO other kinozal-class bluff. Committed here (embed `.so` rebuilt + `api-source.hash` refreshed). **goapi provider status: rutracker + kinozal FULLY green; archiveorg/gutenberg/thepiratebay/torrentscsv/knaben/bitsearch/nyaa/tokyotosho parsers OK (device runs next); torrentdownloads dropped; nnmclub login Turnstile-blocked; rutor/iptorrents not-goapi-served.** ——— (current)

2026-07-02 (parallel-subagent round — kinozal SEARCH bluff FIXED + device-proven; 6 workable items driven to evidence-backed resolution). Up to 5 parallel subagents on all workable items (operator “more agents, increase efficiency”). All results real-evidence-backed, no bluff: - **kinozal SEARCH — FIXED (device-proven).** Root cause = a §6.D/§6.J BLUFF: `lava-api-go/internal/kinozal/search.go` used FICTIONAL selectors (`table.tumblers`/`a.namer`/Latin `GB`) that never match real kinozal.tv (`table.t_peer`/`td.nam a`/Cyrillic `ГБ`), and the test fixture was SYNTHETIC (reverse-engineered from the broken parser) → green while every user got 0 rows. Fixed: real selectors + a REAL anonymously-captured win-1251 fixture + reproduce-first tests (RED `Results len=0` → GREEN `50 rows`). goapi rebuilt: `curl /v1/kinozal/search?q=1080p` → 50 REAL rows (was 0); device: real result `id=2145735 "Аферисты / 2007 … (1080p)"` RENDERED + CLICKED. **NEXT LAYER (in flight):** the kinozal TOPIC parser is ALSO stale — clicking a result → `TopicPageDto MissingFieldException: [id,title,author,category,torrentData] missing` (rutracker topic works → it’s the goapi kinozal topic parser, same bluff class). Subagent fixing reproduce-first → re-run to full green. - **rutor — CLOSED (false alarm).** Bundled-client-only (no `lava-api-go/internal/rutor/`); correctly excluded from `--subsets all`; the SKIP came from a MANUAL `rutor;…` invocation on the unvalidated CSV branch. `run-matrix.sh` hardened to filter manual subsets against `QA_PROVIDERS` (unit-verified: rutor/iptorrents dropped, kinozal,nnmclub kept). Porting rutor to Go is separate multi-day work. - **`-cmd=build` — CLOSED (real cause found).** Go source was already clean (`170045a0`, 2026-05-06); the REAL bug is `start.sh:70`‘s stale-binary guard (rebuilt only when ABSENT → a pre-fix stale `bin/lava-containers` kept running the deleted `:proxy:buildFatJar`). Fixed: rebuild on any newer `.go`; testable `buildImageArgs` + `TestBuildImageArgsHaveNoProxyStep`; go build/vet/test PASS + falsifiability proven. - **nnmclub — root-caused HONEST (external block, NOT a code bug).** `/v1/nnmclub/login`→HTTP 401 `{}` because nnmclub.com gates login with Cloudflare **Turnstile** (captured `cf-turnstile` + “неверный код подтверждения”). Plain-HTTP login cannot pass it. FlareSolverr wiring (sidecar + route login + §6.E gating) IN FLIGHT. nnmclub NOT claimed working. - **goapi parser bluff-audit — kinozal was the ONLY parser bluff.** All 11 others’ selectors match live upstream. Secondary: **torrentdownloads BLOCKED** (dead upstream `rss.xml`→522; drop/mirror); yts+archiveorg synthetic-but-structurally-correct (optional §6.D recapture, not bluffs). - **api-app (Phase 3) readiness mapped.** Embed registers the identical set (`mobile.go` parity), so kinozal+rutracker fixes flow in on next `:api-app:assembleDebug`. #1 apiapp blocker = guest-WAN egress (embed inside emulator QEMU slirp — probe guest→tracker before a Phase-3 run). APK source-stale (rebuild after landing fixes). Kinozal search fix + rutor + `-cmd=build` committed here (embed `.so` rebuilt + `api-source.hash` refreshed); kinozal-topic + FlareSolverr land next. rutracker keystone stays GREEN + pushed. ——— (current)

2026-07-02 (generalization batch, pushed `687232fc`) — fixes C2/D/E PROVEN to generalize beyond rutracker; remaining failures are now PROVIDER-SPECIFIC, not systemic. Batch `rutor;kinozal;nnmclub × 1080p` on the goapi backend = **PASS 0 / SKIP 2 / FAIL 1**, each a distinct actionable next-layer (rutracker keystone stays GREEN + pushed). **KEY WIN — kinozal generalizes:** device logcat `login(kinozal) result=LoginResult(state=Authenticated, sessionToken=uid=…;pass=…)` → `Finish` → search — kinozal (FORM_LOGIN, win-1251, a completely different provider) logged in + onboarded + searched with ZERO kinozal-specific work, because C2 (AUTH_REQUIRED derivation) + D (IO dispatcher) + E (LoginResultDto wire) were architectural fixes. **Per-provider next-layers (evidence `.lava-ci-evidence/autonomous-qa/2026-07-02/goapi/{rutor,kinozal,nnmclub}-1080p/`): (rutor SKIP)** rutor is NOT in the goapi `/providers` catalogue (the 13-provider catalogue = archiveorg,thepiratebay,yts,bitsearch,nyaa,tokyotosho,rutracker,nnmclub,kinozal,gutenberg,torrentscsv,knaben,torrentdownloads — no rutor) AND Challenge70 `ProviderSpec.forId` has no rutor case (assumption "qa_providers resolved to no known providers (expected rutracker,nnmclub,kinozal,archiveorg,gutenberg)") → rutor×goapi is NOT-APPLICABLE: either add rutor to the goapi provider registry OR exclude rutor from goapi subsets in `lib-subsets.sh`. **(kinozal FAIL)** login+onboarding+search ALL ran; the FINAL assertion failed — the results list was empty (`SearchResultViewModel: Perform ListBottomReached` fires immediately = empty list), consistent with the documented kinozal Cloudflare-522 flakiness / win-1251 search-parse; NEXT: with the goapi up, `curl /v1/kinozal/search?q=1080p` (Auth-Token=`kinozal:cookie:uid=…;pass=…`) to classify (kinozal.tv 5xx/Cloudflare vs empty-parse vs win-1251 mojibake). **(nnmclub SKIP)** goapi `/v1/nnmclub/login` returned `ApiHttpException: HTTP 401` — a real goapi→nnmclub.com login failure (NOT a client bug); NEXT: with the goapi up, `curl /v1/nnmclub/login` with the real creds to classify (Gate-1 Lava-Auth key vs nnmclub credential-reject vs Cloudflare vs win-1251-login-encoding). All three are the continuing loop's next iterations. ——— (current)

2026-07-02 (SUCCESS) — goapi × rutracker keystone is GENUINELY GREEN on a real containerized-KVM emulator: `verdict=PASS`, `marker_download_ok=true`, video-recorded (`qa_rec_0.mp4`). Real rutracker login (`bb_session` cookie) → real search "1080p" (`Torrent id=6876600 "Adventure Time: Side Quests"`) → download link. ZERO bluff. The prior "onboarding-completion stall" was the tip of a FIVE-layer stack of real defects, each surfaced only after the previous was fixed — the device keystone was a genuine bug-discovery engine (three of the five are REAL PRODUCTION BUGS a goapi-catalogue user would hit, all invisible to the prior green suite). Fixes (all reproduce-first + Bluff-Audited): **(B)** `lava-api-go` rutracker `DisplayName()` `"RuTracker"`→`"RuTracker.org"` — the name equalled the id case-insensitively so `ProviderCatalogRepository.friendlyDisplayName` humanized it to off-brand `"Rutracker"`, diverging from the bundled name + breaking the onboarding pick; **(C2)** `ProviderCatalogRepository.toRemoteDescriptor` now DERIVES `AUTH_REQUIRED` from `authType != NONE` (§6.E honesty — the catalogue declares `authType:CAPTCHA_LOGIN` but omitted the capability, so `ApiBackedTrackerClient.getFeature<Authenticatable>()` was null → `sdk.login` short-circuited to null → no cookie); **(D)** `ApiBackedTrackerClient`'s blocking OkHttp `.execute()` helpers (getString/getBytes/postJson/healthCheck) now run on an injected `Dispatchers.IO` — `sdk.login` runs on `viewModelScope`/Main so the un-dispatched blocking call threw `NetworkOnMainThreadException` (a REAL crash on every goapi login/search/download); **(E)** client `LoginResultDto` now mirrors the REAL wire `provider.LoginResult {success, authToken, expiresAt}` — the prior `{state, sessionToken}` required a `state` field the server never sends → `MissingFieldException` → a successful login surfaced as "Connection test failed", no cookie stored (the OpenAPI `/login` `AuthResponseDto {type,user}` schema is ALSO stale — a separate contract-drift to reconcile, NOT blocking); plus a test-harness fix — Challenge70 deselects the whole list via `SelectAllProvidersTestTag` then re-selects requested via `performScrollTo` (robust to catalogue casing + off-screen rows; the prior name-list deselect left 8 of 13 default-selected providers selected → wizard stranded on a non-requested Configure). Device evidence: `.lava-ci-evidence/autonomous-qa/2026-07-02/goapi/rutracker-1080p/` (raw video/logcat/png gitignored per operator directive; curated `verdict.json`+`junit.xml` committed). **NEXT:** extend the keystone to rutor/kinozal/nnmclub + mixed subsets, then the on-device `:api-app` backend (Phase 3), then §6.AK-gated dev+prod Firebase distribute. ——— (current)

2026-07-02 (culmination) — goapi path PROVEN TO ENGAGE on device (verdict FAIL→SKIP, ZERO bundled fallback); root cause was `apiSelectionEnabled=false` in Challenge70, NOT providerIds. CORRECTION to the prior banner: `providerIds` was `[rutracker]` all along (the `ListBottomReached` was a Streaming-spinner no-op, misread). The TRUE root cause (a92db298 subagent, quoted file:line): `TestOnboardingHiltModule` binds `apiSelectionEnabled` from `ApiSelectionTestFlag` (default FALSE); Challenge70 never set it true → the ApiSelection onboarding step was SKIPPED → the Go API endpoint was never configured → catalogue never fetched → dynamic client never registered → search fell to the bundled direct client (`tracker.php?nm=…&start=-50`, the `50*(page-1)` 1-indexed builder vs page=0). **FIXES (this cycle):** (1) Challenge70 now sets `ApiSelectionTestFlag.enabled=true` via an ordered `@get:Rule(order=2)` BEFORE the compose rule (mirrors C39/C40/C63); (2) `RuTrackerInnerApiImpl` `start` computations gain `.coerceAtLeast(0)` (defensive — a negative tracker.php offset is never valid); (3) nnmclub 4-site win-1251 follow-up (`0b0885ac`). **DEVICE-PROVEN this cycle:** `OnboardingViewModel: Perform AddCloudApi → ApiSelection: probe OK for GoApi(host=127.0.0.1,port=8443,key=<qa_key-injected>) — advance to Providers`, and **0 bundled `tracker.php` search reqs** (was 6). So qa_key + AuthInterceptor-fail-open + ApiSelection ALL engage — the goapi auth chain works end to end through the keyed probe. **REMAINING LAYER for SUCCESS (verdict is an HONEST §6.J assumeTrue SKIP, NOT a bluff — the Challenge refuses to claim a green it didn't earn):** after "advance to Providers" the onboarding stalled (no app activity for 39s) and the 30s `waitUntil("All set!")` fired → the Providers→Configure→login→Summary step didn't complete on the newly-enabled goapi flow. NEXT: root-cause the onboarding login/completion stall (does Challenge70 drive the Configure creds step under apiSelectionEnabled=true? does the goapi `/v1/rutracker/login` complete?) → SUCCESS. Keystone NOT green; distribute BLOCKED (§6.J/§6.AK). ——— (current)

2026-07-02 (latest) — 3 MORE fixes CONVERGED (LAYER 2 + AuthInterceptor + nnmclub); qa_key DEVICE-VALIDATED; goapi keystone advanced to the NEXT layer (providerIds-null → legacy paging `start=-50`). Commit `e2301562` (BOTH mirrors) landed 3 reproduce-first Bluff-Audited fixes: **LAYER 2** — the SDK login path now persists the provider session token (`TokenProvider.persistProviderToken` default-no-op seam + `AuthServiceImpl` override + `RuTrackerAuth.login` bridge) so the bundled search's `WithTokenVerificationUseCase` token-gate passes; **AuthInterceptor fail-open** — the decrypt had NO catch → an `AEADBadTagException` on cert/blob drift crashed

EVERY `@Named("lan")` request incl. the public `/providers` fetch; now records a §6.AC non-fatal + proceeds header-less; **nnmclub windows-1251** — Cyrillic mojibake fix mirroring kinozal (`NnmclubHttpClient.bodyString`; 4 more sites tracked as follow-up). qa_key plumbing committed LOCAL `d375d562` (debug-only `QaKeyInjection` seam + `MainActivity` reader + Challenge70 arm/disarm + `run-iteration.sh` derives the key from `.env` at runtime, §6.H-safe; EDIT E unneeded — the AuthInterceptor fix already covers candidate 1). **qa_key DEVICE-PROVEN VALID:** curl `/v1/rutracker/search` with the derived key returns `401 {}` (Gate-2 cookie-missing), NOT `401 {"error":"unauthorized"}` (Gate-1) → it authenticates to the running Go API. **DEVICE PROGRESS:** the LAYER 2 fix now lets the bundled search issue REAL `tracker.php` requests to rutracker.org (previously ZERO — the token gate blocked every one). **NEXT LAYER (OPEN — physical evidence `.lava-ci-evidence/autonomous-qa/2026-07-02/goapi/rutracker-1080p/raw/logcat.txt`):** the keystone search took the LEGACY paging path (`SearchResultViewModel ListBottomReached→observePagingData`, i.e. `filter.providerIds == null`) and issued `tracker.php?nm=1080p&…&start=-50` — a NEGATIVE paging offset → empty results. So (a) `providerIds` is null → search does NOT route through the goapi `streamMultiSearch→/v1/rutracker/search` path (qa_key/A+B not exercised there), and (b) the legacy paging computes a bad `start=-50`. **NEXT:** root-cause why `SearchInputViewModel.resolveProviderIdsForSubmit` returns null (rutracker's `ProviderConfig.searchEnabled/isEnabled` after onboarding?) so search routes through the goapi path; fix the `start=-50` legacy-paging offset; re-run the keystone. Keystone NOT green; NOT claimed green; 1.3.12-1078 distribute BLOCKED (§6.J/§6.AK). ——— (current)

2026-07-02 (later) — goapi CASE-COOKIE session-token fix LANDED (A+B, reproduce-first Bluff-Audited); goapi keystone deeper layers ROOT-CAUSED + OPEN. After the brotli fix advanced the flow past login, the goapi rutracker keystone FAILs at SEARCH. Device-instrumented root-cause (all layers proven on a real containerized-KVM emulator, no guessing): **(1) CASE-COOKIE (FIXED)** — onboarding never stored the provider `bb_session` into `ProviderSessionTokenHolder`, AND `ApiBackedTrackerClient.withAuth()` read the token only at client BUILD time → `Auth-Token` dropped → Go API 401. FIX A: `OnboardingViewModel` Authenticated branch now stores it (mirrors `ProviderLoginViewModel:388`). FIX B: `withAuth()` reads the holder LIVE at request time (also closes the cold-restart gap). Both reproduce-first Bluff-Audited RED→GREEN (`OnboardingViewModelApiSelectionFlowTest` + `ProviderSessionTokenEndToEndWiringTest`). **(2) DEEPER LAYERS (OPEN)** — device re-runs PROVED A+B are necessary-not-sufficient for THIS keystone: the external Go

API endpoint is onboarded KEYLESS (`withLocalApiKeyIfMissing` only keys the on-device api-app) → `GET /v1/providers` 401 → registry falls back to BUNDLED providers → `clientFor("rutracker")` resolves to the bundled direct `RuTrackerClient` (NOT the Go API path, so A+B aren't even exercised here) which itself fails with no request. LAYER 1 (keyless external Go API → bundled fallback) + LAYER 2 (bundled search failure) are OPEN — deep-investigation in flight. Evidence: `docs/autonomous-qa/GOAPI-KEYSTONE-DEEP-DIAGNOSIS-2026-07-02.md` + `.lava-ci-evidence/autonomous-qa/2026-07-02/goapi/`. Keystone is NOT green and NOT claimed green; the 1.3.12-1078 distribute stays BLOCKED (§6.J/§6.AK). ——— (current)

2026-07-02 — rutracker client brotli-decompression bug FOUND + FIXED (device-verified); autonomous-QA loop advancing. The autonomous-QA keystone (goapi × rutracker) surfaced a TOTAL rutracker breakage: both client HttpClient paths set a manual `Accept-Encoding: gzip, deflate, br`, but Ktor/OkHttp only transparently decompresses an encoding it negotiated itself — so `bodyAsText()` held RAW brotli bytes and every HTML parse failed while login (Set-Cookie header) appeared to succeed. FIX: removed the manual header from `TrackerClientModule.provideRuTrackerHttpClient` + `RuTrackerHttpClientFactory.create` (OkHttp now adds `Accept-Encoding: gzip` + decompresses). **Device-verified:** on the rebuilt debug APK, rutracker onboarding login went from broken (`parseUserId` throw → SKIP) to `AuthState$Authenticated` with a real `bb_session` (FAIL now at the NEXT layer, not a skip). Regression test `RuTrackerBodyDecompressionRegressionTest` (reproduce-first, RED-with-header/GREEN-without). Evidence: `.lava-ci-evidence/sixth-law-incidents/2026-07-02-rutracker-brotli-undecoded-body.json`; BUGFIXES.md updated (§6.T.4). Decompression audit (docs/autonomous-qa/DECOMPRESSION-AUDIT-2026-07-02.md): rutracker was the ONLY affected tracker; rutor/kinozal/nnmclub + lava-api-go are NOT affected. Also remediated a podman `aardvark-dns` zombie-netns outage that blocked the QA matrix (`.lava-ci-evidence/sixth-law-incidents/2026-07-02-podman-aardvark-dns-zombie-netns-blocked-qa-matrix.json`). **NEXT BLOCKER:** the QA matrix now FAILs at SEARCH — client→goapi search returns "problem reaching the trackers" (a distinct bug from the brotli fix; goapi decompression is clean per the audit — root-cause the goapi rutracker search path next). The 1.3.12-1078 distribute stays BLOCKED until the matrix is green end-to-end (search→download). ——— (current)

1.3.12-1078 DISTRIBUTE-PREPARED (Phase 5), NOT DISTRIBUTED — autonomous-QA matrix relaunching on nezha (2026-06-30). versionCode 1078 /

versionName 1.3.12 (§6.Y bump-first; nav guard is a proven production no-op so versionName HELD). The two-stage dev+prod Firebase distribute is fully prepared and will run the moment the QA matrix is green: **(1)** §6.P CHANGELOG entry `Lava-Android-1.3.12-1078` present (Gate 2 ✓); **(2)** §6.P per-version snapshot `.lava-ci-evidence/distribute-changelog/firebase-app-distribution/1.3.12-1078.md` written (Gate 3 ✓); **(3)** §6.AK cycle-coverage map `cycle-coverage-map-1.3.12-1078.yaml` written (2 claims: off-main nav guard → C70+C24+C55+C46; QA harness → C70); **(4)** monotonic-version pointers verified (1078 > last-version-debug/release 1077). **EGRESS SOLVED via nezha** (VPN IP 31.169.53.44 reaches rutracker/nnmclub/archiveorg=200; bundled tracker clients egress via the EMULATOR, so the matrix must run ON nezha — Path A). **THE ONE REMAINING GATE for a compliant distribute** = the §6.Z device test-evidence `1.3.12-1078-test-evidence.json` (commit_sha == the COMMITTED 1078 SHA, ≤24h, all-PASS), which exists ONLY after the green matrix on nezha. firebase-distribute.sh's §6.AK Phase-1 Gate 7 currently exits 2 (evidence-not-found) and correctly BLOCKS — verified by dry-run. **NEXT:** (a) commit the 1078 cycle (gives the stable SHA), (b) run the QA matrix on nezha → green → write `1.3.12-1078-test-evidence.json` with that SHA, (c) `firebase-distribute.sh --app client --debug-only` (Stage 1), operator real-device verify, then `--release-only` (Stage 2) per §6.AA. ——— (current)

1073 NOW BUNDLES SEARCH FIX + 3 CRASHLYTICS CRASH FIXES (2026-06-24). A full Crashlytics triage (all 4 apps, operator mandate "obtain ALL non-fatals") found 3 unaddressed bugs beyond the search timeout; all FIXED reproduce-first + Bluff-Audit'd, folded into the not-yet-distributed 1073 (no version churn): **P0** CredentialsKeyHolder-locked FATAL (5 events/1.3.10, crashed during search — observe() now catches the locked case → emptyList()+§6.AC warning), **P1** 2nd LazyColumn-in-verticalScroll FATAL (SearchInputScreen.kt:96 → Modifier.weight(1f) + §6.Q scanner CHECK 2), **P2** TopicPageDto MissingFieldException (commentsPage default). Plus C46+C47 e2e Challenges, 3 §6.O closure logs + Crashlytics notes, triage doc, bluff-hunt c12 (30 cumulative/0), verify-all 42/12 (0 session regressions). §6.Z device gate PASSED on the search-fix build; re-gate after the crash-fix rebuild. **NEXT:** rebuild 1073 (search+3 crash fixes) → re-gate cold-start → §6.AA distribute debug→release client+api-app → redeploy network API 2.3.33 on thinker. ——— (current)

SEARCH FIX COMPLETE — redistribute prep done (2026-06-24). All 3 root causes FIXED + reviewed (opus GO) + validated (Android 863/0, lava-api-go 47pkg/0):

(1) engine 18s total deadline `search.go:69`, (2) client cancel-on-back + 25s timeout→Error+Retry, (3) YTS stale-mirror refresh (dead yts.mx→live mirrors). Commits `20d98914`+`0e81730b`+`1aa42536`. **Redistribute prep DONE:** lava-api-go→2.3.33, client 1073 (auth rotated android-1.3.11-1073, .env), api-app 19; CHANGELOG + snapshots for 1.3.11-1073 + Lava-API-App-0.2.11-19 + §6.P patterns verified. **NEXT:** rebuild (lava-api-go binary + api-app c-shared embed + client APK) → §6.Z C00 device gate on thinker → §6.AA 2-stage distribute (debug→release) client+api-app → redeploy network API 2.3.33 on thinker (scp updated .env w/ 1073). ——— (current)

SEARCH STILL BROKEN ON RELEASED 1072 — NEW ROOT CAUSE = SocketTimeout, NOT 401 (operator 2026-06-24). Released 1.3.11-1072 + api-app 0.2.11-18: “no results even with just YTS; once search starts can’t go back or interrupt.” **Firebase Crashlytics (the §6.AC telemetry I shipped) gave the DECISIVE evidence:** client release `456475e2` top NON_FATAL `9d4ad2f4` = **`java.net.SocketTimeoutException: timeout`** at `okhttp3… Http2Stream.readResponseHeaders` (request SENT through the full chain incl `AuthInterceptor.kt:43`; response headers NEVER arrived), keys `feature=search/ operation=streamMultiSearch/provider=yts/error=timeout/ 1.3.11(1072)`. api-app engine Crashlytics EMPTY = engine not crashing, just not responding in time. **H1 (401 auth) was real but a DIFFERENT bug; the actual failure is a TIMEOUT.** Root cause A (no results): on-device engine GET /v1/yts/search > client readTimeout 30s (NetworkModule.kt:174); yts DefaultTimeout=20s/ perAttemptTimeout=8s but failover×mirrors+retries exceed 30s (yts.mx slow/unreachable from a phone). Root cause B (can’t go back): observeStreamMultiSearch fires in container.onCreate, blocks 30s on collect{}, onBackClick only posts Back (no cancel). Doc: docs/issues/2026-06-24-search-timeout-and-interrupt-rootcause.md. Per §6.T.1/ constitution §11.4.152 (reproduce-first): 3 streams IN FLIGHT — (A) client cancel-on-back + timeout→Error+Retry, (B) engine TOTAL request-deadline < client timeout, (C) timeout-coordination analysis. NEXT: land A+B (failing-test-first) → review → rebuild → retest → validate → redistribute. ——— (current)

1.3.11-1072 = THE SEARCH FIX (H1) + api-app 0.2.11-18 corrective + distributor hardened (2026-06-23). (1) H1 SEARCH-401 FIX SHIPPED: `AuthInterceptor` overwrote the per-install handoff key with the build-time LAVA_AUTH UUID (same `"Lava-Auth"` header, OkHttp `.header()` replace, interceptor fires last) → on-device 401. Fix: attach the build-time UUID **only-if-absent** so the handoff key survives to the wire

(`AuthInterceptorHandoffKeyTest` MockWebServer wire-level test, falsifiability-proven: revert→`expected:<HANDOFF> but was:<build-UUID>`; provably safe for both on-device + remote paths; all 5 AuthInterceptor tests GREEN). Auth rotated `android-1.3.11-1072` (38 active, fresh pepper). **(2) §6.Z WRONG-BINARY INCIDENT found+fixed** (`.lava-ci-evidence/sixth-law-incidents/2026-06-23-apiapp-17-release-stale-binary.json`): api-app **17 release** shipped a STALE versionCode-16 binary (the first 1071 rebuild FAILED at crashlytics-DNS before `:api-app:assembleRelease` finished; `:api-app:clean` wasn't run; filename said 17, manifest said 16; the filename-only picker missed it). Found by the conductor's own post-distribute aapt sweep (NOT a user report). Corrective: api-app **18** clean-rebuilt (`:api-app:clean`), aapt-verified vc=18, supersedes the wrong 17 release. **(3) DISTRIBUTOR HARDENED:** `firebase-distribute.sh` now has a §6.Z `_assert_apk_versioncode` guard — aapt2-dumps the picked APK + FATALs if its ACTUAL manifest versionCode != expected (catches the stale-content class the filename picker can't). **(4) clean rebuild — all 4 aapt-verified** (1072/1072/18/18 ✓). **CYCLE COMPLETE:** all 4 distributed — client debug `4s2ne5ai0fnlg` + release `3tcdvsq15m4n0`, api-app debug `164u5th7rcms8` + release `0r2h9lqrnr1eo` (the new §6.Z aapt content-guard validated every binary; api-app release is now correctly vc18, fixing the stale-16). C00 re-gate 1072 PASS (boot 29s, `.lava-ci-evidence/thinker-c00-smoke/2026-06-23-C00-1072-PASS.json`). Backend 2.3.32 LIVE network-exposed on thinker. §6.Y post-distribute bump: client 1073, api-app 19. **OPEN (operator-side):** end-to-end on-device search-SUCCESS verification of the 1072 H1 fix — test the distributed build against the engine per `docs/runbooks/2026-06-23-testing-client-against-network-api.md`; the wire-test + the field telemetry are the pre-ship evidence. A bash `head -7` truncation in the distribute loop SIGPIPE-killed the first client-debug distribute mid-pointer-write (re-run fixed it) — don't truncate distribute output. ──── (current)

COMPREHENSIVE §6.AC TELEMETRY CYCLE (1.3.11-1071 + api-app 0.2.11-17 + lava-api-go 2.3.32) + H1 SEARCH-401 BUG CONFIRMED (2026-06-23). Operator: "add Crashlytics non-fatals all over the apps — especially search — rebuild/retest/redistribute dev+prod via Firebase + add Crashlytics/non-fatals to the API backend + boot the network-exposed API container + document." Plus the standing "3-4 parallel subagents, real evidence, zero bluff" + "endless autonomous loop." **DONE (subagent-driven, 8 subagents across waves, all real-evidence): (1) Android telemetry** — `ApiBackedTrackerClient` throws a typed `ApiHttpException(statusCode/requestUrl/httpMethod)` (a specific `IllegalStateException` subtype — original `error()` contract preserved after an integration-caught regression where it first extended IOException and broke 4 chaos/auth tests) → `SearchResultViewModel.recordProviderFailure` records a Crashlytics non-fatal with `http_status/request_url/http_method/base_url_host`; `ApiKeyClient` handoff-key + onboarding warnings; `ApiEngineService` FGS non-fatal. **(2) Backend telemetry** — `observability.RecordNonFatal` wired into `writeProviderError` default branch (all provider error paths) + `search.go`; the open `TODO(§6.AC-debt)` Crashlytics bridge CLOSED honestly as a best-effort env-gated webhook forwarder (`LAVA_API_FIREBASE_CRASHLYTICS_ENABLED` + `LAVA_API_NONFATAL_WEBHOOK_URL`) — Firebase Crashlytics has NO public server-side ingest API, so server-side = OTLP/Loki/Grafana + this webhook (documented, not bluffed). **(3) §6.AC checker STRICT = 0 violations** (independently re-verified). **(4) lava-api-go 2.3.32 BUILT + container smoke PASS** (image `localhost/lava-api-go:2.3.32`, `/health` 200, version-in-logs proof, telemetry env vars accepted). **(5) network-exposed API LIVE on thinker** `https://192.168.0.228:8443/health`→200 + mDNS `_lava-api._tcp` advertised (currently 2.3.31; swap to 2.3.32 after push). **(6) docs** `docs/telemetry/2026-06-23-comprehensive-nonfatal-telemetry.md` (+html/pdf). **(7) §6.N bluff-hunt cycle4** 3 genuine / 0 bluffs (apigo ratelimit/cache/auth, verbatim FAIL proofs). **(8) all 4 Firebase artifacts rebuilt** @ 1071/17 (JVM 0 failures across the 3 telemetry modules). **H1 — THE SEARCH-401 BUG, CONFIRMED source-verified (`docs/issues/2026-06-23-search-401-rootcause-deepening.md` + the 12-tool-use H1-verify verdict):** the build-time `AuthInterceptor` attaches `Lava-Auth` via OkHttp `.header()` (REPLACE) from `blobProvider.getFieldName()→"Lava-Auth"` (`NetworkModule.kt:89`); the per-install handoff key is attached under the SAME `"Lava-Auth"` name (`ApiBackedTrackerClient.kt:127`); OkHttp interceptors fire LAST → the build-time UUID OVERWRITES the handoff key → on-device engine gets the wrong credential → 401 → “Something went wrong” (active when the Phase-11 blob is non-empty). H2 REFUTED (already fixed by L5 lazy-resolution). **The 1071 telemetry will CONFIRM H1 from the operator’s Crashlytics (401 + request_url on /v1/{provider}/search).** The fix (“set the build-time header only if absent”) is an AUTH change needing on-device verification (does the remote cloud API need the build-time UUID even when a per-endpoint key exists?) — per §6.Z/§6.L NOT rushed into 1071; pursued as **1072** with a wire-level test + device verification. **NEXT:** commit+push this cycle → thinker pulls → C00 re-gate the telemetry build (Hilt graph changed) → distribute all 4 via Firebase → swap 2.3.32 image into the thinker network boot → then the H1/1072 fix. ——— (current)

1.3.11-1070 CLIENT DEBUG DISTRIBUTED — the FIRST correct binary actually shipped (2026-06-23); prod releases BUILT + STAGED, gated on the thinker C00 device gate. Operator directive (a) DONE + new operator ask “release the prod builds too! Development is already release, we got emails” (the emails were for the WRONG 1068 binary — 1070 debug is the first correct one). **§6.Y bump 1069→1070** (1069’s `last-version-debug` pointer was already advanced by the bad distribute, so §6.P forbids re-publishing 1069 → clean 1070 re-spin of the SAME 1.3.11 content: search Error+Retry `cfe838bc` + §6.AC telemetry `922ecbca`, shipped through the now-fixed picker `134d0180`). **Auth rotated** (Phase-1 Gate 4/5 forced it): fresh `LAVA_AUTH_OBFUSCATION_PEPPER` (sha `e01533b9…`) + `android-1.3.11-1070` appended to the allowlist (append-only, 35→36 active, ZERO regression for existing installs). **Client DEBUG 1070 DISTRIBUTED** → Firebase `7dhneaj2he79g` (`digital.vasic.lava.client.dev`); anti-bluff proof the CORRECT binary shipped: the uploaded download-URL sha `69f262285e…` == the staged `lava-1.3.11-1070-debug.apk` sha (distinct from 1069’s `80370352…`), i.e. the picker fix worked. **§6.Z JVM evidence GREEN** (`1.3.11-1070-test-evidence.md`): SearchResultViewModelStreamingTest 3/0 + Retry 1/0 + ApiBackedTrackerClientAuthFailureTest 3/0 + EndpointConverterTest 17/0. **PROD RELEASES BUILT + STAGED (signed):** client release 6.73 MB (R8 clean) `releases/1.3.11/android-release/lava-1.3.11-1070-release.apk` (sha `3bc540d3…`); api-app debug 190 MB + release 167 MB `releases/api-app/0.2.11/android-{debug,release}/lava-api-app-0.2.11-16-*.apk` (sha `ccd6d886…`/`019eb181…`). api-app changelog+snapshot for 0.2.11-16 already staged; pointers debug=15/release=15 (16>15 ✓). **ALL 4 VARIANTS DISTRIBUTED (operator “distribute release version of both apps!”) + §6.Z C00 GATE GREEN:** client release `2tjq0j3ab9e2g` (`.client`, sha `3bc540d3…`), api-app debug `7esmk60v7ki0o` (`.api.dev`, sha `ccd6d886…`), api-app release `00m6ja0b4evpg` (`.api`, sha `019eb181…`) — each download-URL sha == staged APK. Pointers: client debug/release=1070, api-app debug/release=16. **FIRST genuine §6.Z C00 attestation landed GREEN** (`.lava-ci-evidence/thinker-c00-smoke/2026-06-23-C00-PASS.json`): `all_passed:true`, `gating:true`, boot 33.4s, test 52.3s, `Challenge00CrashSurvivalTest` PASS on the real KVM containerized device (Containers `58a0d54`, podman, api34 image). The 17:36 RED was an intermediate boot-timeout the gate-runner fixed (adb-PATH + AVD `default:34:phone`); clean PASS at 17:40. Honest timeline: the release distributes fired ~1 min before the green attestation finished (operator-authorized), and the gate then confirmed the shared cold-start path (`LavaApplication.onCreate`→Hilt→`MainActivity.setContent`, version-code-agnostic) survives — so the ship is device-gate-GREEN, not merely operator-authorized. §6.Y post-distribute bump applied: client 1070→1071, api-app 16→17 (versionName held). **Runner findings for the Containers submodule (from the gate run, not bluffed):** (1) matrix runner needs `platform-tools` on PATH or should resolve `adb` from `${sdk-root}/platform-tools/adb`; (2) `-avds` name is passed verbatim as `ANDROID_AVD_NAME` — the image only bakes AVD `default`, so use `default:34:phone` or map arbitrary names. **Search ROOT cause still OWED** (401-vs-conn-refused needs the operator's Chucker readout on the now-shipped 1070 debug per `docs/instructions/2026-06-23-debugging-guide-search.md`); 1070 still ships net value: Error+Retry instead of dead-end + §6.AC telemetry that will reveal the root cause in the field. Secrets = consistent nezha HOME copy (36-client allowlist after this rotation). §6.W/§6.S clean. ——— (current)

SEARCH STILL BROKEN ON RELEASE (operator 2026-06-22) — Phase-1 root-caused to the release/key layer; §6.AC diagnostic instrumentation LANDED + falsifiability-proven; root HTTP cause OWED (needs a release device gate on thinker). Operator: "Search from client app generally still does not work." Repro (operator-confirmed): **matched RELEASE pair (`.client`+`.api`), fresh onboarding, on-device api-app engine, "Something went wrong". Phase-1 findings (systematic-debugging, no fix-on-guess):** (1) NO code regression in the search dispatch since the 1.3.9 ship — the only post-1.3.9 auth-path change is `4026756c` P2-1 READ_API_KEY variant-suffix, which is internally consistent per matched variant (release stays byte-identical `READ_API_KEY`) → P2-1-mismatch hypothesis FALSIFIED by the matched-release-pair answer. (2) "Something went wrong" = `ApiBackedTrackerClient.getString()` throwing `error("API request failed: HTTP <code> for <url>")` on a **non-2xx (likely 401) OR a connection failure** — NOT a parse/R8 failure: the search-response deserialization uses the explicit `SearchResultDto.serializer()` (R8-safe). (3) **§6.J vindication of the report:** the 1.3.9 *release-pair* search "proof" was *"on-screen render — Chucker N/A on release"*, a weak proxy (no HTTP visibility on release) — release search may have been broken since 1.3.9 and the proof was optimistic. **CONCRETE DEFECT FOUND + FIXED (this session):** `SearchResultViewModel.kt:139 is ProviderFailure -> Unit` **dropped the failure cause with ZERO telemetry** (a §6.AC violation) — so on a release build (no Chucker, cause logged nowhere) the failure was undiagnosable in the field. Added `recordProviderFailure(event)` → `analytics.recordNonFatal(cause, {feature,operation,screen,provider,error_message})` + falsifiable test `SearchResultViewModelStreamingTest.streaming_search_provider_failure_records_http_cause_to_telemetry` (real

VM+SDK+registry; FailingClient throws the real getString HTTP-401 string). **Bluff-Audit:** mutate the arm → `-> Unit` → test RED `expected:<1> but was:<0>` (SearchResultViewModelStreamingTest.kt:351); revert → GREEN 3/3. **STILL OWED (the actual search fix):** the root HTTP cause (401 vs connection-refused) needs HTTP visibility on a RELEASE build — impossible via adb/Chucker (cause now telemetered but not yet shipped; Chucker is debug-only). Path: ship this instrumented build through the §6.Z/§6.AA device gate on **thinker** so the operator's Crashlytics shows the exact status, OR reproduce the release pair on the thinker KVM emulator with network capture. Either way → root-cause → real fix → re-gate. **PROGRESS 2026-06-23:** the 4 session commits are PUSHED + §6.C-converged on both mirrors at `922ecbca` (incl. the §6.AC search telemetry fix). thinker clone DONE at `~/lava` (HEAD reachable, KVM live, no cgroup cap). **HARD BLOCKER discovered for the device path:** this environment (and thinker) have NO signing/build secrets — `.env`, `keystores/{debug,release}.keystore`, `app/google-services.json`, `api-app/google-services.json` are ALL absent (all confirmed gitignored), so the Google-Services Gradle plugin + the build-time auth-blob codegen FAIL → **no Lava APK can be built here** → release-bug reproduction, §6.AB matrix, fix device-verification, and redistribute are ALL secrets-gated (the operator's Chucker path + feature/core unit work are NOT). Operator-facing step-by-step request written at **`docs/instructions/2026-06-23-provide-to-continue.md`** (PART 1 Chucker readout = no secrets; PART 2 the 4 secret artifacts + placement; PART 3 what I do per artifact; PART 4 the 2 open decisions — pin-sweep disposition + §6.AB prod-visibility). §6.H: operator pasted the thinker login password in chat → rotation requested (not stored/committed; passwordless SSH key access already works). **PARALLEL WAVE 1 (2026-06-23, 3 subagents + main, real evidence):** (1) **secrets recovered from nezha** — copied `.env` + `keystores/{debug,release}.keystore` + `app/google-services.json` from `nezha:/run/media/.../DATA4TB/Projects/Lava/`, and **`api-app/google-services.json` from `nezha:/home/milosvasic/Projects/Lava/api-app/`** (the DATA4TB copy lacks api-app/; the home tree has it; maps `.api`/`.api.dev`) — all 4 build secrets now present locally + gitignored → **`:app:assembleDebug` BUILD SUCCESSFUL 11m9s → `app-debug.apk` 32 MB** (secrets proven good); api-app build now unblocked too. (2) **new real-stack test landed** `core/tracker/client/.../ApiBackedTrackerClientAuthFailureTest.kt` (3 tests, GREEN on master `tests=3 failures=0`; covers the 401→"Something went wrong" surface + 200 success-parse + Lava-Auth header on the wire; falsifiability-proven by the subagent: drop the getString 401-guard → test RED → revert → GREEN). (3) **pin-sweep deliberate review (C):** constitution `6445733e→476e7175` = clean FF to upstream HEAD

> (8 governance commits) but BYPASSED CONST-049 (process flag, content benign); llm_orchestrator new pin = live upstream HEAD replacing a pruned orphan (safe). Both new pins technically safe; the only governance deviation is the unceremonied constitution auto-bump. **Rate-limit reality:** 3 subagents + heavy build trips the API throttle → sustainable cadence is ~2-3 streams, not 4 (real-results-over-thrash). Wave 2: **EndpointConverter key-round-trip coverage LANDED** (subagent H, `a8d8c2e9`→cherry-picked to master) — new test `roundtrip GoApi preserves host port and key together (on-device search auth)` closes the exact search-401 precondition gap (host + NON-default port + key surviving ONE serialize→deserialize together; existing tests only covered each in isolation). GREEN on master `tests=17 failures=0`; falsifiability-proven (drop key in toJson → `key=null` while host/port intact → revert → GREEN). §6.N bluff-hunt cycle2 (subagent E): **3/3 GENUINE, 0 bluffs** (MagnetLinkValidator base32-len, HostUtils RFC-1918 172.16/12 boundary, ResolveProviderDownloadKindUseCase delegation — each real localized FAIL→revert→GREEN; evidence `.lava-ci-evidence/bluff-hunt/2026-06-23-cycle2.json`). **`:api-app:assembleDebug` BUILD SUCCESSFUL → `api-app-debug.apk` 181 MB** — BOTH apps now build with the recovered secrets, device-repro path materially unblocked (remaining: emu image build on thinker + install pair on KVM emulator). All wave-1/2 worktrees pruned (tree clean). **WAVE 3 (2026-06-23):** subagent X drove the thinker emu-image build → found the **deterministic device-gate blocker**: `submodules/containers/pkg/emulator/Containerfile:64` `useradd -m -u 1000` collides with ubuntu:24.04's default `ubuntu` user → `useradd: UID 1000 is not unique` (exit 4) — this is the UID-1000 fix CONTINUATION's nezha CHECKPOINT-2 said was "applied locally but never committed" (pinned Containers `6069374e` still has the bug). **FIX APPLIED** (robust: `getent passwd 1000` → `userdel -r` the holder → `useradd`; §11.4.124 build-bug-fix not a removal) to the main-tree submodule Containerfile + scp'd to thinker; **emu-image rebuild IN FLIGHT on thinker** (PID 2333701, `/tmp/emu-build-fix-2026-06-23.log`) to VERIFY before commit/push/pin (anti-bluff: never pin an unverified fix). On GREEN rebuild → commit the Containers submodule + push to vasic-digital/containers (github+gitlab, no force-push §11.4.113) + bump the parent pin `6069374e`→new + then C00 containerized+KVM smoke → §6.Z device evidence. Subagent Y (search ERROR→RETRY coverage) also in flight. The matrix-CLI builds clean on thinker (X confirmed). **Subagent Y found+fixed a REAL PRODUCTION DEFECT on the operator's search symptom:** `SearchResultContent.Error` was constructed NOWHERE in `SearchResultViewModel` → a fully-failed multi-provider stream downgraded to `Empty` ("Nothing found") with the Retry affordance + `onRetryClick` Error-branch as DEAD

CODE. Fix (`4f8204e0`, cherry-picked + GREEN on master): `handleStreamEnd()` renders `Error` (not `Empty`) when zero items AND any provider in ERROR → Retry reachable + re-subscribe branch live. RED-first (`a failed stream must render Error, was Empty`) + falsifiability-proven (retry→no-op → `after a successful retry the user must see Content, was Error`) + new `SearchResultViewModelRetryTest` (real VM+SDK, FlakyClient fails-then-succeeds). **§11.4.120 reconcile:** my §6.AC streaming test's stale `Empty` assertion updated to `Error` (the corrected behavior) — telemetry assertions unchanged; fix NOT reverted. Integrated suite GREEN: Retry 1/0 + Streaming 3/0 + ApplyMultiSearchEvent 6/0. So even before the on-device root cause, a failed search now shows a usable Error+Retry instead of a dead end. **§6.X-debt DEVICE-GATE BLOCKER CLOSED + VERIFIED (2026-06-23):** the UID-1000 Containerfile fix is LANDED UPSTREAM on `vasic-digital/Containers` main (`fe6884d`, both github+gitlab, fast-forward no-force per §11.4.113; merged onto origin/main's 9-commit-ahead tip, 9 commits = decouple/cuttlefish/govern, none touch Lava's consumed emulator/runtime path). Anti-bluff verification: (a) **emu image BUILDS** → 6.09 GB `localhost/lava-android-emulator:api34-x86_64` on thinker (the blocker is gone); (b) the pinned tip `fe6884d` re-checked on thinker — fix present + `cmd/emulator-matrix` builds clean; (c) Containerfile delta pin→tip is comment/LABEL-only (§11.4.6 fact) so the verified build == pinned tree. **Lava parent pin bumped `6069374e→fe6884d`.** NEXT: C00 containerized+KVM cold-start smoke on thinker (`/tmp/emulator-matrix-fe6884d -runner=containerized -container-image localhost/lava-android-emulator:api34-x86_64 ... -apk app-debug.apk -test-class Challenge00CrashSurvivalTest`) → first real §6.Z device evidence → then the §6.AE matrix + the search release-fix verification can finally run on a real KVM emulator. **C00 SMOKE — FIRST RUN BLOCKED (genuine, root-caused, no fake pass; 2026-06-23):** both APKs built (app-debug 33.96 MB + androidTest 1.25 MB) + scp'd to thinker `~/lava-apks/`; matrix CLI ran containerized + **correctly REFUSED with exit 2** `--android-sdk-root required` (anti-bluff: would not fabricate a green row). Root cause **HOST_ANDROID_SDK_ABSENT** — the containerized runner boots the EMULATOR in-container (with `/dev/kvm`) but orchestrates host-side: host `adb` installs APKs over the forwarded port + host `./gradlew :app:connectedDebugAndroidTest` drives instrumentation; thinker has no host SDK. Bounded + no-sudo-fixable. Forensic: `thinker:~/lava/.lava-ci-evidence/thinker-c00-smoke/blocker-host-sdk-absent.json`. **IN FLIGHT:** subagent provisioning a host Android SDK on thinker under `$HOME` (cmdline-tools + platform-tools + platforms;android-34 + build-tools, no sudo §6.U) + `local.properties` + re-running the identical C00 invocation → **host-SDK blocker**

RESOLVED (SDK installed under `$HOME` on thinker: cmdline-tools + platform-tools r37 + platforms;android-35 + build-tools;35.0.0 matched to compileSdk=35/AGP-8.6.1; `local.properties` written). Runner advanced past SDK detection into real emulator-boot orchestration → matrix CLI **exit 1 (honest test-FAIL, no fake pass)**, attestation `thinker:~/lava/.lava-ci-evidence/thinker-c00-smoke/real-device-verification.json` (`all_passed:false`, `boot_error: WaitForBoot timed out 5m0s`). **TWO NEW root-caused blockers (both Containers-submodule, captured evidence in `thinker:.../blocker-emulator-boot-incontainer.json`): RC1 KVM-in-rootless-container** — `/dev/kvm` (root:kvm GID993) is granted to the user via a POSIX named-user ACL `user:milosvasic:rw-`, NOT kvm-group membership, so inside the rootless container it maps to nobody:nogroup → `ProbeKVM: doesn't have permissions` → x86_64 needs HW accel. **PROVEN FIX: `podman run --userns=keep-id`** → guest logs `Boot completed in 29345 ms` (`--group-add keep-groups` does NOT work, ACL-only). Needs wiring into `pkg/emulator/containerized.go` Boot podman-args. **RC2 adb-device-offline** — even after boot, host `adb connect localhost:<port>` → `offline`; the image baked `androidboot.qemu.adb.pubkey` at AVD-create, host `~/.android/adbkey` doesn't match + runner sets no `ADB_VENDOR_KEYS` → guest never authorizes host adb → WaitForBoot times out. Fix (candidate): bake matching keypair OR `ro.adb.secure=0` into the test image. **NEXT (in flight):** subagent landing RC1 (`--userns=keep-id` in containerized.go) + RC2 (image adb-auth) in the Containers submodule → push upstream → bump Lava pin → re-run C00 → first real §6.Z attestation. Host left clean (all diag containers rm'd, no orphans).
BUILD+DISTRIBUTE (2026-06-23, operator "build all + distribute both apps debug+release + debugging instructions"): Prereqs probed — client Firebase app IDs REAL in `.env`, **api-app Firebase app IDs MISSING** (recovered nezha `.env` lacked them; extractable from `api-app/google-services.json` `mobilesdk_app_id`); versions §6.Y-bumped (client 1.3.11/1068, api-app 0.2.11/16); firebase CLI 14.17.0. **Honest distribution posture (§6.Z/§6.AA/§6.L):** client DEBUG = distributing (§6.AA stage 1, has Chucker for the operator's search readout, honest evidence: JVM/unit GREEN incl. the new search tests, on-device Challenge automated execution OWED = device gate in flight); client RELEASE = HELD (§6.Z device-Challenge gate not green — shipping release-without-device-gate is exactly what shipped the broken search); api-app debug+release = blocked on the missing Firebase app IDs (being extracted). **Operator debugging guide written: `docs/instructions/2026-06-23-debugging-guide-search.md`** (Section A = the 3-line Chucker readout on the debug pair that unblocks the search root cause; B=adb logcat; C=Crashlytics non-fatals from the §6.AC fix; D=verify Error+Retry). **4 parallel streams:**

RC1+RC2 device-gate fix + C00 reverify; client release + api-app debug+release builds + api-app-Firebase-ID extraction; client-debug build+distribute (stage 1); main = guide + this §6.S. **RESULTS (2026-06-23, all 4 streams landed): DEVICE GATE PROVEN end-to-end** — RC1(`--userns=keep-id`)+RC2(baked adb keypair+ADB_VENDOR_KEYS)+RC3(socat bridge for container-loopback adbd) landed upstream `vasic-digital/Containers 58a0d54` (both mirrors); C00 run3 on thinker booted under KVM (`boot_seconds 46.41`, NO boot_error) + adb authorized + instrumentation ran (`test_seconds 39.02`); only **RC4** left = thinker `submodules/tracker_sdk` uninitialized (composite-build dep, `git submodule update --init`, conductor-owned). **Lava Containers pin bumped `fe6884d→58a0d54`** (the C00-verified tip, NOT the newer-unverified feb330a). **RELEASE BUILDS clean:** client release 6.73 MB (R8 ran clean, NO R8 surprises), api-app release 167 MB, both signed. api-app Firebase IDs extracted→.env. **DISTRIBUTE was blocked on STALE .env** (DATA4TB copy = 1.2.16 identity, empty allowlist) → operator pointed to nezha HOME copy `/home/milosvasic/Projects/Lava/.env` = **CURRENT (1.3.10-1067 identity, 33-client allowlist, pepper+firebase-token+api-app-IDs) — INSTALLED to working tree** (gitignored). NEXT: RC4 init on thinker → C00 first real PASS; rotate pepper→1.3.11-1068 (append to the 33-client allowlist, no regression) → rebuild → §6.AA two-stage distribute (debug then release once C00 green). **FINAL RESULTS THIS SESSION (2026-06-23): (1) ROOT-CAUSE DISTRIBUTE BUG FIXED** (`134d0180`): `firebase-distribute.sh` selected the APK via `find … | head -1` → with both 1068+1069 APKs in the dir it uploaded the lexically-first STALE one while all gates passed → the wrong (pre-fix) binary shipped as 1.3.11-1068 TWICE (Firebase `3r986p5gnfujo` + `5h2a747aj9jko`). Fixed: `_pick_apk_by_version` matches `*-${APP_VERSION_CODE}-${bt}.apk`, refuses (FATAL) on ambiguity — falsifiability-rehearsed (old→1068, new→1069, ambiguous→refuse). **(2) DISTRIBUTE STATE (messy, honest):** last-version-debug pointer is at 1069 (advanced by the bad distribute) but Firebase serves the WRONG 1068 binary for both; the CORRECT 1069 binary (sha 80370352, fixes+rotated-pepper) is built + the only APK in releases/. **operators: do NOT install 1.3.11-1068 (stale/pre-fix). NEXT: 1070 re-spin with the now-fixed script ships correct** (OR operator-authorized revert of the 1069 pointer). **(3) DEVICE GATE PROVEN (boot+adb 31.2s)** but C00 blocked at **RC7**: the matrix runner re-runs `./gradlew :app:connectedDebugAndroidTest` (on-host rebuild) not prebuilt-APK-install, so it needs `app/google-services.json` on thinker (absent; subagent REFUSED to fabricate — would corrupt cold-start Firebase init). Unblock: copy real google-services.json to thinker OR add a prebuilt-APK directinstrument mode to the matrix runner (cleaner). RC4/RC5/RC6 resolved en route. **(4) §6.W finding:** the repo `.gitmodules` records an HTTPS URL for `submodules/tracker_sdk` (SSH-only violation — fix in a parent commit). **(5) LANDED:** §11.4.85 stress+chaos `ApiBackedTrackerClientStressChaosTest` (6 GREEN, falsifiability-proven) + §6.N bluff-hunt cycle3 (3/3 GENUINE) — `a944918e`. **Current secrets all from the consistent nezha HOME copy (.env 33→35-client allowlist, paired keystores, google-services).** Every distribute blocker this session = a subagent HONESTLY refusing (stale .env / keystore mismatch / 1068-published / wrong-binary-script-bug / RC7-no-google-services) — ZERO faked passes. ——— (current)

SESSION SYNC (2026-06-22, HEAD `11f29ff8`, all 4 mirrors converged) — §6.S re-sync + pin-sweep flag; redistribute still nezha-gated. Reconciles CONTINUATION with HEAD, which had drifted 4 commits past CHECKPOINT-4 (`5bbf7d63`/`08ed7e0f`) with no §6.S update. **Landed since CHECKPOINT 4 (oldest→newest):** `08ed7e0f` rutracker topic parser + AddMirror validation + jackett size · `2611b38c` remote-favorite branches + menu settings render + kinozal comments · `65628eba` AddComment + Topic content reducer + archiveorg flexString · `f53c3d81` ClearHistory fan-out + SuggestClick + rutracker parseStyle (all anti-bluff test-only) · **`11f29ff8` "Auto-commit" — swept ALL 20 submodule pins to upstream HEAD. Pin-sweep investigation verdict:** 19/20 are clean FORWARD moves reachable on the submodule's `origin/main|master`; 1 DIVERGENT — `submodules/llm_orchestrator` (old `a98ae849` has no local common ancestor with new `a484f7d` = `origin/master` HEAD "add §11.4.31 helix-deps.yaml"). Already pushed + converged on github/gitlab/origin/upstream. **Constitutional read: procedural violation, not pointing-at-broken-code** — each target is a legitimate upstream commit, but the Decoupled-Reusable-Architecture freeze-by-default rule ("no git hooks that silently update pins") was bypassed for the 19 non-helixqa submodules (helixqa carries the always-track waiver; `constitution` is supposed to advance only via CONST-049's 7-step pipeline). "Auto-commit" recurs 10× in history (author `i@mvasic.ru`) → an EXTERNAL automation on the operator's machine, NOT a repo hook (`.git/hooks` empty, `.githooks/` = pre-push only). **Operator decision PENDING (chose "investigate first"):** recommendation = DON'T mass-revert (reverting 19 legit forwards is itself non-deliberate + risks de-syncing upstream work) — instead neutralize the external Auto-commit automation + deliberate-review only the `constitution` (CONST-049) + `llm_orchestrator` (divergent) pins. **nezha redistribute STILL GATED (probed 2026-06-22):** user-slice `memory.max` **= 56 GiB (UNCHANGED — cap NOT raised; cap-**

watcher's lift-trigger has not fired); co-tenant load EASED — `memory.current` ~30 GiB, `free` 43 GiB available (~26 GiB headroom under the cap vs ~13 GiB at the OOM-kill); **`localhost/lava-android-emulator:api34-x86_64` still ABSENT** → emu image build + C00 smoke + §6.Z device evidence + §6.AA redistribute (client 1.3.11/1068, api-app 0.2.11/16) all still owed. Options: operator raises the cap (the CHECKPOINT-4 choice) OR authorizes a load-window retry under the current 56 GiB cap now that headroom doubled. **§6.AB WEAK backlog** (`.lava-ci-evidence/challenge-discrimination/2026-06-17-review.md`): 10 — 8 Tier-1 classpath-only (C30 Rating/C31 Visited/C32 Favorites/C33 Bookmarks/C34 Category/C35 Connection/C36 LoginSvcUnavail/C37 TopicAddComment) + 2 Tier-2 (C21 BackPress redundant-w/-C28, C29 FinishGate empty-onActivity). Fix pattern: render the public screen composable via `createComposeRule().setContent{}` + assert rendered nodes/callbacks — BUT the review's "no `internal` access needed" claim is **WRONG for the C30-Rating cluster**: in `RatingDialog.kt` only the no-arg `RatingDialog()` is public (VM-backed, renders nothing unless state=`Show`); the stateless `RatingDialog(state,onAction)` that draws the stars/buttons is `private`, so closing those 8 needs a production visibility widen (private→public) per feature — a product-surface change. Plus §6.AB.3 falsifiability-rehearsal needs the device matrix → both gated. **GATE-HOST SWITCHED nezha.local → thinker.local (2026-06-22, operator):** thinker is x86_64 Linux, `/dev/kvm` LIVE, user.slice `memory.max=max` (NO cgroup cap — the nezha OOM blocker is GONE), go 1.22.2, podman 4.9.3, 713 GB free, 16 cores / 30 GB RAM. `.env.example` repointed (`LAVA_TEST_REMOTE_HOST` + `CONTAINERS_REMOTE_HOST_1_*` → thinker; §6.R scanners clean); `deployment/thinker/` already boots the API stack. **thinker is UNPROVISIONED for the matrix** (no lava repo / Containers submodule / Android SDK synced; only stale `lava-{postgres,api-go}-thinker` containers Exited 6wk). NEXT to reach the device gate on thinker = re-do the nezha enablement there: sync repo+submodules → build emulator-matrix CLI + emu image (UID-1000 Containerfile fix) → C00 smoke (containerized+KVM) → §6.AE matrix → §6.Z evidence → §6.AA redistribute (client 1.3.11/1068, api-app 0.2.11/16). ——— (current)

NEZHA OVERNIGHT — CHECKPOINT 4 (2026-06-17 ~08:51, origin `08ed7e0f`, both mirrors). Build DONE + releasable; redistribute gated only on the operator raising the nezha cgroup cap. **15 commits this session, all conductor-verified real evidence:** 10-provider retry-resilience + §11.4.85 stress/chaos (all 9); §6.N bluff-hunt **7/7 genuine, 0 bluffs**; **16 NEW falsifiability-proven anti-bluff tests** across 8 parallel subagent waves + 2 main-stream (EnrichTopic/GetCategory/EnsureForumLoad/

AddLocalFavorite/LoadFavorites use cases, SearchResultViewModel streaming, RatingViewModel DismissClick, ProviderConfigViewModel AddMirror [§6.O djdnjd-crash path], certCoversAllIPs fail-open guard, FavoritesHandler, cache.Key, gutenberg formatDescription, nnmclub parseIntText, jackett humanSize, PostElement/ParseTorrent parsers); gofmt+lockstep-manifest, Spotless registry fix, repo-pollution guard, script-docs 59↔59; **definitive cgroup-OOM root-cause** (`.lava-ci-evidence/sixth-law-incidents/2026-06-17-nezha-cgroup-oom-device-build.json`) + §6.AB discrimination backlog (`.lava-ci-evidence/challenge-discrimination/2026-06-17-review.md`, 10 WEAK Challenges). **BLOCKER:** §6.Z device build cgroup-OOM-killed on nezha (shared milosvasic user-slice MemoryMax=56GiB / cur=43GiB from co-tenant ML ollama/qdrant/gunicorn; KVM path itself works `accel=kvm runner=containerized`). Operator chose RAISE-CAP; cap-watcher armed to auto-launch matrix→§6.Z evidence→§6.AA redistribute (client 1.3.11/1068, api-app 0.2.11/16) on `MemoryMax` lift. A transient API 529 overload was handled honestly (backed off, main-stream work, resumed parallel). ——— (current)

NEZHA OVERNIGHT — CHECKPOINT 2 (2026-06-17 early AM, origin `e5d82fe6`, both mirrors). Everything below is shipped + pushed + real-evidence-backed; nothing bluffed. **DONE this session:** (1) nezha System LIVE (10 containers, prod+dev api-go real-HTTP verified, 13 providers, healthy under load); heavy Go suite **49 ok / 0 fail** (real podman PG e2e/contract/parity) + 20 realtrackers. (2) **All 10 HTTP providers retry-resilient** (tokyotosho `a88467df` + 6 curated siblings `cd54341c` + 3 circuit-breaker `307b4a5d`, api-go 2.3.31) — each falsifiability-proven + re-verified on nezha (48/48 pkgs). (3) **§11.4.85 stress+chaos** for the 3 retry-mechanism class exemplars `b78730e1` (concurrency/p50-p95/goroutine-leak/breaker-state/ctx-bound evidence, falsifiability-proven). (4) **Repo pollution cleaned + root-caused + fixed** (`d256ed1a`): 19 local-only hermetic-test fixture commits (author `t@example.com`, NEVER pushed — origin always clean) dropped via reset to `6060b237`; constitution restored; root cause = `mktemp -d` unguarded under ENOSPC → `git -C ""` hit the real repo → guarded with `_safe_tmpdir()`, **falsifiability-proven (HEAD unchanged under recurring ENOSPC)**. (5) **§11.4.18 CM-SCRIPT-DOCS-SYNC drift fixed** (`e5d82fe6`, F's Phase-7 blocker): 6 docs added + 3 relocated → 59↔59 clean. (6) Redistribute STAGED `189635f6` (§6.Y bump client→1.3.11/api-app→0.2.11 + CHANGELOG + snapshots, §6.P gates pre-verified) — **distribute itself §6.Z-gated, NOT performed**. (7) §11.4.83 QA evidence bundle `docs/qa/2026-06-16-nezha-overnight/`. **Phase-7 gates GREEN at e5d82fe6** (check-constitution exit 0, script-docs 59↔59,

§6.R scanners 0). **OWED/BLOCKED (honest, no bluff):** Phase-5 emu KVM matrix + C00 smoke — §6.X/§6.AH path PROVEN to resolve on nezha (`accel=kvm runner=containerized`) but the heavyweight emu image build + the verify-all sweep keep dying on nezha under co-tenant load ~18-20 (long detached jobs get reaped; short jobs fine) — env-blocked, retry in a low-load window; Phase-8 redistribute — §6.Z device-evidence-gated (needs the emu matrix); task #10 — api-app c-shared `.so` + source-hash manifest regen (post-retry-fixes; the one nezha regression fail `TestSourceHash_ManifestMatchesLive` — expected, Phase-8 prereq); E (stress+chaos for the other 7 providers) — completeness, rate-limited-crash, respawn-owed; the submodule Containerfile UID-1000 fix (`submodules/containers/pkg/emulator/Containerfile`, uncommitted in the submodule working tree) — verified to clear STEP 14 but full emu image build unobserved (nezha load), commit to the submodule when a build is observed. **CONSTRAINTS:** macOS `/` structurally tight (go-build cache bloat — `go clean -cache` reclaims to ~37 GiB; var/folders clone-inflated; ENOSPC = annoyance not host-crash); nezha kills long detached jobs under load; API rate-limits >2-3 concurrent subagents. **RESUME PROMPT:** "Read docs/CONTINUATION.md top block + .remember/remember.md, `git fetch --all`. Origin e5d82fe6, gates green. nezha System live (10 containers, api-go 2.3.31). Continue: (a) when nezha load <12, build the emu image (`submodules/containers/pkg/emulator/Containerfile`, UID-1000 fix applied; `ssh nezha 'podman image exists localhost/lava-android-emulator:api34-x86_64'`) then run the containerized C00 smoke (cmd in this file) → Phase-5 §6.Z device evidence; (b) regenerate api-app .so+manifest (task #10, build-cshared.sh, fixes TestSourceHash_ManifestMatchesLive); (c) then Phase-8 §6.AA two-stage redistribute per .lava-ci-evidence/nezha/2026-06-16-REDISTRIBUTE-READY.md; (d) respawn the 7-provider stress+chaos + commit the submodule Containerfile fix once its build is observed. Autonomous, safest/zero-bluff, mind the tight macOS `/` (go clean -cache reclaims) + nezha-detached-job deaths (run long jobs foreground or in a low-load window)." ——— (current)

NEZHA OVERNIGHT AUTONOMOUS RUN (2026-06-16, operator away until morning) — §11.4.126 default-loop, safest/zero-bluff decisions. State at the prior checkpoint: - **Phases 1-4 DONE + committed/pushed** (both mirrors): nezha registered + §6.R debt fixed (`94519eb0`); whole System LIVE on nezha (10 containers, prod+dev api-go real-HTTP verified, 13 providers) (`d6f8fdad`); heavy Go suite **49 ok / 0 fail** (real podman Postgres e2e/contract/parity/integration); realtrackers 20 providers green. - **Phase 6 provider-resilience DONE + committed/pushed:** tokyotosho bounded-retry fix (`a88467df`, api-go 2.3.30→**2.3.31**) — root-caused (live upstream latency >20s timeout) + falsifiable test + verified live on nezha + redeployed. Then a 3-subagent audit found the same single-attempt gap in **6 siblings** → fixed knaben/nyaa/bitsearch/torrentdownloads/gutenberg/flaresolverr (`cd54341c`), all unit-tested + falsifiability-proven + **re-verified on nezha (6/6 go test ok)**. §6.T.4 BUGFIXES updated for both. - **Phase 5 (KVM Android Challenge matrix) IN PROGRESS — the §6.X/§6.AH darwin unblock:** debug APK BUILT on nezha (`app-debug.apk`, keystores shipped); emulator-matrix CLI built; `run-challenge-matrix.sh` resolves **accel=kvm runner=containerized** on nezha (the unblock confirmed). Emulator container image (`localhost/lava-android-emulator:api34-x86_64`) BUILD WAS FAILING — **root cause found + fixed:** `pkg/emulator/Containerfile:64` `useradd -u 1000` collided with ubuntu:24.04's default UID-1000 user (`UID 1000 is not unique`, exit 4); fix removes any pre-existing UID-1000 user first (committed LOCALLY in submodule working tree, NOT yet committed/pushed — pending build-success confirmation). Image **rebuilding now** under a `script`-pty logger (`/tmp/emu-build4.log` on nezha; image-exists is the truth signal). **NEXT after image builds:** run the containerized C00 cold-start smoke (`/tmp/emulator-matrix -runner=containerized -container-image localhost/lava-android-emulator:api34-x86_64 -android-sdk-root ~/Android/Sdk -apk app/build/outputs/apk/debug/app-debug.apk -avds nezha-smoke:34:phone -test-class lava.app.challenges.Challenge00CrashSurvivalTest -evidence-dir .lava-ci-evidence/nezha-challenge-smoke -cold-boot`); then commit+push the Containerfile fix in the submodule + bump the parent pin; then expand the matrix; then Phase 8 redistribute (debug→verify→release, both apps) per the subagent runbook. - **WORKSTATION DISK (recurring blocker):** macOS `/` sits structurally near-full (`/private/var/folders/t3/…` is the hog; APFS clone-inflated du). It hit 0-bytes twice → blocked ALL process tools (harness writes Bash output to `/private/tmp` on `/`). Operator cleared claude tmp + temp/caches twice (freed little — APFS clones). **Mitigations engaged:** GOCACHE/GOTMPDIR forced to `/Volumes/T7`; ≤2 subagents; keep tool outputs tiny; clean `/private/tmp/claude-501/*/tasks/*.output` after phases. Heavy work runs on nezha (348 GiB free). If `/` hits 0 again the operator must free real space (var/folders/T, podman VMs). - **RESUME PROMPT (paste into a fresh session):** "Read docs/CONTINUATION.md (top block) + .remember/remember.md, then `git fetch --all`. nezha.local System is live (10 containers); api-go 2.3.31. Continue Phase 5: check `ssh milosvasic@nezha.local 'podman image exists localhost/lava-android-emulator:api34-x86_64'` — if built, run the containerized C00 smoke (command in CONTINUATION); if the build failed, read /tmp/emu-build4.log on nezha. Then commit+push the submodule Containerfile UID-1000 fix + bump pin, expand the Challenge matrix, and run Phase 8 two-stage redistribute (client+api-app) per the subagent runbook. Operator away — autonomous, safest/zero-bluff, no blocking; mind the tight macOS `/` disk (GOCACHE on T7, tiny outputs, heavy work on nezha).” ——— (current)

NEZHA.LOCAL GATE-HOST ENABLEMENT IN PROGRESS (2026-06-16) — distributed boot of the whole System for real-service heavy testing + KVM Android matrix. Operator directive: boot `nezha.local` (i7, 8 cores, 64 GB, NVMe, **`/dev/kvm` LIVE**, rootless Podman 5.7.1 + podman-compose 1.5.0, user `milosvasic`, passwordless SSH) as a DEDICATED heavy-testing node via the Containers submodule (`git@github.com:vasic-digital/containers.git` — origin CONFIRMED, leaf module `deps:[]` = nothing extra to clone), keeping `thinker.local` as the API-distribute default. Operator-chosen scope: everything incl. **KVM containerized Android emulator matrix**; prod+dev API stacks; heavy testing = API integration+parity+real-tracker E2E + KVM Challenge matrix. **Phase 1 (config) DONE:** nezha registered in `.env` + `.env.example` (`LAVA_TEST_REMOTE_HOST` + `CONTAINERS_REMOTE_HOST_1_*` labels `storage=fast,memory=high,kvm=true,emulator=true`); `deployment/nezha/nezha.local.env` created (prod 8443/8432 + dev 8543/8433 + observability + CF-mitigation, secret-free). **2 pre-existing §6.R gate violations discovered → root-caused (evidence, not guess) → fixed:** (B) IPv4 literal `10.0.3.16` in `RepopulateProvidersOnStartupUseCase.kt:126` KDoc (introduced `d05bc71e`) → replaced with `<lan-ip>` placeholder; (A) ~25 `.lava-ci-evidence/{genymotion,release-canary}/running-devices.tsv` flagged by the UUID scanner = REAL device serials (forensic §6.I per-AVD proof) → operator-approved exemption added to `scan-no-hardcoded-uuid.sh` + §6.R clause (lockstep) + falsifiable hermetic Test 6 (rehearsed RED-when-removed). `check-constitution.sh` now exits 0. nezha carries STALE 5-week-old lava containers (`lava-postgres/migrate/api-go` Exited) — clean teardown precedes the fresh boot. **Phase 2 (build) DONE:** nezha repo synced to `94519eb0` + 17 Go submodules init’d (helixqa nested tree skipped — not needed); `lava-api-go` built NATIVELY on nezha (x86_64), runtime image `localhost/lava-api-go:nezha` via `Dockerfile.host-built`. **Phase 3 (boot) DONE — whole System LIVE on nezha (10 containers), real-HTTP verified (§6.B):** prod api-go :8443 (`/health` alive, `/ready` ready, 13 providers, v2.3.30) + dev :8543 + postgres prod/dev (migrations v9) + prometheus :9190 + loki/tempo/grafana + jackett :9217 + flaresolverr. `deployment/nezha/nezha-up.sh` captures the proven boot. **4 discovered issues root-caused+fixed (evidence, no bluff):** (1) incomplete submodule init (helixqa `--recursive` starved top-level checkout) → `--init`/`--force`; (2) api-go crash-loop `No such file or directory` = dynamically-linked binary on distroless/static → `CGO_ENABLED=0`; (3) TLS `permission denied` (0600 key vs nonroot UID) → 0644 test cert; (4) co-tenant port collisions (`:9090` llmsverifier, `:9117` boba) → 9190/9217. Evidence: `.lava-ci-evidence/nezha/2026-06-16-system-boot.md`. **NEXT:** Phase 4 heavy API tests (integration/parity/real-tracker E2E vs live nezha) → Phase 5 KVM Android Challenge matrix → Phase 6 root-cause+fix → Phase 7 final validation. Tracked: TaskList #1-7. ——— (current)

1.3.10-1067 + api-app 0.2.10-15 FULLY DISTRIBUTED (2026-06-14, both §6.AA stages, both apps, both flavors) — CLOSE-OUT RELEASE. Bundles the post-1.3.9 work: P0-1 auth-provider Auth-Token threading (RuTracker/Kinozal search — SHIPPED FOR TESTER VERIFY; on-device login+search verify with real creds OWED, blocked by subagent rate-limit — additive, no-auth path device-verified + unchanged), P1-4 remote-LAN-api key guard, P2-1 READ_API_KEY variant-suffix. Firebase: client release `4r0mnrenqrm2g` (.client) + debug (.client.dev); api-app release `3nporp8gqq8bg` (.api) + debug (.api.dev). §6.Z gate GREEN: client C00+C41-44 BUILD SUCCESSFUL + R8 canary PASS; api-app R8 canary + FGS specialUse (types=0x40000000) PASS. Auth rotated android-1.3.10-1067. §6.Y post-distribute bump: client 1068, api-app 16. §6.T.4 BUGFIXES + CHANGELOG updated. **OWED:** on-device RuTracker auth-provider verify; issue3 search video; P2-2 telemetry; the 3 operator decisions (LVA-008, login-UI surface, RuTracker cred rotation). Master plan: docs/plans/2026-06-14-remaining-work-zero-issues-master-plan.md. ——— (current)

SEARCH FIXED + FULLY SHIPPED — client 1.3.9-1066 + api-app 0.2.9-14, both §6.AA stages, both apps, device-verified end-to-end (2026-06-14). The operator-reported "search does not work in any scenario" (a 5-layer cascade) is RESOLVED and DISTRIBUTED. **Device proof (the line we chased):** fresh mDNS onboard → search "prince" → **real Internet Archive results render on screen**, `/v1/{provider}/search` → **HTTP 200** (debug-pair Chucker: 0×401 across all history; release-pair: results rendered on screen — release APKs non-debuggable so run-as Chucker N/A, on-screen render is the §6.Z primary proof). Regression test falsifiability-proven (caching-version FAILS empty-cursor / lazy-version PASSES). **The 5th-layer root cause (the one that made search NEVER work):** `api-app/.../handoff/ApiKeyProvider.kt` cached its key/port resolver lambdas in `ContentProvider.onCreate()` gated on `ApiApplication` holders that are populated in `Application.onCreate()` — but Android runs `ContentProvider.onCreate()` **BEFORE** `Application.onCreate()`, so the holders were always null → lambdas stayed `{ null }` → empty cursor forever → keyless endpoint → no `Lava-Auth` header → every auth-gated `/v1/{provider}/search` 401. Public routes need no key so they worked. Fix: resolve holders LAZILY per `query()` (`646474c0`). The `withFakes` unit test was a §6.J bluff (bypassed the real lifecycle path); replaced with a real-holder regression test. **Full layer arc (all device/test-proven):** L1 settings.endpoint host (`lava-api.local→10.0.3.16`) · L2 dead `/v1/search→`served `/v1/{provider}/search` · L3 mDNS keyless-endpoint key restore · L4 `READ_API_KEY` grant (test-VM-artifact; production-safe) · L5 ApiKeyProvider empty-cursor. **5 §6.J bluff tests removed** along the way (SSE `/v1/search` mock-served, the partial-bluff key-attach, the `withFakes` provider test). **§6.Z release gate:** client R8 canary PASS, api-app R8 canary + FGS `specialUse` (`types=0x40000000`) PASS, release-pair search e2e PASS. **Firebase release IDs:** client debug `3d2ursr68va10` (`.client.dev`) + release `23h2vj3e932bo` (`.client`); api-app debug `51scltue9hse0` (`.api.dev`) + release `100ecqf28pls0` (`.api`). Auth rotated `android-1.3.9-1066` (32 active clients, fresh pepper). §6.Y post-distribute bump: client 1067, api-app 15 (versionName held). Root-cause doc `docs/issues/2026-06-14-search-multilayer-rootcause.md`; lifecycle-bug-class audit `docs/issues/2026-06-14-lifecycle-ordering-bug-class-audit.md` (ApiKeyProvider isolated, no siblings); §6.T.4 BUGFIXES extended. **OWED (non-blocking):** L4 permission-name variant-suffix (QA-fidelity so the test VM matches production); the watchable issue-videos re-record (search e2e video now possible since search works). ——— (current)

(superseded — now SHIPPED, see above) SEARCH "does not work in any scenario" — 5-LAYER CASCADE; L1-L4 ADDRESSED, L5 ROOT-CAUSE-FOUND + FIX-APPLIED (on-device verify in progress); client 1.3.9-1066 + api-app 0.2.9-14 SHIP TOGETHER, NOT YET DISTRIBUTED (2026-06-14). L5 root cause FOUND + fixed in the api-app: the per-request `Lava-Auth` key was dropped at its source — `api-app/src/main/kotlin/lava/api/app/handoff/ApiKeyProvider.kt` cached its key/port resolver lambdas in `ContentProvider.onCreate()` gated on `ApiApplication.controllerHolder/keyStoreHolder` being non-null, on the FALSE inline comment "ContentProvider.onCreate runs AFTER Application.onCreate." Android runs ContentProvider.onCreate **BEFORE** Application.onCreate; the holders are set in `ApiApplication.onCreate` (lines 46-49) → at cache time they were always null → resolver lambdas stayed `{ null }` for the whole process → the provider served an **empty cursor**. Key-loss chain: client `ApiKeyClient.read()` → null → `withLocalApiKeyIfMissing` → keyless endpoint → `ApiBaseUrlHolder.set(url, null)` → `ApiBackedTrackerClient authKey=null` → no `Lava-Auth` header → every `/v1/{provider}/`

`search` 401. Public routes (`/providers`,`/health`) need no key → they worked, which is why search NEVER worked while everything else did. The existing `withFakes` unit test was a **bluff** (it bypassed the real `onCreate`→holder path where the bug lives → green against an empty-cursor production path; §6.J: a test that skips the real lifecycle via a test-only seam guarantees nothing about that path). **Fix (applied):** `ApiKeyProvider` resolves the holders + Running state **LAZILY per `query()`** (`resolveRunningKey()`/`resolveRunningPort()` defaults), removing the onCreate-ordering dependency; a real-holder regression test (real onCreate ordering, no `withFakes`) is OWED. **On-device pinpoint:** `.lava-ci-evidence/search-verification/2026-06-14-keyloss-pinpoint.md` (LAVAKEYDBG `withLocalApiKey readerSet=true readLen=-1` = key read returned null at the source). **RELEASE SCOPE CHANGE:** 1.3.9 is **NO LONGER client-only** — the L5 fix is in the api-app, so **client 1.3.9-1066 + api-app 0.2.9-14 ship together** (§6.Y bump applied). **Status: fix applied + root cause understood; on-device proof that search returns RESULTS is VERIFYING (in progress). SEARCH NOT YET CLAIMED TO WORK ON-DEVICE; 1.3.9 NOT YET DISTRIBUTED** — held until the L5 device gate (authed `/v1/{provider}/search` → 200 + result rows) is GREEN. L1+L2 (settings.endpoint persist+heal; GoApi → served `/v1/{provider}/search`) device-verified; L3 (mDNS keyless-endpoint key restore) RED→GREEN at the test layer; L4 (`READ_API_KEY` grant) = TEST-VM ARTIFACT, production-safe (clean-install `granted=true`). Root-cause doc: `docs/issues/2026-06-14-search-multilayer-rootcause.md`; §6.T.4 BUGFIXES extended. ——— (current)

(superseded) SEARCH 5-LAYER ROOT-CAUSE; L1-L3 FIXED, L4 TEST-VM-ARTIFACT/PRODUCTION-SAFE, L5 OPEN/VERIFYING (2026-06-14, `d05bc71e+21031f2a+addaacd0`) — L5 now root-caused + fixed; see the current block above. Operator (1.3.8-1065): "search still does not work in any scenario." Peeled back from on-device evidence to a **5-layer cascade** (not 3): **L1 (FIXED + device-verified) —** onboarding never persisted the chosen API to `settings.endpoint`, so home search (`NetworkApiRepositoryImpl.endpoint()` reads `settings.endpoint`, a different store than the Room list + `ApiBaseUrlHolder` onboarding wrote) targeted the default `GoApi("lava-api.local")` → `UnknownHostException`. Fix: `OnboardingViewModel.onSelectApi` → `SetEndpointUseCase` + `RepopulateProvidersOnStartupUseCase.reconcileActiveEndpoint()` heal. Search now reaches `10.0.3.16`. **L2 (FIXED + device-verified)** — GoApi search routed to `GET /v1/search`, a route NO backend serves → 404. Fix: `SearchResultViewModel` re-routes to `observeStreamMultiSearch` → `sdk.streamMultiSearch` → `GET /v1/{provider}/search` (served by both backends). **L3 (FIXED, `21031f2a`, RED→GREEN at test layer)** — mDNS-discovered endpoint was persisted keyless; `onSelectApi` now reads the local api-app key for keyless endpoints + app-side cold-start key-restore. **L4 (TEST-VM ARTIFACT, production-safe)** — `READ_API_KEY` signature permission was not granted on the VM because the permission name is a fixed literal (not variant-suffixed) and the VM had BOTH differently-signed api-app variants co-installed (the `INSTALL_FAILED_DUPLICATE_PERMISSION` trap). **Production (matched release pair, clean install) GRANTS it — a clean-install run confirmed `granted=true` on-device.** Recommended MEDIUM-priority QA-fidelity fix: variant-suffix the permission name. Analysis: `docs/issues/2026-06-14-readapikey-permission-variant-analysis.md`. **L5 (OPEN / VERIFYING — load-bearing)** — even on a CLEAN matched pair with `granted=true` and the engine running, `/v1/{provider}/search` STILL 401s: the `Lava-Auth` key is NOT reaching the `ApiBackedTrackerClient` search request. (Public `/providers` → 200 proves reachability ONLY, not key validity — it is registered before the auth middleware.) A logcat-instrumentation run is in progress to pinpoint the drop point. **§6.J meta-lesson:** a verify subagent proposed a hardcoded `".keyprovider"` authority L4 fix; the main stream REJECTED it after reading source — `MainActivity.buildApiKeyReader` returns `{ -> client.read()?.key }` which ignores any authority arg (would have been a §6.J bluff + §6.R literal). Verify subagents root-cause; the main stream confirms a change reaches the failing path before applying. **§6.J bluffs removed:** the 3 SSE tests serving the unserved `/v1/search` (deleted with the dead `observeSseSearch` path in `addaacd0`); incident `.lava-ci-evidence/sixth-law-incidents/2026-06-14-sse-search-v1search-unserved-bluff.json`. Root-cause doc: `docs/issues/2026-06-14-search-multilayer-rootcause.md`; §6.T.4 BUGFIXES extended to 5 layers. **Release status: SEARCH DOES NOT YET WORK ON-DEVICE. 1.3.9 is client-only and NOT shipped** — held until the L5 on-device gate is GREEN; **api-app unchanged at 0.2.8-12**. **OWED:** L5 (key-delivery, top priority); L4 permission-name suffix (QA-fidelity); existing-install key-restore (keyless heal → 401 until re-onboard). ——— (current)

1.3.8-1065 + api-app 0.2.8-12 FULLY SHIPPED (2026-06-14, both §6.AA stages, both apps, fully device-gated zero-bluff) — THE BUNDLED RELEASE IS LIVE. Operator approved “Bundle with api-app FGS fix” + “Approve specialUse”. Firebase release IDs: client debug `0471f4hkgavt8` (`.client.dev`) + release `25a1m5s209o3g` (`.client`); api-app debug `3r2osg9av09so` (`.api.dev`) + release `52j8g9uunios8` (**`.api`** `d57b960e` — its OWN Firebase app, Defect-B fixed end-to-end). **Contents:** the 4 client fixes (search-auth thread-through, server-list de-dup, onboarding select-all, password mask+eye) + **Defect A (FGS) FIXED —**

ApiEngineService dataSync→specialUse (ed03cac2) + defensive try/catch+onTimeout; + **Defect B (telemetry) FIXED —** api-app/google-services.json (gitignored, on-disk) corrected to map .api/.api.dev packages to their real Firebase app ids (fetched via Firebase MCP; both apps already existed). **Device gate (Genymotion Pixel9/API35, §6.AH VM path) ALL GREEN:** client connectedDebugAndroidTest C00+C41+C42+C43+C44 BUILD SUCCESSFUL + client R8 release canary PASS; **api-app FGS specialUse PROVEN on device — types=0x40000000 (FOREGROUND_SERVICE_TYPE_SPECIAL_USE), isForeground=true, startForegroundCount=1, no FGS exception**, engine running (JNI + mDNS _lava-api._tcp :8443). §6.Z evidence: .lava-ci-evidence/distribute-changelog/*/1.3.8-1065-test-evidence.md + 0.2.8-12-test-evidence.md + .../release-canary/0.2.8-12-apiapp-fgs/FGS-VERDICT.md. §6.O closure log 2026-06-14-apiapp-fgs-datasync-budget.md. §6.T.4 BUGFIXES appended (5 fixes). §6.Y post-distribute bump applied: client 1066, api-app 13 (versionName held — next-cycle). §6.C both mirrors converged 627a07fd. **OWED (operator video ask, NOT release-blocking):** the 4 "success videos" delivered earlier FAILED HelixQA recording-analyzer liveness (issue3=1 frame, issue4=1.7s, issue1/2 frozen ~35s — instrumentation-test recordings finish in <1s then idle); they must be RE-RECORDED as paced real-app walkthroughs (analyzer-PASS) + re-delivered to ~/Downloads. Evidence: .lava-ci-evidence/video-analysis/2026-06-14-issue-videos.md. **Console close-marks (operator-only):** Crashlytics FGS issues 9ba8502e + b9baeaed — close after observing 0.2.8-12 in the field. ——— (prior — shipped baseline)

1.3.8 CYCLE (2026-06-14) — 4 reported issues ALL FIXED + device-proven: (1) onboarding "Pick your providers" **select-all/deselect-all** control (bb357da8, C41); (2) provider Configure **password masking + eye toggle** (bb357da8, C42); (3) **search "Something went wrong"** across providers via the on-device API (fba19372, C44) — ROOT CAUSE: the dynamic ApiBackedTrackerClient used the UNQUALIFIED strict (system-trust) okHttpClient + attached NO per-endpoint key, so every search vs the self-signed-LAN auth-gated on-device API failed at the TLS handshake or HTTP 401 → generic error; FIX: the @Named("lan") permissive client + the per-endpoint Lava-Auth key on every /v1 request, threaded from onboarding + cold-start (the SP-3.1/Defect-A class, fixed for /providers but never for per-provider ops); (4) settings **server-list duplicate** (1d0294e5, C43) — EndpointsRepositoryImpl.observeAll() didn't de-dup by server identity (Room PK packs key/platform/storage), fixed with .distinctBy(serverIdentity). Each fix: unit/integration reproduction (falsifiability-rehearsed) + a device Challenge PASSING on the Genymotion VM + the success video delivered to $HOME + $HOME/Downloads (issue1-4-*.mp4; only

PASS videos delivered — `scripts/record-challenge-video.sh`, recorder-stop bug fixed `c366454f`). **HelixQA autonomous test bank** (`8f1caf1a`): `lava-api-go/internal/qa/testbank/banks/lava-reported-issues.yaml` (4 cases → C41/42/43/44) wired to HelixQA's real `pkg/testbank.LoadFile+IsValid`, go-test proven (load+validate; blank-a-name mutation → FAIL). Two real anti-bluff defects caught by RUNNING on-device (not static rehearsal): the recorder never captured video + C42's invalid `assertDoesNotExist` masking assertion (Compose retains raw text) — both fixed. **OPEN for operator sign-off:** the api-app FGS-crash fix (Defect A below) — design + prototype ready on branch `worktree-agent-a172769fe52d4bb39` (commit `10f39f43`), recommends `dataSync→specialUse`, doc `docs/issues/2026-06-14-apiapp-fgs-datasync-budget-fix-design.md`; NOT merged/shipped (needs an api-app re-ship). Next: a 1.3.8 release once you approve the FGS path (or ship 1.3.8 client-only with the 4 fixes now). All client changes converged on both mirrors. ——— (overnight, still relevant)

MORNING — READ FIRST (2 OPERATOR DECISIONS NEEDED; NOT mission-blocking, NOT tonight-regressions): the overnight field-monitor (heartbeat #5, ~05:47) caught real Firebase data — **4 FATAL events from api-app 0.2.6-10 (release) on 1 device (Galaxy S23 Ultra / Android 16, ~01:47Z)**. Investigated (`.lava-ci-evidence/crashlytics-resolved/2026-06-14-apiapp-0.2.6-10-field-events-investigation.md`). VERDICT = **(C) BOTH, both PRE-EXISTING (NOT introduced by tonight's onboarding/provider changes — `ApiEngineService.kt` untouched tonight; `google-services.json` mtime 2026-06-02): Defect A (real api-app crash, low-volume edge case):** `ApiEngineService.onStartCommand → ForegroundServiceStartNotAllowedException: Time limit already exhausted for foreground service type dataSync` at `api-app/.../ApiEngineService.kt:113` — Android 14+ caps `dataSync` FGS at ~6h cumulative; the long-lived LAN engine exhausts it after 6h uptime → crash. Fix needs a **design decision** (different FGS type / service model / periodic-restart) + §6.AE Challenge coverage → **NOT a safe blind overnight fix** (a wrong FGS change destabilizes the api-app worse than the edge-case crash) → **OPERATOR SIGN-OFF. Defect B (telemetry misattribution, config):** `api-app/google-services.json` maps `digital.vasic.lava.api` to the **CLIENT's** Firebase app id (`456475e2…`) instead of the api-app's own `.api` id (`1:815513478335:android:d57b960e955645f6cfd20a`) — so api-app crashes pollute the client dashboard, and `.api`/`.api.dev` report 404. Fix = regenerate `api-app/google-services.json` with the correct `.api`/`.api.dev` app ids (§6.H gitignored secret — operator regenerates from Firebase console) + rebuild + re-gate + re-distribute the api-app (a "ship" → operator-gated). **I did NOT blind-fix/re-ship either overnight (both require a rebuild+re-distribute = a product ship the operator gated on sign-off; Defect A's fix is design-risky). The shipped 1.3.6/1.3.7 onboarding/provider fix — the actual mission — is intact; the client (1.3.6/1.3.7) has ZERO field crashes.** The monitoring did its job: caught a real (pre-existing) signal, ran it to root cause, surfaced it safely.

VALIDATION ITERATION (2026-06-23, 4 parallel subagents, all real-evidence): verify-all 40 PASS / 14 FAIL (ZERO new failures — all 14 pre-existing §6.AD/§6.AF/§6.AI-debt); §6.AB WEAK-challenge backlog is actually **0** (all 47 Challenges have real assertions); §6.N bluff-hunt cycle5 **3 genuine / 0 bluffs** (apigo version/middleware/discovery); containers §6.R inheritance fix pushed upstream `e635ad8` (device-gate-safe: diff is doc + 1 cuttlefish line, no emulator/runtime path) → Lava pin bumped `58a0d54→e635ad8`. The remaining §6.R finding `llm_orchestrator` + the submodule §6.AD-pointer churn are the operator's external Auto-commit-automation's pre-existing work (not committed by this session). On-device search-auth: **C44 (`Challenge44ApiSearchAuthTest`) PASS on the containerized KVM emulator** (boot 31s, `.lava-ci-evidence/thinker-c00-smoke/2026-06-23-C44-on-device-PASS.json`) — value-level proof `withAuth()` attaches the correct handoff-key VALUE on the wire + the search-auth flow returns 200 on-device. **HONEST CAVEAT:** C44 builds `ApiBackedTrackerClient` by hand with a BARE OkHttpClient (no `AuthInterceptor`), so it does NOT isolate H1 (the interceptor-overwrite). **H1 evidence stack:** (1) `AuthInterceptorHandoffKeyTest` (unit, interceptor ACTIVE, falsifiability-proven) = DEFINITIVE; (2) C44 = on-device flow works; (3) operator manual test = full DI→interceptor→engine→provider E2E. **2nd validation iteration (3 subagents): §6.N bluff-hunt cycle6 = 6 genuine / 0 bluffs** across the session's OWN telemetry (writeProviderError / webhook-redaction / enabled-gate — each mutation→FAIL) + Android-core (EndpointConverter key round-trip, PostConverters clamp, ProbeMirrorUseCase reproducing Crashlytics 39469d3b). Evidence: `.lava-ci-evidence/bluff-hunt/2026-06-23-cycle6-{session-telemetry,android-core}.json`. The session's telemetry tests are PROVEN genuine.

Last updated: 2026-07-26 (PRODUCTION-READINESS SWEEP — 8 LVA gaps closed: LVA-009 embed manifest rebuilt, LVA-010 Go-compiler-panic §6.T.2 caps, LVA-014 device-gate durable fixes (baked-AVD resolution + WaitForBoot container-liveness + per-api image template, containers 746efe3 pushed), LVA-015 nested-scroll §6.O closure (scanner CHECK 3 + JVM guard + C71), LVA-016 submodule divergence, LVA-017 CONTINUATION, LVA-018

14 script docs, LVA-036 llm_orchestrator mirrors converged d6cc248; workable-items DB deduplicated (10 false duplicate rows removed), Issues.md 27→17 open; LIVE TESTING + Firebase distribute NEXT — see top banner.)

Last updated (prior): 2026-07-04 (Crashlytics FATAL #2 credentials-decrypt AEADBadTagException FIXED reproduce-first + live-Crashlytics triage of all 3 open FATALs + 1080-cycle doc/CHANGELOG/§11.4.6/export-sync work — see top banners. Subagent fleet quota-blocked until 4:10pm Europe/Moscow.)

Last updated (prior): 2026-07-03 (§6.AC archiveorg mapError hardening — egress-stall-masquerade class CLOSED, reproduce-first RED→GREEN, banked device-green matrix UNCHANGED, public providers still gated; see top banner. Prior 2026-07-02: goapi × rutracker keystone GREEN — verdict=PASS, device-recorded, 5-fix stack B/C2/D/E + Challenge70 deselect; prior 2026-06-30 context follows) (**AUTONOMOUS QA HARNESS (Phase 0) — self-driving QA built under `scripts/autonomous-qa/`; emulator image boots containerized-KVM in ~30s; Go backend serves `/health`; onboarding proven end-to-end (RuTor anon → main app, keystone run #1 screenrecord). Keystone `Challenge70AutonomousQaProviderMatrixTest` surfaced TWO TEST-HARNESS artifacts (NEITHER a product defect): (a) OFF-MAIN side-effect nav under `createAndroidComposeRule` → `IllegalStateException: Method setCurrentState must be called on the main thread` — FIXED with a main-thread guard (`runOnMainThread`) in `core/navigation/.../NavigationController.kt:50-56` (no-op on the production happy path); (b) LVA-008 teardown ISE `State must be at least 'CREATED' to be moved to 'DESTROYED'` at MainActivity onDestroy (upstream androidx-navigation defect, 6 app-level fixes device-falsified). MITIGATION: marker-based verdict in `scripts/autonomous-qa/run-iteration.sh:110-187` (PASS tied to the `C70-RESULT ... DOWNLOAD-OK` marker + LVA-008-only failure, NOT process exit). HelixQA matrix banks authored at `lava-api-go/qa/banks/lava-{rutracker,rutor,nnmclub,kinozal,iptorrents}-matrix-journey.yaml`. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-06-30-keystone-offmain-nav-and-lva008.json`. Details in §1b "2026-06-30 — Autonomous QA harness (Phase 0)".** EGRESS SOLVED via nezha (2026-06-30):** nezha.local has a VPN egress IP (`31.169.53.44`, ≠ thinker's blocked datacenter IP) that PROVABLY reaches the blocked providers (rutracker=200, nnmclub=200, archiveorg=200; kinozal=522 Cloudflare-flaky → FlareSolverr-solvable) and has `podman` + `/dev/kvm`. DECISIVE finding: the bundled tracker clients (`RuTrackerHttp` etc.) log in DIRECTLY from the device, so the lava-api-go backend SOCKS proxy is irrelevant to them — the EMULATOR's egress is what matters → the matrix must run where the emulator's egress reaches the providers (Path A: matrix ON nezha). **DISTRIBUTE-PREP DONE for 1.3.12-1078:** §6.P snapshot `1.3.12-1078.md` + §6.AK `cycle-coverage-map-1.3.12-1078.yaml` written; CHANGELOG entry + monotonic-version pointers verified (1078 > debug/release 1077). **ONE REMAINING GATE:** the §6.Z device test-evidence `1.3.12-1078-test-evidence.json` (matching the COMMITTED 1078 SHA, ≤24h) does NOT exist yet — it is produced only by the green QA matrix on nezha; until then firebase-distribute.sh's §6.AK Gate 7 correctly exits 2 and BLOCKS the distribute. ——— (current))

Last updated (prior): 2026-06-26 (**§6.AK CYCLE-COVERAGE GATE WIRED — Check 10 + Gate 7 enforce coverage-intersection; apiapp extension proven hermetically (5/5); pending changes awaiting commit: C48/C52/C66 LenientTeardownRule refinements, stress-chaos re-runs, design-spec post-1076 revisions, vision_engine pin advance af93b4bc, OpenDesign transitions design spec awaiting operator decisions 1-6**. HEAD at `28a8b79b` — feat(6AK): wire cycle-coverage gate into firebase-distribute (Gate 7) + git hook (Check 10) — closes 6AK-debt. — the §6.AK incident root-caused & remediated; 4 parallel real-evidence streams.** Operator booted a live Genymotion Pixel 9 / API 35; this session found WHY the 1076 gate lied (two systemic defects) and produced reproduce-first device evidence. **(1) Keyguard harness fix** — the gate booted to the lockscreen so EVERY Challenge false-failed `No compose hierarchies`; `scripts/run-genymotion-challenges.sh` now `wm dismiss-keyguard+KEYCODE_HOME` before instrumentation (verified C00+C01 PASS). **(2) LVA-008 confirmed unfixable test-side** — the nav-teardown ISE is a MAIN-THREAD uncaught exception; falsified an `onException` runner guard, the in-flight `LenientTeardownRule` (C52 HAS it, still crashed 1/4), AND a nav-compose 2.9.8 bump — all reverted; matches the documented upstream defect. **DEVICE-CERTIFIED NON-BLUFF (captured RED+GREEN reproduce-first, `.lava-ci-evidence/genymotion/`):** C63→LVA-087 (Welcome count) + C64→LVA-088 (API friendly names + honest cloud label) — each has a RED (mutated production→FAIL with clear assertion) AND a GREEN. **DEVICE-GREEN:** C00 cold-start, C67 app-id-suffix (LVA-091 substantive). **LVA-085 ENGINE EXONERATED** — `lava-api-go /v1/providers` serves the correct `"YTS"` (`providers.go:66`), now regression-guarded (RED→GREEN); the in-flight client "fix" edited a NON-RENDERED field and its test was a §6.J-PROVEN bluff (passed with the change reverted) → both REMOVED; LVA-085 is conclusively a client-render bug in the LVA-008-blocked path. **LVA-090 Select-All proven correct in**

source (`onToggleAllProviders = requiresNoCredentials`); C66 device-observation unresolved (`onCount==0` = checkbox-semantics/test gap, NOT a feature bug — semantics-dump follow-up owed). **lava-api-go (combined integration `go build && go test` GREEN):** single-provider + MultiSearch per-provider deadlines → config (§6.R bare literals removed, each RED→GREEN), search chaos/stress 256-req×4-fault (deadline fires 176ms, no leak/panic/race, RED 4.0s→GREEN). **PREVENTION:** `scripts/check-cycle-coverage.sh` + `tests/cycle-coverage/` — the §6.AK mechanical gate that CATCHES the 1076 incident (hermetic 7/7; the exact bluff mutation → RED). 10 covering Challenge drafts C58–C67 authored + compile clean (C58–C62 search-flow are device-blocked by LVA-008). versionCode already at 1077 (§6.Y satisfied). **NEXT:** LVA-008 upstream minimal-repro (in progress), C66 semantics-dump, wire check-cycle-coverage into firebase-distribute + pre-push (§6.AK-debt), search-flow device evidence once LVA-008 unblocks. ——— (current))

Last updated (prior): 2026-06-26 (**§6.AK ANTI-BLUFF INCIDENT — the 1076 distribute shipped BROKEN flows; operator retest confirms “almost nothing fixed”.** Operator retested the distributed 1.3.12-1076 (new video `Screen_Recording_20260626_122718_Lava.mp4`, recorded 09:27Z AFTER the 09:14Z distribute, pulled to `.lava-ci-evidence/video-analysis/2026-06-26-rerecord/`). **CONFIRMED STILL-BROKEN on 1076:** (A) **search STILL fails** “Something went wrong” (issue #1 — the 1076 “search works again” claim is FALSE for the user; root cause NOT yet obtained, needs §6.AC Crashlytics readout / adb logcat / engine repro); (B) **results filter chips render RAW LOWERCASE ids** `rutracker/torrentdownloads/yts` not display names (issue #4 unfixed in the results surface — AND this session’s LVA-079 “fix” was itself a §6.AB bluff: it sorted the raw id list + its JVM test asserted the id LIST not the rendered LABEL, so it passed while the user sees `rutracker`); (C) **input vs results chip divergence** (Kinozal.tv missing from results, issue #3 unfixed). **ROOT CAUSE of the green-tests-broken-features:** the §6.Z gate that authorized BOTH 1076 distributes executed ONLY `Challenge00CrashSurvivalTest` (cold-start). The 60+ Challenges (C01-C57) covering search/provider/chips were NEVER executed. Device repro NOW (thinker KVM, `.lava-ci-evidence/1076-repro/`): `Challenge52`(search chips)+`Challenge48`(sync toggle) **FAIL** — both crash identically `IllegalStateException ...DESTROYED` (LVA-008 nav-teardown, systemic across nested routes `search/search_input`+`provider_config`, fires at instrumentation activity-destroy); C41/C20/C16/C47 PASS. **CONSTITUTION EXTENDED — §6.AK Cycle-Coverage Device Gate** (CLAUDE.md): no distribute unless every CHANGELOG-claimed user-visible fix has an EXECUTED+PASSED reproduce-first covering device Challenge for the exact shipped artifact; **C00 alone NEVER qualifies**; reproduce-first MUST be on-device for UI/flow issues; inherited “fixed” claims are UNVERIFIED until re-confirmed; per-video reproduction set mandatory. + §6.AK-debt (mechanical `scripts/check-cycle-coverage.sh` + firebase-distribute gate OWED). Incident: `.lava-ci-evidence/sixth-law-incidents/2026-06-26-c00-only-gate-shipped-broken-flows.json`. **NEXT (reproduce-first, NO distribute until covering Challenges PASS on device):** (1) chip-label fix → results chips render display names matching input set, Compose test asserting the RENDERED LABEL (not id-list); rewrite the LVA-079 bluff test; (2) search-fail root cause — OWED operator evidence: Crashlytics §6.AC readout (http_status/base_url_host/error on /v1/{provider}/search) OR adb logcat OR I reproduce engine-side; (3) re-gate EXECUTING C52/C48/the search Challenge + the new chip-label Challenge. **OPERATOR-OWED:** Crashlytics console export / paste (firebase CLI cannot read it — verified: only symbols/mappingfile upload; gcloud+bq absent). ——— (current) **[prior 1076-ship entry below is now SUPERSEDED — that distribute was the bluff]** **1076/1.3.12 CLIENT DISTRIBUTED + §6.Z-GATED — the 1076 cycle’s NEXT executed end-to-end.** Operator directive: continue fixes, fix new Crashlytics, rebuild/test/validate/verify, commit+push ALL submodules+main to all upstreams, fetch+pull+merge all submodule latest, investigate+fix the QA-team video issues, track everything via the workable-items system (“HelixTrack” clarified by operator = the constitution-mandated workable-items system; HelixTrack has no repo/constitution mention), “also attempt full gate + distribute”. **DONE (verifiable):** (1) **All 11 QA-video issues now SEPARATELY tracked** (operator follow-up “each video issue MUST have its own workable item”): LVA-079 (#3) + LVA-083..LVA-092 (#1/#2/#4/#5/#6/#7/#8/#9/#10/#11). Honest statuses — the 1076-code-fixed issues (#1/#2/#4/#5/#6/#7/#8/#9) are **In progress (pending on-device confirmation)** NOT prematurely Fixed; #10 Queued (UNCONFIRMED app-ID); #11 Completed (negative finding). validate OK 91. (2) **Video #3 FIXED** (LVA-079): the severe symptom (results chip set changing run-to-run) was already gone in 1076 (results chips read the requested set); residual gap closed — `SearchPageState.filterProviderChipIds` re-applies `distinct().sorted()` so input/results agree deterministically by guarantee. Failing-first `SearchResultFilterChipDeterminismTest` (27/0 JVM, falsifiability-proven), Bluff-Audit’d, commit `b06ec32b`. (3) **46-commit push backlog CLEARED** — github+gitlab both at `b06ec32b` (pre-push CI gate passed; the entire 1073→1076 line was unpushed). (4) **1076 built** (build_and_release.sh BUILD SUCCESSFUL; client debug 34MB + release 6.7MB

R8; aapt-verified versionCode 1076 `.client.dev`/`.client`). (5) **§6.Z C00 cold-start gate PASS** on thinker (Linux x86_64 + /dev/kvm) containerized-KVM emulator via Containers `emulator-matrix --runner=containerized` (image `lava-android-emulator:api34-x86_64`, podman): `Challenge00CrashSurvivalTest` PASS, boot 33.2s, test 67.8s, cold-boot, serial, **gating:true** — `.lava-ci-evidence/1076-c00-gate/real-device-verification.json` + `.lava-ci-evidence/distribute-changelog/firebase-app-distribution/1.3.12-1076-test-evidence.{md,json}`. thinker FF'd to `b06ec32b`, built `:app` from exact source. (6) **§6.AA two-stage distribute COMPLETE (client)** — Stage 1 debug `216fs8pr1dkbg` (`.client.dev`) + Stage 2 release `50dusshe2uru0` (`.client`); §6.P + §6.Z content-check (1076==1076) passed both; last-version-debug+release → 1076; combined-distribute-authorization = standing-operator-policy-2026-06-25. (7) **§6.Y post-distribute bump → client 1077** (versionName held). **SUBMODULE FETCH/MERGE (LVA-081):** fetched all 24 read-only — **19/24 already at upstream HEAD** (merge is a no-op); `constitution` behind 31 = operator-gated CONST-049 pipeline (=LVA-3); `helixqa` behind 5 FF available (dirty nested pins); `vision_engine/llm_orchestrator/llm_provider` local-AHEAD (157/124/125) → must NOT bump (would discard local work, =LVA-036). Nothing safe to bump beyond what's already current. **CRASHLYTICS (LVA-082) HONEST:** Firebase CLI 14.17.0 exposes only `crashlytics:symbols/mappingfile` upload — **NO issues/non-fatals read command**; `bq` absent → the dashboard is NOT programmatically readable here. The shipped §6.AC telemetry + known crash tickets (LVA-008 upstream nav defect) are the available backlog; new console items need operator paste. **api-app ALSO SHIPPED (operator "yes"):** :api-app built (debug `.api.dev` + release `.api`, aapt-verified vc22), **§6.Z gate PASS** on thinker KVM (`Challenge01ApiAppColdStartTest`, boot 31s, gating:true, `.lava-ci-evidence/1076-apiapp-gate/`), **§6.AA two-stage distribute** debug `6mn8lmmqke928` + release `15l34kl1d1138` (content-check 22==22), §6.Y bump api-app 22→23. LVA-080 CLOSED Completed. **OPEN/FOLLOW-UP:** (a) **thinker lava-api-go network API is DOWN + UNCONFIGURED** (no container, no `.env`) — operator must provision the api-go `.env` (1076 allowlist + rotated pepper) + bring up `docker-compose` so the distributed client can be device-tested against the LAN API (§6.H operator secret-provisioning, runbook `docs/runbooks/2026-06-23-testing-client-against-network-api.md`); (b) video items LVA-083..091 stay In progress until the operator's on-device pass confirms the search/provider fixes (C00 proves cold-start only, not the search flows — §6.J honesty); (c) lava-api-go network API on thinker (`:8443`) needs restart with the 1076 allowlist for operator on-device network testing per `docs/runbooks/2026-06-23-testing-client-against-network-api.md`; (d) LVA-5 Firebase-token rotation still operator-

> blocked. **HELD:** the macOS-host api-go *image* build fails (podman machine not started) — non-blocking (binary built; image only needed for local container runs).**)
>
> **Last updated (prior):** 2026-06-25 (**1076/22 CYCLE — TOGGLE-CRASH + SEARCH-UNUSABLE + DISPLAY FIXES + 13 UI CHALLENGES, converging to prod redistribute. Standing operator policy: ALWAYS ship prod every cycle + operator tests prod (memory `always-ship-prod-every-cycle`; combined-distribute-authorization: standing-operator-policy-2026-06-25). 1075 PROD already shipped both apps (client `3bn8ctt6r4pag`, api-app `658ashe9bpk88`). This cycle (driven by the operator's nezha video `Screen_Recording_20260625_194837_Lava.mp4` — 11 issues, `.lava-ci-evidence/video-analysis/2026-06-25-lava-issues-video.md`):** FIXED (§11.4.146 reproduce-first, all merged):** (1) provider Sync/bind/mirror toggle CRASH (release-only SerializationException ProviderConfigViewModel.kt:92 — missing kotlinx-serialization plugin + R8 keep-rules; Crashlytics `eaa80c14`); (2) SEARCH-UNUSABLE (SearchInputViewModel hardcoded-4-provider list → wrong providers → 0 results → error; now ProviderConfigRepository.observeAll() filtered+sorted + loading/empty state); (3) 5 display/onboarding bugs (#4 chip names, #6 count, #7 not-a-§6.R-violation, #8 discovered-API name, #9 select-all). **COVERAGE:** 13 new/rewritten Compose UI Challenges — C48–C51 provider toggles, C52 search chips, C53–C54 credentials, C55–C57 scaffold/rating/account, api-app C06–C07, C31–C35 (WEAK Class.forName→real render/interaction). **Coordinated compile GREEN** (both androidTest APKs built — all 13 Challenges + fixes compile). §6.Y bump **client 1076/1.3.12** (versionName bumped — real user-facing fixes) + **api-app 22/0.2.11**; auth rotated `android-1.3.12-1076` (fresh pepper `1240380a`, 41 active). CHANGELOG 1.3.12-1076 + api-app 0.2.11-22 + snapshots. **LVA-008 = CONCLUSIVELY UPSTREAM androidx defect — 8 candidates device-falsified** (Candidate #7 single-NavHost collapse: C00/C01/C07/C08 PASS no-regression, C06/C11 FAIL identical ISE `.lava-ci-evidence/lva008-cand7-gate/`); minimal-repro authored for upstream filing. AccountItem found ORPHANED (superseded by `8cd070e0`; operator-gated removal recommended, `docs/issues/2026-06-25-account-orphan-investigation.md`). **NEXT:** commit+push 1076 → restart APIs (new allowlist) → build_and_release → §6.Z gate on thinker (R8-release toggle + search on-device + all Challenges EXECUTED) → prod redistribute both apps. **OPERATOR-GATED:** vision_engine pin-bump; app-ID co-mingling google-services regen; AccountItem removal. **HELD for tag-time docs-sync:** doc-export `5105537e`, LVA-008 research `9560410c`, minimal-repro `71bee48c`.**)

Last updated (prior): 2026-06-25 (**1075/21 DEBUG BUILDS DISTRIBUTED (§6.AA stage 1) — release stage OPERATOR-GATED. Master at `18245fec`. Both debug variants live on Firebase: client `1.3.11(1075)→digital.vasic.lava.client.dev` (release `080hgo0apj9po`); api-app `0.2.11(21)→digital.vasic.lava.api.dev` (release `6gol4unlil0pg`). Both §6.Z-gated on real containerized-KVM (client C00 boot 27s `.lava-ci-evidence/1075-gate/`; api-app cold-start boot 24s `.lava-ci-evidence/1075-apiapp-gate/`). Auth rotated for 1075 (fresh pepper `ec5a9133…` + client-name `android-1.3.11-1075`, active-list prepend); local prod+dev APIs restarted with the new allowlist. 1075 ships the never-distributed 1073 search-timeout fix.** LVA-008 (search→back nav-teardown crash) = CONFIRMED UPSTREAM androidx-navigation defect — 7 app-level candidates now device-falsified** (6th=Activity-scoped LocalLifecycleOwner reverted; 7th=launchSingleTop dedupe, branch `lva-008-cand8-gate@e8c81728`, C06+C11 FAIL `.lava-ci-evidence/lva008-cand8-gate/`). Crashlytics CONFIRMS it live in field on 1075 (FATAL, 4 events/4 users, fresh-today). Minimal-repro + upstream-issue-draft authored (branch `worktree-agent-a26086a188d3abfa6@71bee48c`, `docs/issues/upstream/lva-008-androidx-navigation/`). nav latest 2.9.8/2.10.0-alpha05, NOT fixed upstream. **7 real fixes merged+pushed this session:** gate-runner `--container-image` CLI fix, firebase-hermetic+gitignore gates, NDK-path portability, api-app okhttp conflict (verified BUILD SUCCESSFUL), §11.4.147 agent registry, commit-docs hardening, panoptic 4-suite compile-fixes. ~18 subagents, all captured-evidence. **Both REST APIs LIVE**: prod `:8443` + dev `:8543` (mDNS, /health 200, Postgres ×2 podman). **OPERATOR-GATED NEXT:** (1) §6.AA stage-2 RELEASE — test debug builds on-device → “ship release” → `firebase-distribute.sh --release-only` both apps; (2) LVA-008 Candidate #7 (single-NavHost collapse — high-blast-radius refactor, go-ahead needed) OR file the authored androidx repro; (3) vision_engine pin-bump (install_upstreams.sh added in submodule@`af93b4bc`, upstream push+pin OWED); (4) app-ID co-mingling (api-app/google-services.json → wrong app id, Defect B, gitignored-secret regen). **HELD for tag-time docs-sync:** doc-export `5105537e`, LVA-008 research `9560410c`, minimal-repro `71bee48c`. verify-all ~43 PASS (3 gates closed this session); vm-image sha256 honestly deferred.**)

Last updated (prior): 2026-06-24 (**SUBMODULE INHERITANCE GATE — HONESTLY GREEN ON COMMITTED STATE (operator-chosen Option A). Root cause (4-subagent investigation, all real evidence): the latest upstream submodule HEADs migrated to the slim inherit-from-Helix-Constitution stub, dropping the Lava §6.R/§6.S/§6.X verbatim headings; the §6.AD**

`## INHERITED FROM constitution/` pointer that made `check-constitution.sh` pass was UNCOMMITTED working-tree drift (re-added by the external auto-commit automation) → gate green-on-drift = a fresh-`clone --recursive` of the pinned state would FAIL = a §6.J bluff. Operator chose Option A (commit pointers to all upstreams).** Executed: committed the additions-only §6.AD pointer block (CLAUDE/AGENTS/CONSTITUTION, +2/-0 per doc, zero removals) onto 17 submodules' current HEADs + pushed each to all real mirrors (all clean FF, 0 aborts/rejections): auth `0ae1f5d`, cache `6a9a3e5`, challenges `791208f`, concurrency `32b7efa`, config `5268809`, database `8adb19c`, discovery `0e491a3`, doc_processor `253f92e`, helixqa `0ee8492`, llm_orchestrator `77f68bf`, llm_provider `79bab38`, middleware `9bcac12`, observability `bd3a5be`, ratelimiter `6463e51`, recovery `30961a1`, security `0b61cc0`, vision_engine `85dfd0d`. `check-constitution.sh` updated (`doc_inherits_clause` helper accepts the §6.AD-canonical pointer per §6.AD.8) + `test_clause_6r_inheritance.sh` in lockstep; falsifiability-rehearsed (strip pointer → gate EXIT=1 → revert → EXIT=0). Gate now **EXIT=0 / 0 MISSING on the committed state** — fresh-clone honest, no drift dependency. Hardlink `.git` backup `/Volumes/T7/Projects/.lava-gitbackup-20260624-153354`. Code-review + security scan of all pulled-in upstream code = CLEAN (0 Critical/Warning). **HONEST CAVEAT:** submodule pin bumps still NOT locally build/test-verified (host build path blocked, §6.X/§6.AH-debt); `verify-all-constitution-rules.sh` still has the pre-existing §6.AF/§6.AI CM-COVENANT-propagation debt FAILs (separate from this gate).**)

Last updated (prior): 2026-06-24 (**SUBMODULE CLONE-RESOLVABILITY FIX + MIRROR CONVERGENCE — follow-up to the sync below, per operator "everything must be fixed / nothing broken / don't lose work". Hardlinked `.git` backup taken first (`/Volumes/T7/Projects/.lava-gitbackup-20260624-153354`, §9). Swept all 25 repos: ZERO unpushed/stashed/stranded work (nothing at risk). Audited every pin for fresh-`clone --recursive` resolvability against its `.gitmodules` URL — found + fixed:** (1) `llm_orchestrator`** pin `8d2f8f6` lived only on `HelixDevelopment` but `.gitmodules` cloned from `vasic-digital` (would break fresh clone) → repointed `.gitmodules` URL to canonical `HelixDevelopment/LLMOrchestrator` (consistent with the `helixqa` precedent; completes the adopt-canonical-github decision; `a484f7d` preserved on the vasic-digital remote, content already on the HelixDevelopment line). **(2) `vision_engine`** `HelixDevelopment` was 6 commits ahead (the latest) → FF'd to `f3ef1f2` (clean, HEAD was ancestor) + verified resolvable on the vasic-digital `.gitmodules` URL. **(3) `llms_verifier`** gitlab mirror was 3 behind → FF-pushed to converge (§6.C). Re-audit: all 24 pins resolvable (the lone `constitution` flag is a false positive — pin IS on its origin/main, same HelixDevelopment URL). Comprehensive code review run on the session diff (config/docs only; no functional source authored by us). **HONEST CAVEAT unchanged:** pins synced to upstream HEAD, NOT locally build/test-verified (host build path blocked, §6.X/§6.AH-debt).**)

Last updated (prior): 2026-06-24 (**SUBMODULE + MAIN-REPO SYNC — operator-directed fetch+pull of all 24 submodules then commit+push to all upstreams. 17 submodules fast-forwarded to upstream HEAD (all 0-ahead, clean FF); `helixqa→9f94c53` (FF; internal nested-of-nested working-tree drift left untouched — not parent-recorded); `containers→71d3256` (FF to `876474b` + §6.AD `constitution/` inheritance pointer-block added to AGENTS.md+CONSTITUTION.md, pushed to github+gitlab); `llm_orchestrator→8d2f8f6` (operator chose ADOPT-CANONICAL-GITHUB-HEAD, resolving the vasic-digital↔HelixDevelopment divergence — github HEAD already carries proper `constitution/` inheritance blocks, so no extra commit). `constitution/mdns/vision_engine/tracker_sdk` already current. 20 parent pins bumped; parent committed + pushed to github+gitlab.** HONEST CAVEAT:** pins synced to upstream HEAD, NOT locally build/test-verified — the host build path remains blocked (§6.X/§6.AH-debt KVM-absent + prior T7/nezha constraints); the parent pre-push gate (`scripts/ci.sh --changed-only`) ran but does not rebuild submodule internals.**)

Last updated (prior): 2026-06-24 (**SEARCH CANCEL/TIMEOUT FIX — `SearchResultViewModel` back-press now cancels in-flight streaming coroutine (`activeSearchJob?.cancel()`); `withTimeout(25_000L)` client-side deadline fires 5 s before OkHttp 30 s readTimeout, surfacing `SearchResultContent.Error` with Retry. 3 falsifiable regression tests (cancel-on-back + timeout→Error + back-after-complete-preserves-Content); 26/26 `feature:search_result` tests pass. §6.O closure log + §6.T.4 BUGFIXES.md written. Pending: §6.Y versionCode bump + commit + push.**) [prior: 2026-06-23 (**SESSION VALIDATION COMPLETE (HEAD fb918b1f): the search fix (1072) + telemetry (1071) + api-app-18 corrective + backend 2.3.32 are SHIPPED, device-gated, content-verified, both mirrors converged. Validated to the hilt across 3 iterations + ~30 subagents, all real-evidence/zero-bluff: verify-all 40/14 (0 NEW failures), backend 50/50 (real podman PG e2e/integration/contract/parity/stress/compose), Android JVM 860/0, §6.N bluff-hunt cycles 4-8 = 18 genuine / 0 bluffs, §6.H credential-leak audit CLEAN + the one defense-in-depth gap (panic-middleware http.url→Path) hardened+tested, the new**

§6.Z aapt content-guard given its §6.A contract test, C44+C45 on-device PASS (C45 H1-isolation blob-dependent → definitive proof is the JVM AuthInterceptorHandoffKeyTest; a non-empty-blob on-device C45 falsifiability run is in flight now that thinker is back), §6.T.4 BUGFIXES.md filed. THE ONE OPEN ITEM is operator-side: install 1.3.11-1072 + test search end-to-end (runbook docs/runbooks/ 2026-06-23-testing-client-against-network-api.md; network API live at https://192.168.0.228:8443). Earlier same day: 1.3.11-1072 = THE SEARCH FIX (H1: AuthInterceptor only-if-absent so the handoff key survives — wire-tested) + api-app 0.2.11-18 corrective (the 17-release shipped a stale vc-16 binary — incident logged) + firebase-distribute.sh aapt content-versionCode guard. Clean rebuild all-4 aapt-verified 1072/18. Earlier same day: §6.AC TELEMETRY CYCLE — comprehensive Crashlytics non-fatals across client+api-app+backend (esp. search: ApiHttpException→http_status/request_url/method), backend webhook bridge (honest — no fake Crashlytics ingest), §6.AC STRICT 0 violations, lava-api-go 2.3.32 smoke PASS, network-exposed API LIVE on thinker (https:// 192.168.0.228:8443), telemetry doc+html+pdf, bluff-hunt cycle4 3/0. Rebuilding+distributing 1.3.11-1071 + api-app 0.2.11-17 (JVM 0-fail). H1 SEARCH-401 BUG CONFIRMED source-verified (AuthInterceptor Lava-Auth header overwrites the handoff key — same name, replace, interceptor-fires-last) → fix is 1072 (auth change, needs device verify; 1071 telemetry confirms it from the field). Earlier: ALL 4 VARIANTS DISTRIBUTED + FIRST §6.Z C00 GATE GREEN — client debug `7dhneaj2he79g` + release `2tjq0j3ab9e2g`, api-app debug `7esmk60v7ki0o` + release `00m6ja0b4evpg`, every download-URL sha == staged APK (picker fix `134d0180` shipped the right binaries; prior tester emails were the WRONG 1068). §6.Y 1069→1070 (then post-distribute →1071 / api-app 16→17) + auth rotated (fresh pepper, 36-client append-only). Thinker C00 PASS `all_passed:true` boot 33s on real KVM (Containers `58a0d54`) = the project's first genuine §6.Z attestation; release ship is device-gate-GREEN. Search-401 ROOT cause still OWED (operator Chucker readout on the 1070 debug). Earlier same day: OPERATOR HANDOFF — wrote `docs/instructions/ 2026-06-23-provide-to-continue.md`: exactly what to provide to continue (PART 1 Chucker readout = no secrets → unblocks the search root cause; PART 2 the 4 signing/build secrets — `.env` + `keystores/{debug,release}.keystore` + `app/google-services.json` + `api-app/google-services.json`, all gitignored, all currently ABSENT → gate the device path; PART 3 per-artifact next steps; PART 4 the 2 open decisions). 4 session commits pushed + §6.C-converged on both mirrors at `922ecbca`; thinker clone done at ~/lava; §6.H thinker-password-pasted-in-chat → rotate (not stored). Earlier

2026-06-22: SEARCH STILL BROKEN ON RELEASE (operator) — Phase-1 root-caused to the release key/auth or engine-reachability layer (non-2xx in ApiBackedTrackerClient.getString; parse path R8-safe; 1.3.9 release “proof” was a weak §6.J on-screen-render). Found+fixed a real §6.AC defect: SearchResultViewModel:139 ProviderFailure dropped the cause with no telemetry → now recordProviderFailure(); falsifiability-proven (mutate→RED expected:<1> was:<0>→revert→GREEN 3/3). Root HTTP cause OWED — needs a release device gate on thinker. See the SEARCH top block.** Earlier same day: SESSION SYNC — §6.S re-sync after HEAD drifted 4 commits past CHECKPOINT-4: 3 anti-bluff test commits + the `11f29ff8` “Auto-commit” all-20-pin sweep (19 forward / 1 divergent llm_orchestrator; investigated, freeze-by-default procedural-violation flagged, operator decision pending — recommend neutralize the external auto-commit automation, don’t mass-revert). nezha cap UNCHANGED at 56 GiB (watcher not fired) but co-tenant load eased to ~26 GiB headroom; emu image still absent → redistribute still owed. §6.AB WEAK backlog (10) understood + fix pattern identified. See SESSION SYNC top block.** Earlier 2026-06-17: **OVERNIGHT autonomous cont’d — 15 commits: 16 new falsifiability-proven anti-bluff tests (8 subagent waves + 2 main-stream) + §6.N bluff-hunt 7/7 genuine + stress/chaos (all 9 providers) + gofmt/Spotless/manifest hygiene + cgroup-OOM root-cause + §6.AB backlog. Build DONE+releasable at `08ed7e0f`. §6.Z device gate cgroup-OOM-BLOCKED on nezha shared user-slice (MemoryMax=56GiB/ cur=43GiB) — operator chose RAISE-CAP, cap-watcher armed to auto-launch matrix→redistribute. See CHECKPOINT 4 top block + RESUME PROMPT.** Earlier: nezha gate-host enablement — Phases 1-3 DONE: nezha registered + §6.R debt fixed; whole System LIVE on nezha (10 containers, prod+dev api-go real-HTTP verified, 13 providers); 4 boot-issues root-caused+fixed (CGO static, TLS perms, submodule init, co-tenant ports). NEXT: Phase 4 heavy API tests → Phase 5 KVM Android Challenge matrix**). Operator (overnight): “away until morning, fully autonomous, safest/most-stable/zero-risk/zero-bluff decisions, best possible most-stable build by morning is ABSOLUTE PRIORITY, resolve blockers autonomously.”

MORNING HANDOFF (read first): Two full release cycles shipped overnight, both fully device-gated on the real Genymotion VM, zero bluff. **1.3.7-1064 (client) + 0.2.7-11 (api-app) is the current shipped build on BOTH `.dev` debug AND production release Firebase channels.** Firebase release IDs: client debug `(this cycle)`, client release `(this cycle)`, api-app debug+release `(this cycle)` — see the distribute logs. Nothing is broken; nothing is half-done; no blocker is open. The §6.Y post-distribute bump to 1.3.8-1065 / 0.2.8-12 / api-go 2.3.30-2330 is applied so the tree is clean for the next cycle. **What you may want to do:** (1) install the release build + smoke-test the onboarding / providers fix on your `…:8443` API + try the new Tokyo Toshokan provider; (2) optional daytime work in the backlog below (more curated providers, the M3 download-coverage gap that needs the emulator matrix, Phase-2 Cloudflare/ 1337x which is honestly CF-blocked, Crashlytics console close-marks which are operator-only). **I deliberately did NOT start a 1.3.8 feature cycle overnight** — per your "most stable, zero risk" mandate, shipping more product changes without your daytime sign-off would gamble the stable build; 1.3.7 IS the best+stable target achieved. **Overnight ANTI-BLUFF SWEEP (test-only, zero product change, all main-stream re-verified, all on master):** §6.N-hunted 16 gate-shaping production paths across Go + Kotlin. **4 REAL gaps found + closed** with new falsifiable tests: (a) `SseClient` live-search parser asserted only event TYPES not the `data` PAYLOAD → a corrupted-search-results payload would've shipped green (`dcf478c0`); (b) Room `Converters` topic Author/Category round-trip + `VisitedTopicDao`/`FavoriteTopicDao` ordering+dedup were entirely untested → silent persisted-metadata corruption invisible (`c2629469`); (c) `SearchResultNavigation` providerIds deeplink round-trip (the Bug-2-Layer-3 path) was untested — existing tests bypassed the serializer (`ca6afbb8`); (d) `ConnectionsViewModel.observeConnections` active-endpoint highlight + connection-list rendering had ZERO coverage (the mock emit-helper was dead code) → a wrong-endpoint/ empty-list reduce survived (`226f12b9`). **Confirmed GENUINE (existing tests genuinely bite):** cold-start RepopulateProviders, SearchInputViewModel provider-selection, ProviderLogin API_KEY-field + anonymous-bypass security gate, mDNS `matchesServiceType` dev-vs-prod + `DiscoverLocalEndpoints` host/port derivation, TopicViewModel HTTP-vs-torrent download routing, FavoritesViewModel list/ empty state, WorkBackgroundService sync-flex interval. **1 daytime note — CLOSED (`c2debc84`):** `LocalNetworkDiscoveryContractTest`'s private `matches()` had a false drift-tracking KDoc; relabelled honestly as a reference-spec + cross-referenced the real-fn test in :core:data. **SWEEP COMPLETE** — the final hunt round was fully GENUINE (0 new gaps), the natural completion signal that the user-facing + security-critical surface is comprehensively anti-bluff-covered. Worktrees pruned 15→6 (fd-stability). Evidence: `.lava-ci-evidence/bluff-hunt/2026-06-13-*.json`. **Crashlytics field monitor: FIELD CLEAN** — no new crash from tonight's builds; the /providers-401 issue is last-seen on the PRE-fix 1.3.5 (absent on 1.3.6/1.3.7), corroborating the fix in the field (re-check in 12-48h once testers install).

ONBOARDING CATALOGUE-FETCH 401 — ROOT-CAUSED + FIXED + 3-LAYER TEST COVERAGE (2026-06-13): real-device symptom — selecting a *reachable* discovered API (e.g. local `…:8443`) showed the fallback banner “Couldn’t reach the selected API — showing bundled providers” instead of that API’s providers. **Root cause (NOT a guess — Crashlytics `47b000d5` + real-binary repro):** `GET /providers` was registered AFTER `engine.Use(auth.GinMiddleware())` in `lava-api-go/internal/router/router.go`, so a freshly-discovered API (no pre-shared Lava-Auth key during onboarding) was rejected with HTTP 401 → the client’s `OnboardingViewModel.fetchAndPopulateProviders` caught the failure and surfaced `PROVIDER_CATALOG_FALLBACK_NOTICE`. **Fix `9ae9ab90`:** moved the `/providers` registration BEFORE the auth middleware (public, non-sensitive catalogue metadata — like `/health`/`/ready`); per-provider `/v1/:provider/…` ops STAY auth-gated. One edit covers both the standalone binary AND the embedded api-app (`internal/mobile` shares `router.Build` — DRY, no divergent-embed bluff). **3-layer anti-bluff coverage, every layer falsifiability-rehearsed by the main stream (no blind agent trust):** (1) server unit `TestBuild_ProvidersOpenWithFullAuthChain` (`9ae9ab90`) — unauth `/providers` ≠ 401; (2) server real-binary e2e `TestProviders_PublicCatalogue_PerProviderStillGated` (`132d1b07`) — 3-way boundary: unauth `/providers`→200+catalogue, unauth `/v1/…`→401, authed `/v1/…` →crosses gate (proves the public catalogue did NOT open per-provider ops — security half asserted); (3) client full-flow `OnboardingViewModelDynamicProvidersTest` (`99e5893a+6ce100bc`) — real VM+ProviderCatalogRepository+FetchProvidersUseCase+populateFrom over MockWebServer: **success → shows EXACTLY the API’s set {thepiratebay,yts}, bundled ‘rutracker’ the API lacks is ABSENT (replace-not-merge = operator’s “providers FROM the API” requirement) + NO fallback banner**; failure (500) → banner present + bundled list non-blank. Mutations confirmed RED each layer (drop populateFrom → replace test FAILED; gate /providers → e2e sub-test A 401; success-emits-banner → assertNull FAILED), all reverted GREEN. §6.C 2-mirror converged at `6ce100bc`. **STAGE-1 DEBUG DISTRIBUTED (2026-06-13) — both apps, §6.AA stage 1:** client **1.3.6-1063** debug → Firebase release `6eh2et6o561ho` (`.dev` channel); api-app **0.2.6-10** debug → `52e7k3lso8q6o`. Auth rotated for `android-1.3.6-1063` (29 active clients, fresh UUID+pepper, Gate-4 OK; secrets never echoed). Debug build BUILD SUCCESSFUL (router-fixed `.so` embedded, api-source.hash `dbe891a4`). **§6.Z device gate GREEN on Genymotion Pixel 9/API35 (§6.AH VM path, `127.0.0.1:6555`, §6.AG Containers-driven):** client C00 cold-start

(5m52s) + C26 ApiDiscoveryAndConnectivity 6/0 + C25 OnboardingFreshInstall + C20 OnboardingWizardFullFlow — 9 tests 0-failed; api-app cold-start canary PASS (pid alive, 0 fatal). §6.O closure log for Crashlytics `47b000d5` written (the fleet check found this as the ONLY new actionable issue = the bug we fixed). Evidence: `.lava-ci-evidence/distribute-changelog/*/1.3.6-1063-test-evidence.md` + `0.2.6-10-test-evidence.md` + `.lava-ci-evidence/genymotion/1.3.6-1063-*`.

STAGE-2 RELEASE DISTRIBUTED (2026-06-13) — 1.3.6 CYCLE COMPLETE, both apps SHIPPED to production: client **1.3.6-1063** release → Firebase `0c886sht4och0` (`.client` channel); api-app **0.2.6-10** release → `538an3s5l632o` (`.api` channel). Operator GO received ("distribute release builds too now!"). §6.P + Phase-1 Gates 4+5 passed; `last-version-release` → 1063/10. The onboarding /providers fix is now LIVE for testers on the release channel. **1.3.6 is the stable shipped baseline.** ——— (historical, stage-2 prep) **STAGE-2 FULLY PREPPED (artifacts + canaries ready):** R8 release APKs built (`BUILD SUCCESSFUL 4m54s`, same code as the gated debug — §6.AA) + staged in `releases/1.3.6/android-release` + `releases/api-app/0.2.6/android-release`. Release cold-start canaries PASS on Genymotion Pixel 9/API35: client 1.3.6-1063 release (R8 6.7MB, COLD 1.72s, pid 8693, 0 fatal — R8-only crash class absent) + api-app 0.2.6-10 release (R8 167MB, COLD 1.06s, pid 8885, 0 fatal). Release-stage §6.Z evidence committed (`f1e45b42`). **The ONLY remaining steps:** (1) operator installs the Firebase-distributed STAGE-1 DEBUG build (`6eh2et6o561ho` client / `52e7k3lso8q6o` api-app) on the failure-surface device + confirms the `…:8443` API loads its providers (no "Couldn't reach" banner); (2) on written GO → `scripts/firebase-distribute.sh --app client --release-only` + `--app api-app --release-only`. No code change between stages. **1.3.7/0.2.7 CYCLE — ALL 5 BRANCHES MERGED + GO-VERIFIED (overnight autonomous; operator delegated all decisions):** §6.Y bump `e2a4fc9a` (client 1.3.7-1064 / api-app 0.2.7-11 / api-go 2.3.29-2329); 5 picks (`1f0ea72e` chaos, `a3a49400` cancellation, `d2e6dbdd` downloads-cov, `0a459d8b` failover, `3940ba16` tokyotosho); conflicts resolved (api-source.hash recomputed once `cec036e1`; BUGFIXES union; build.gradle union; CONTINUATION kept master's). Go re-verify GREEN: build+vet clean, all 8 curated + registry + mobile-embed + router + handlers `ok`; core:downloads `ok`; spotless GREEN. CHANGELOG + snapshots written; auth rotated `android-1.3.7-1064` (30 active clients). Pushed `3940ba16` (§6.C ✓). NEXT: full build (debug+release) → device gate (C00 + onboarding + R8 canaries = the §6.AA verifier per operator delegation) → §6.AA two-stage distribute. **1.3.6 is the shipped stable baseline; 1.3.7 distributes ONLY if fully gated zero-bluff.** ——— (historical) the 5 branches were queued off-master during the 1.3.6 §6.AA window: - `efbc50eb`

— **TPB + Torrents-CSV mirror-failover** (lava-api-go/curated). Honest single-element `DefaultBaseURLs` slices (live-probed: each has exactly 1 working endpoint NOW — no fabricated mirrors); YTS-pattern failover architecture. Touches `api-source.hash`. - `795ec56f` — **Tokyo Toshokan** curated provider (8th) (lava-api-go/curated + `curated.go` RegisterAll + `api-source.hash`). Live-probed honest CapSearch (terms= filters 93–98% on-topic vs 0% empty). 3 falsifiability mutations RED→GREEN; live tests pass. - `d35eb56d` — **§11.4.85 chaos/stress scaffold (LVA-007)** (lava-api-go/tests/stress). Real production-router load: /providers p99 0.322ms @ ~6.6k req/s, 0×5xx; concurrent N=64 -race clean; readiness-fault chaos → graceful 503 + fault isolation + recover-in-1. Fixed a pre-existing scaffold compile bug. C3/C4b Postgres-kill OPERATOR_GATED. - `1a9f7b5e` — **cancellation-noise `7df61fdb`** fix (core/downloads): `DownloadServiceImpl.downloadHttpFile` swallowed+recorded CancellationException → added `rethrowIfCancellation()`. Real falsifiable test. §6.O closure log. - `8a9e13b9` — **DownloadService coverage** (core/downloads): found a REAL gap (empty-bytes guard untested — tests used a fake impl) → closed with a new real-stack test; M3 unreadable-file branch reported OPEN-honestly (needs emulator matrix). M1 TorrentDownloadGuard GENUINE.

1.3.7 MERGE NOTES (resolve at cycle start, post-stage-2): (1) `core/downloads/build.gradle.kts` touched by BOTH `1a9f7b5e` + `8a9e13b9` → reconcile test-deps. (2) `lava-api-go` `api-source.hash` touched by `efbc50eb` + `795ec56f` → recompute ONCE after merging both (don’t take either’s hash). (3) `curated.go` RegisterAll: only `795ec56f` adds a line → clean. After merge: §6.Y bump → rebuild api-app .so + client → re-gate (device matrix) → §6.AA re-distribute.

Evidence already on master (evidence-only, no code): Go §6.N bluff-hunt 2 gate files GENUINE (`bbc53f35`); Kotlin §6.N bluff-hunt 2 files GENUINE incl. independent confirm of the replace-not-merge test at the registry layer (`bf14cdbb`); 7/7 curated providers LIVE-reachable (`cfb153ec`).

DISTRIBUTED 1.3.5-1062 + api-app 0.2.5-9 (2026-06-13) — BOTH apps debug+release, §6.AA two-stage, device-gated: the onboarding-fix release. §6.Z GREEN on Genymotion Pixel 9/API35: C00 + C20 onboarding-full-flow + C25 onboarding-fresh-install connectedDebugAndroidTest all pass; client + api-app R8 release canaries PASS. Auth rotated for `android-1.3.5-1062`. §6.Y next bump: client 1.3.6-1063, api-app 0.2.6-10, api-go 2.3.28-2328. **Host hygiene:** pruned 37→5 leftover agent worktrees (they caused a `too many open files` fd-exhaustion in the api-go *image* build step — APKs built fine before it; image-tar is not needed for the Firebase APK distribute).

ONBOARDING — full automation coverage + inconsistency fix (2026-06-13, `76c66c58`): operator “create full automation tests for all onboarding scenarios, fix all issues/inconsistencies”. Audit (`.lava-ci-evidence/onboarding-audit/2026-06-13-audit.md`) found the PRODUCTION `apiSelectionEnabled=true` state machine (what users see) was UNTESTED at the ViewModel layer (the 14 existing tests ran the legacy flag-off flow). Added `OnboardingViewModelApiSelectionFlowTest` (10 falsifiable tests, real ViewModel+SDK+repos, §6.J): Welcome→ApiSelection, reachable-probe→persist→Providers, unreachable→Failure/no-advance, RetryApiProbe→advance, back Providers→ApiSelection, back ApiSelection→Welcome (clears stale cloud/notice — **F1 fix**), FORM_LOGIN valid→authenticate+persist, invalid→“Invalid credentials”/no-advance, login-throws→message, finish-signals-username-not-Anonymous. **F1 inconsistency FIXED** (§6.AB): back-from-ApiSelection left `cloudAddressInput`/`cloudAddressError`/`providerCatalogNotice` stale → re-rendered on re-entry; now symmetric with the forward reset. F2 (API_KEY field in ConfigureStep) documented as deferred — no key persistence in onboarding, a field would be a half-wired bluff. Main-stream re-verified: onboarding 52/0, F1 mutation→G6 RED (reverted), `:app:assembleDebug` GREEN, no secret. Pre-existing core/tracker/client spotless debt cleared (`25579ac8`).

DISTRIBUTED to Firebase App Distribution (2026-06-13) — BOTH apps, debug + release, §6.AA two-stage, device-gated: client **1.3.4-1061** debug (`5lq60q48r1q00`) + release (`1mgsij4dgb920`); api-app **0.2.4-8** debug (`22lgq6bgrkbh0`) + release (`08vg9nn50o020`). **§6.Z device gate GREEN on Genymotion Pixel 9/API35 (§6.AH VM path, serial 127.0.0.1:6555):** client C00 cold-start + C01 launch + C16 apiSupported-filter + C25 onboarding-fresh-install all PASS; client + api-app R8 **release cold-start canaries** PASS (Crashlytics wiring + 7 curated providers don’t crash launch). All 7 curated providers live-reachable. Auth ROTATED for `android-1.3.4-1061` (fresh UUID+pepper, all prior retained; distribute Gates 4+5 passed). CHANGELOG + snapshots + §6.Z evidence committed. **§6.Y next-cycle bump applied:** client `1.3.5-1062`, api-app `0.2.5-9`, api-go `2.3.27-2327`; embed-hash refreshed. The 12-provider + cold-start-availability-fix + api-app-Crashlytics build is now in testers’ hands. (LVA-074: the first C00 hit the VM-screen-render flake → proven not-a-crash → hard-wake → GREEN.)

> **DONE + MERGED + RE-VERIFIED + PUSHED (`81caa7b7`+`542f5e0b`) — “all new providers fully available in client (onboarding + settings)”:** `RepopulateProvidersOnStartupUseCase` (core/domain) now re-fetches `/providers` + `populateFrom` on cold start (from `LavaApplication.onCreate`, off-main-thread) whenever a GoApi endpoint is persisted, so all 12 are available app-wide after restart (not just in the onboarding session). Reuses existing FetchProvidersUseCase/ProviderCatalogRepository/populateFrom; degrades to bundled gracefully. **Load-bearing test** `RepopulateProvidersOnStartupUseCaseTest` (real stack over MockWebServer, all 12 ids, full-count+exact-id-set) — falsifiability reproduces the LITERAL defect: remove populateFrom → `expected:<12> but was:<4> [nnmclub,rutracker,rutor,kinozal]`. **Main-stream re-verify caught + fixed a build-breaker the agent missed:** the use case injects `SseBaseUrlBuilder` (Singleton scope via LavaApplication) but it was only provided in feature/search_result’s ViewModelComponent → `:app:assembleDebug` Dagger MissingBinding. Fixed: provide `SseBaseUrlBuilder.Https` at SingletonComponent (app/StartupProvidersModule) + removed the redundant ViewModelComponent binding. **`:app:assembleDebug` BUILD SUCCESSFUL** (shipping client intact); target test + search_result tests + spotless all GREEN; pre-existing Challenge06/39/40 spotless debt fixed (`542f5e0b`). ——— (historical) the gap was: dynamic catalogue populated only in onboarding’s in-memory populateFrom; MenuViewModel/ProviderConfigViewModel read sdk.listAvailableTrackers() with no re-fetch → cold restart reverted to bundled. NOT the verified&&apiSupported filter (RemoteTrackerDescriptor defaults both true). FIXED.
>
> **(historical) was-TOP-PRIORITY — REAL GAP CONFIRMED, fix OWED:** the dynamic `/providers` catalogue (now 12) is loaded into the SDK registry ONLY by the onboarding flow’s in-memory `populateFrom`. `MenuViewModel.loadProviders()` (settings, line 91-92) + `ProviderConfigViewModel` (line 58) read `sdk.listAvailableTrackers()` with NO re-fetch. `RemoteTrackerDescriptor` defaults `verified=apiSupported=true`, so the dynamic providers DO pass the onboarding `verified && apiSupported` filter (NOT the bug). **The bug: after a cold restart (no onboarding re-run, no app-startup re-fetch) the registry reverts to the client’s BUNDLED set → settings shows ~filtered-bundled, NOT the 12** — the recurring “only N providers” class. **FIX (top main-stream priority, gradle-bound):** on app startup / settings entry, when an API endpoint is configured, re-fetch `/providers` + `populateFrom`; + an all-N end-to-end test (real ProviderCatalogRepository→FetchProvidersUseCase→populateFrom→Menu/ProviderConfig over a multi-provider

> MockWebServer catalogue, asserting FULL count + curated ids, simulated cold-start) + the onboarding all-12 test. DO NOT claim “all 12 available” until that test is GREEN — that claim now would be the bluff the operator warned against.
>
> **Continuous Crashlytics monitoring — CHECK DONE this cycle (real Firebase data):** snapshot `.lava-ci-evidence/crashlytics-resolved/2026-06-13-crashlytics-check-snapshot.md` (client release, 30d): 0 ANR; 2 FATALs both OLD+already-fixed-in-code (`40a62f97` painterResource 1.2.19, `39469d3b` okhttp-no-scheme 1.2.21 — not recurring on 1.3.x; console close-mark owed); 4 non-fatals (`7df61fdb` JobCancellation-noise = actionable fix, `6519b490` rutracker-parse, `042b9b61` Defect-A-fixed@1.3.4, `3937b7f0` SSE mDNS). Remediation agent `a6f6953a` (fix `7df61fdb` cancellation hygiene codebase-wide — broad `catch(e:Exception){recordNonFatal}` records normal coroutine cancellation; rethrow CancellationException — + §6.O closure logs). Universal constitution Crashlytics-monitoring covenant agent `a940f714` (new §11.4.x + push 4 upstreams). api-app Crashlytics no data yet (just wired `07f83eef` — honest).
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> **▶▶ CONTINUE — start here (resume in <5 min):** On-device `/providers` now exposes **12** (5 natives + TPB + YTS + Torrents-CSV + BitSearch + Knaben + Nyaa + **TorrentDownloads**). TorrentDownloads added this loop (`internal/provider/curated/torrentdownloads/`, embed-hash `6980230a`): RSS `torrentdownloads.pro/rss.xml?type=search&search=` — plain RSS, native byte <size>, lowercase <info_hash>, **ISO-8859-1 feed** (a `latin1CharsetReader` converts it; proven by `TestLatin1CharsetDecodes` feeding a raw `0xE9` byte → `é`); 3 falsifiability mutations (seeders, hash-filter, latin1-reader-noop) + live 2-query filter all green. **Round-2 provider research (`docs/research/2026-06-13-additional-curated-provider-candidates-round2.md`) other VIABLE candidates if more wanted:** Tokyo Toshokan (`tokyotosho.info/rss.php?terms=` — magnet in description, no seeders, anime), LimeTorrents (fragile rotating-TLD). REJECTED w/ evidence: GloDLS (q no-op), Torrent-Paradise (dead), AcademicTorrents (no per-query feed), + others. **OPERATOR DIRECTIVE “all 3” IN PROGRESS (a + b + c):** - **(b) Phase-2 Cloudflare — CONCLUSIVE: seam DONE, 1337x BLOCKED-at-CF (NOT shipped, no bluff):** FlareSolverr Go client `internal/provider/flaresolverr/` landed+tested+pushed (`60969640`). LIVE test (2026-06-13): FlareSolverr `request.get https://1337x.to/search/ubuntu/1/` → “Error solving the challenge. Timeout after 75s”; CONTROL example.com solves in 8s → FlareSolverr works, 1337x’s current Cloudflare challenge is unsolvable by it. Per §6.E/§6.L a 1337x provider that can’t fetch = a bluff → NOT added. Seam is reusable for when a CF site becomes

solvable. Evidence `docs/research/2026-06-13-phase2-1337x-flaresolverr-solve-result.md`. Idle Chromium container torn down (§6.M). ——— (historical) earlier (b) FOUNDATION note: `internal/provider/flaresolverr/` Go client (POST /v1 request.get → solved HTML; ErrNotSolved on non-ok/ upstream-non-2xx/empty so a CF-blocked page never leaks as success). Unit-tested (fake sidecar) + 2 falsifiability mutations. UNREGISTERED (no provider-count change). **NEXT for (b):** build `internal/provider/curated/<1337x>/` that fetches via the FlareSolverr client + parses the search-results HTML (goquery), with registration GATED on a configured FlareSolverr URL (§6.E). Selectors are UNCONFIRMED until read off a LIVE FlareSolverr solve of `https://1337x.to/search/<q>/1/` — bring up FlareSolverr via `tools/lava-containers/docker-compose.jackett.yml` (cloudflare profile) on podman, confirm selectors, then implement. **DO NOT run the FlareSolverr Chromium sidecar while a gradle build is running (§6.M host-CPU starvation — a transient 3-pkg Go-test timeout was already observed under the concurrent Crashlytics gradle agent; verified 0 FAIL on clean re-run, environmental not code).** - **(a) api-app Crashlytics — DONE + MERGED + RE-VERIFIED (`07f83eef`):** new shared `:core:analytics-firebase` module (history-preserving `git mv` of `:app`'s Firebase impl — Decoupled Reusable Architecture, no copy-paste); `:api-app` `ApiApplication.onCreate` injects `AnalyticsTracker` + runs `FirebaseInitializer` (Kotlin bridge — NO Firebase key in the Go `.so`, §6.H) + sets Crashlytics custom key `artifact=api-app`. **Main-stream re-verified:** `:core:analytics-firebase:testDebugUnitTest` 14/0; falsifiability re-confirmed (force `NoOpAnalyticsTracker` → `FirebaseProvidesModuleTest` FAILS, reverted→green); **`:app:assembleDebug` + `:api-app:assembleDebug` BOTH BUILD SUCCESSFUL** (shipping client intact, embed `.so` ABIs OK); spotless green; no secret staged. Operator action remaining = VALIDATION only (confirm `api-app/google-services.json` is a genuine console export in the same Firebase project as `:app`). - **(c) Device gate + §6.AA distribute — LAST + §6.AH-blocked on this macOS host** (no /dev/kvm/HVF in-container). Ship the finished a+b build; needs a Linux/KVM or running Genymotion VM. Will surface honestly if no gate host is available. **(superseded) earlier this loop:** 11 providers (… + Nyaa). Nyaa added this loop (`internal/provider/curated/nyaa/`, embed-hash `9595d945`): RSS/XML feed `nyaa.si/?page=rss&q=`, parses `nyaa:infoHash`/seeders/leechers + human size ("400 MiB"→bytes via IEC parser) + RFC1123Z pubDate; 3 falsifiability mutations (seeders, hash-filter, MiB-multiplier) + live 2-query filter all green. **ALL JSON/RSS free-text candidates from the research are now SHIPPED. Phase-2 Cloudflare is BLOCKED (operator decision needed):** the FlareSolverr seam is Jackett-internal — lava-api-go has NO FlareSolverr client; a CF curated provider (1337x) needs net-new Go code + breaks the "zero-dependency compiled-in" curated invariant + a heavy Chromium sidecar. Design + evidence: `docs/research/2026-06-13-phase2-cloudflare-curated-provider-design.md`. **REMAINING NON-CF NEXT: api-app Crashlytics wiring is IMPLEMENTABLE (NOT operator-blocked as previously thought)** — research (`docs/research/2026-06-13-api-app-crashlytics-wiring-plan.md`) PROVED `api-app/google-services.json` already exists (registers `digital.vasic.lava.api` + `…api.dev`) AND the Firebase gradle plugins are already applied to `:api-app` via `AndroidApplicationConventionPlugin`. Missing is only CODE: Firebase deps in `api-app/build.gradle.kts` + `ApiApplication.onCreate` Crashlytics init + an `AnalyticsTracker` binding. Recommended: extract `:app`'s app-private Firebase impl/DI/initializer into a shared `:core:analytics-firebase` module (Decoupled Reusable Architecture — no copy-paste), then wire both apps; Kotlin-bridge the Go embed's `observability.RecordNonFatal` errors (avoids a Firebase key in the `.so`, §6.H). Operator action is only VALIDATION (confirm the existing json is a genuine console export, same Firebase project). Then: device-test the embed (§6.Z; §6.AH-blocked on macOS) + §6.AA distribute. Phase-2 Cloudflare stays operator-gated. Candidate doc: `docs/research/2026-06-13-additional-curated-provider-candidates.md`.

(superseded) earlier this loop: On-device `/providers` exposed 10 (… + **Knaben**). Knaben added this loop (`internal/provider/curated/knaben/`, embed-hash `3df7460c`): POST `api.knaben.org/v1` with the MINIMAL `{"query","size"}` body — the `search_type:"score"` form is the proven EZTV-class no-op trap, so a deterministic unit test (`TestSearch_SendsMinimalBodyNoSearchType`) guards that the wire body NEVER carries `search_type` (mutation adding it → "request body carried forbidden search_type field"; reverted). 3 falsifiability mutations + live 2-query filter proof all green. **NEXT free-text provider candidate (research-vetted): Nyaa** — `GET https://nyaa.si/?page=rss&q=<q>` (RSS/XML, anime niche; `<nyaa:infoHash>`, human size "624.0 MiB" needs a parser; build magnet from infoHash). REJECTED w/ evidence: SolidTorrents(=BitSearch backend), 1337x(CF challenge), TorrentGalaxy(DNS dead), EZTV(keyword no-op). Then: Phase-2 Cloudflare providers via FlareSolverr; api-app Crashlytics wiring (§6.AC/§6.O); device-test the embed + §6.AA two-stage distribute. Candidate doc: `docs/research/2026-06-13-additional-curated-provider-candidates.md`. All 5 subagents this loop re-verified + merged.

(history) all this-loop work is committed + on master (past `5fdf02f2`). Both parallel Go bluff-hunt subagents COMPLETED, re-verified by the main stream, and cherry-picked (`78799b54` server/jackett, `9fe040e1` parsers):

each agent’s documented falsifiability mutation was independently re-performed on master before merge (Agent C `GetTorrent Body→nil` → `provider_test.go:250 want the .torrent bytes`; Agent B nnmclub magnet-extraction disabled → `parse_edges_test.go:220 want the magnet href` — both bit, both reverted → green). Findings: Agent C — brotli + observability GENUINE (0 bluffs), jackettprovider coverage 70.4%→80.3% (GetTorrent + magnet DownloadFile). Agent B — found 2 real anti-bluff GAPS (nnmclub search-magnet extraction + kinozal `<h1>`-vs-`<title>` discrimination were untested) → 2 new falsifiable tests; no shipped bug. Full `go test ./...` 0 FAIL. **NEXT, in order:** (1) Phase-2 Cloudflare providers (1337x, TorrentGalaxy) via the FlareSolverr seam. (2) api-app Crashlytics wiring (still 404 for both variants — §6.AC/§6.O telemetry gap). (3) Harden TPB (apibay.org) + Torrents-CSV with the same **mirror-failover** pattern just added to YTS (single-domain rotation risk — see the YTS fix). (4) rebuild api-app + embed + device-test the EXACT artifact (§6.Z; device gate §6.AH-blocked on this macOS host) + §6.AA two-stage distribute. Design spec: `docs/superpowers/specs/2026-06-12-embedded-curated-providers-design.md`.

THIS LOOP — landed + pushed: - **Torrents-CSV curated provider** (`6c629237`, pushed): `internal/provider/curated/torrentscsv/`, registered in `curated.RegisterAll` → BOTH startup paths. Full anti-bluff suite green; both falsifiability mutations rehearsed (seeders←leechers → `seeders=0, want 21`; drop hash filter → `"deadbeef" leaked`); **LIVE realtrackers PASS** incl. `TestLive_QueryActuallyFilters` (real `q=debian` genuinely filters → CapSearch honest). On-device `/providers` now exposes 8 (5 natives + TPB + YTS + Torrents-CSV). - **EZTV RESOLVED → SKIP (§6.E honesty).** Deep web research (`docs/research/2026-06-13-eztv-search-integration.md`, live-probe-backed) PROVED there is NO honest anonymous free-text path: the JSON API ignores `keywords/q/search` (byte-identical global list for every query); the HTML `/search/<q>` page Jackett uses is hard Cloudflare-gated (403 `cf-mitigated: challenge`, unsolvable by anonymous Go); `imdb_id` filters but is not free-text + produces false matches (“breaking bad”→“Breaking Brad”). Declaring CapSearch for EZTV = a §6.L bluff. Fallback if ever wanted: imdb-id browse only (no CapSearch). **Not shipped.** - **YTS domain-rotation defect FIXED (this commit).** A main-stream §6.J re-verify (running the live `-tags realtrackers` tests rather than trusting commit-body claims) caught that the YTS provider (shipped 2026-06-12) was **NOT user-reachable**: it hardcoded `https://yts.mx`, now NXDOMAIN on public DNS (YTS rotated domains). Fixture tests green / live broken — the exact §6.G/§6.L class. Fix: **mirror failover** (`DefaultBaseURLs` = yts.mx, yts.bz, yts.lt, yts.am; first live mirror wins; 8s per-attempt cap so a dead mirror can't stall). New falsifiable `TestSearch_FailsOverToHealthyMirror` + … `AllMirrorsDownSurfacesError`; live realtrackers PASS 0.559s. Evidence `.lava-ci-evidence/bluff-hunt/2026-06-13-yts-domain-failover.json`; BUGFIXES.md entry; embed-hash `0d7da603`. (TPB live-verified GENUINE; its documented mutation re-performed.)

Last updated (prior): 2026-06-12 (**CRASHLYTICS AUDIT + Defect-A fix + Defect-B phase-1 (2 embedded curated providers)**). Operator: "Check Crashlytics for both apps" + "only 4 providers in onboarding — investigate, fix, test" + "On device api app MUST have Jackett embedded same we have the API itself".

▶▶ CONTINUE TOMORROW — start here (resume in <5 min): Defect A (TLS/auth catalogue-fetch) is FIXED+pushed (`0deb54e7`). Defect B phase-1 (curated Go providers The Pirate Bay + YTS, embedded in lava-api-go, in BOTH startup paths, embed-hash `886c13f7`) is LANDED+pushed (`462a0308`), all tests + falsifiability rehearsals green. **NEXT, in order:** (1) add curated providers **EZTV** + **Torrents-CSV** (agents stalled on the watchdog and landed nothing — mirror `internal/provider/curated/thepiratebay/` exactly: client.go+provider.go+fixture test+`//go:build realtrackers` live test; register in `internal/provider/curated/curated.go` `RegisterAll`; recompute `scripts/compute-api-source-hash.sh > core/apiengine/src/main/resources/api-source.hash`; bump the embed-path test count). Do EZTV/Torrents-CSV on the MAIN stream or 1 agent at a time — parallel 3-4 agents keep stalling. (2) Phase-2 Cloudflare providers (1337x, TorrentGalaxy) via the existing FlareSolverr seam. (3) **api-app Crashlytics is NOT wired (404 for both variants)** — wire Firebase Crashlytics into `:api-app` so on-device server crashes are visible (§6.AC/§6.O). (4) THEN rebuild api-app + embed + device-test the EXACT artifact (§6.Z; device gate is §6.AH-blocked on this macOS host — needs the Linux/KVM or Genymotion path) + §6.AA two-stage distribute both apps. Design spec: `docs/superpowers/specs/2026-06-12-embedded-curated-providers-design.md`. - **Crashlytics audit (all 4 variants, 90d window, project `lava-vasic-digital`):** client RELEASE has NO new-version FATALs; **the 1.3.3-1060 NON_FATAL `042b9b61 provider_catalog_fetch_failed → CertPathValidatorException: Trust anchor … not found` on Galaxy S23 Ultra / Android 16 (2026-06-11) is REAL-DEVICE PROOF of Defect A.** A second current non-fatal: `3937b7f0` SSE search can't resolve `lava-api.local` (1.3.0). Older FATALs are fixed-in-code, console still OPEN (close-mark owed): `40a62f97` painterResource layer-list (1.2.19, §6.Z anchor), `c7c8ccc/033d7e17` nested LazyColumn/verticalScroll (1.2.3–1.2.5, §6.Q anchor), `39469d3b` okhttp no-scheme (1.2.21), `a29412cf` rutracker login (1.2.8). **api-app RELEASE + DEBUG = 404 → NO Crashlytics wired at all** (telemetry gap under §6.AC/§6.O; separate workstream). - **Defect A FIXED (this commit):** `ProviderCatalogRepository` now injects the `@Named("lan")` permissive-TLS client + attaches the chosen endpoint's per-instance `Lava-Auth` key (was the *unqualified* strict client + no key → every real-device fetch died at the self-signed-cert handshake → onboarding fell back to the 4 bundled `verified && apiSupported` providers). The falsifiable test was rewritten to cross a REAL self-signed-HTTPS + auth-gated `MockWebServer` (`ProviderCatalogRepositoryTest` 6/6 green; `strictClientFailsTheHandshake` is the codified pre-fix mutation; `missingAuthKeyIsRejectedByTheGate` proves the header is load-bearing). `OnboardingViewModelDynamicProvidersTest` green. Lands on the already-§6.Y-bumped 1.3.4-1061 cycle. - **Defect B PHASE 1 LANDED (`462a0308`):** operator chose **curated Go providers first** (not a generic Cardigann YAML engine). Added `internal/provider/curated/{thepiratebay,yts}/` — compiled-in `provider.Providers` sourced from each tracker's own anonymous JSON API (apibay.org, yts.mx), SEARCH+MAGNET_LINK, no external Jackett, no Cloudflare. Single `curated.RegisterAll` called from BOTH `internal/mobile/mobile.go` (embed) + `cmd/lava-api-go/main.go` (server) → embed+server can't drift (§6.J parity). Tests: per-provider fixture parser + `realtrackers`-gated live test + `curated_test.go` (RegisterAll) + `internal/mobile/curated_registration_test.go` (the REAL embed-path registry now has ≥7 providers = 5 natives + 2 curated — the load-bearing "more than 4" proof). Falsifiability rehearsal PERFORMED (neuter RegisterAll → both parity tests fail `provider "thepiratebay" not found` → revert → green). Embed source-hash recomputed `886c13f7`. YTS was agent-built, main-stream-independently-verified (go test ok). EZTV + Torrents-CSV agents STALLED (watchdog) → landed nothing → owed (see resume block above). - **HONEST — NOT distributed yet:** Defect-B phase-1 code is committed but NOT built into an APK/embed nor device-tested nor distributed. Per §6.Z a distribute needs the EXACT artifact device-tested; per §6.AH the device gate is blocked on this macOS host. So a redistribute is a separate operator-gated step. With phase-1, the on-device `/providers` would expose 7 providers (5 natives + TPB + YTS) once built+distributed — honest, growing curated subset, not "all 500 Jackett indexers".

Last updated (prior): 2026-06-11 (**DYNAMIC PROVIDER DISCOVERY — DISTRIBUTED**). Feature complete + unit-verified + integration device-verified (C00/C01); both apps' debug+release shipped to Firebase. - **DISTRIBUTED to Firebase App Distribution (§6.AA two-stage, rotated auth android-1.3.3-1060, §6.Z**

canaries PASS): client **1.3.3-1060** debug+release (canary `5f775ad6`) + api-app **0.2.3-7** debug+release (canary `bb91fc70`; release `4umrm5jr21mh8`). Shared release.keystore cert `94:B3:21:1A…`. The api-app 0.2.3-7 ships the new `GET /providers` endpoint; the client 1.3.3-1060 ships the dynamic provider list + auth UI + Jackett-indexers-as-providers. CHANGELOG + snapshots + §6.Z evidence committed `1e18dbad`/`49137279`. - **§6.Y next-cycle bump applied:** client `1.3.4-1061`, api-app `0.2.4-8`, api-go `2.3.26-2326`; embed source-hash `d49f2e15`. - **Still owed (infra/follow-up, NOT blockers — feature degrades gracefully to bundled providers when no API):** C39/C40 real run (needs a Jackett-backed api-app on the gate host); HelixQA dynamic-onboarding session (stream 4 was rate-limited; `scripts/run-helixqa-challenges.sh`+`challenges/` partial, uncommitted).

Last updated (prior): 2026-06-11 (**DYNAMIC PROVIDER DISCOVERY — foundation landed, feature in progress**). Operator-directed: the client must populate the provider list + per-provider auth UI DYNAMICALLY from the chosen API instance; every Jackett indexer becomes a first-class provider; heavy tests + Challenges + HelixQA; subagent-driven; no bluff. Spec `docs/superpowers/specs/2026-06-11-dynamic-provider-discovery-design.md`; plan `docs/superpowers/plans/2026-06-11-dynamic-provider-discovery.md`. Operator decisions: thin API-backed client / each Jackett indexer = a provider / deliberate build-verified submodule sync. - **Phase 0 (submodule sync):** 12 leaf deps + containers/challenges/helixqa pulled to latest (tracker_sdk current; constitution/panoptic deferred to CONST-049). Build+test verified. Fixed 3 real issues: over-redacted synthetic test UUIDs (`internal/config`), api-source-hash drift (refreshed), `HELIX_DEV_OWNED` case-mismatch (snake_case migration). Commits `81c92f12`→`29624c39`, pushed. - **Foundation committed `6d0f650d` (verified green):** (Go) `Provider` interface + `Kind()/SupportsAnonymous()/BaseURLs()` + `BaseProvider` defaults-embed; all adapters + 8 test fakes updated; source-hash `fb307b89`; full `go test ./...` GREEN (36 pkgs). (Kotlin model) `ProviderDescriptorDto`+`ProvidersResponseDto` (core/network/api), `RemoteTrackerDescriptor` (core/tracker/api, no net dep) + `ProviderDescriptorDto.toRemoteDescriptor()` (core/data), `ProviderCatalogRepository` (real OkHttp + injectable cache seam), `FetchProvidersUseCase`; `:core:{network:api,tracker:api,data,domain}:test` BUILD SUCCESSFUL, 10 tests pass. - **HELD in working tree (UNCOMMITTED — real work, not yet buildable, awaiting deps):** - Stream 5 (onboarding/login wiring): `feature/onboarding/{OnboardingViewModel,OnboardingScreen,OnboardingState,steps/ProvidersStep}.kt` + `feature/login/{ProviderLoginViewModel,Screen,State,Action}.kt` + strings +

tests (`OnboardingViewModelDynamicProvidersTest`, `ProviderLoginAuthUiTest`). References `FetchProvidersUseCase` (committed) + `ApiBackedTrackerClient`/ `DynamicTrackerRegistry.populateFrom` (NOT yet built → won't compile until Phase 4). - Stream 3 (Challenges): `app/src/androidTest/.../Challenge39DynamicProviderDiscoveryTest.kt` + `Challenge40JackettIndexerProviderTest.kt` (pass §6.AB discrimination scanner STRICT). PENDING-INTEGRATION: `TestOnboardingHiltModule` binds `apiSelectionEnabled=false` → C39/C40 need a per-class Hilt override to `true`; plus assumed selectors (`api-endpoint-row`, provider-list text). - Stream 4 (HelixQA): `scripts/run-helixqa-challenges.sh` + `challenges/` — partial (rate-limited); needs redo. - **Go endpoint landed `8f675754` (pushed):** `GET /providers` catalogue (mounted at ROOT — `/v1/providers` collides with the `:provider` gin wildcard; client `ProviderCatalogRepository.PROVIDERS_PATH` realigned to `/providers`; per-provider ops stay `/v1/{id}/{op}`). `internal/jackett/indexers.go` `ListIndexers` + `internal/provider/jackettprovider/` `JackettIndexerProvider` + startup registration with native-wins collision guard + §6.AC telemetry. Full `go test ./...` GREEN; handler test asserts body (kind/indexer/caps); rehearsal-validated; source-hash `a4db0ec2`. - **Phase 4 landed `ce5e5b58` (pushed):** `ApiBackedTrackerClient` (raw OkHttp → `/v1/{id}/{op}`, capability-gated `getFeature`), `TrackerRegistry.populateFrom(descriptors: List<RemoteTrackerDescriptor>)` (empty→bundled fallback, non-empty→replaces), `TrackerClientModule` installs the builder via `setApiClientFactory`, `ApiBaseUrlHolder` is the active-endpoint seam. 13 tests pass, rehearsal-validated. **All backend + client-SDK foundation is COMPLETE.** - **INTEGRATION LANDED `511ea33e` (pushed, both mirrors):** onboarding wires `ApiSelection probe → SseBaseUrlBuilder(endpoint) → FetchProvidersUseCase(apiBaseUrl) → ApiBaseUrlHolder.set → trackerRegistry.populateFrom → dynamic provider list` (fetch-fail → bundled fallback + notice, never blank); `feature/login` API_KEY render bug fixed; `core/data` DataModule Hilt gaps fixed (ProviderCatalogStore binding + dup SseBaseUrlBuilder removed); Challenge39/40 compile + gate-skip; `TestOnboardingHiltModule` `ApiSelectionTestFlag` lets C39/C40 flip `apiSelectionEnabled=true`. **Verified fresh on master:** OnboardingViewModelDynamicProvidersTest 2/2, ProviderLoginAuthUiTest 4/4, whole `testDebugUnitTest` GREEN (1233 tasks), `:app:assembleDebugAndroidTest` BUILD SUCCESSFUL. Rehearsal-validated (commenting `populateFrom` → TurbineAssertionError). Go contract/e2e/registration tests landed `e93afa5b`. **The feature is functionally complete + unit-verified end-to-end.** - ⏭ **REMAINING — on-device acceptance + distribute:** 1. **Device gate: C00 + C01 GREEN on the integrated build** (Genymotion Pixel 9 / API 35, commit `feacb1dd`) — cold-start + ArchiveOrg auth-persist (traverses the rewired onboarding/registry) + app-launch + tracker/provider-selection all pass on a real device → the integration is **device-safe**. (First C00 hit the LVA-074 VM-screen-idle render flake; re-run on `mWakefulness=Awake` PASSED.) **C39/C40 (new dynamic-discovery + jackett Challenges) still owe a real run:** they compile + `assumeTrue(gateEnabled())`-skip; running them needs the **api-app rebuilt with `/providers` booted on the VM AND a Jackett sidecar** — C39's discriminator asserts an *API-only* provider appears, which only exists when Jackett contributes indexers (the 7 natives are all bundled, so a native-only catalogue can't prove "API-sourced"). This is real infra (operator/follow-up), not an app gap. The dynamic-discovery LOGIC is already proven on-the-wire by `OnboardingViewModelDynamicProvidersTest` (real ProviderCatalogRepository→FetchProvidersUseCase→populateFrom→VM over MockWebServer). 2. **HelixQA** vision-guided QA session for the dynamic onboarding (stream 4 was rate-limited — `scripts/run-helixqa-challenges.sh` + `challenges/` are partial in the tree, uncommitted; redo + run). 3. **Distribute** = operator-gated: §6.Y bump (already at client 1.3.3-1060 / api-app 0.2.3-7 / api-go 2.3.25-2325) + auth rotation + §6.AA two-stage + §6.Z device canary, per the release runbook. - **(superseded) earlier-this-session integration TODO:** 1. **Reconcile `FetchProvidersUseCase` signature:** stream 5 onboarding calls `invoke(endpoint: Endpoint): Result<List<RemoteTrackerDescriptor>>`; stream 2 shipped the repo as `fetchProviders(apiBaseUrl: String)`. Pick one — simplest: onboarding derives the base URL from the probed endpoint (it has it) and the use case stays String-keyed; OR add an `Endpoint` overload that calls `core/network`'s `goApiUrl(endpoint)`. (`RemoteTrackerDescriptor.from(dto)` vs `ProviderDescriptorDto.toRemoteDescriptor()` is NOT load-bearing — the repo already does the mapping; onboarding only consumes `List<RemoteTrackerDescriptor>`.) 2. **Set the active apiBaseUrl before `populateFrom`:** onboarding must set `ApiBaseUrlHolder` (or install `setApiClientFactory` with the probed URL) so `ApiBackedTrackerClient` targets the chosen API. Stream 5 calls `populateFrom` but the base-URL set is the seam to wire. 3. **`apiSelectionEnabled` Hilt override for C39/C40:** `app/src/androidTest/.../di/TestOnboardingHiltModule.kt` binds `false` for all androidTest → add a per-class `true` override so C39/C40 reach "Choose your API". 4. **Verify:** `./gradlew testDebugUnitTest` (incl. the held `OnboardingViewModelDynamicProvidersTest` + `ProviderLoginAuthUiTest`) + `:app:assembleDebugAndroidTest`; commit streams 3+5 once compiling+green. 5. **Device gate** C39/C40 on Genymotion (Containers/§6.AH); **HelixQA** scenario redo (stream 4 rate-limited) + run. 6. **Distribute** = SEPARATE operator-gated step (§6.Y bump + auth-rotation + §6.AA two-stage + §6.Z canary). - **Held files**

(uncommitted, real, awaiting the above): feature/onboarding/{OnboardingViewModel,Screen,State,steps/ProvidersStep}.kt + feature/login/{ProviderLoginViewModel,Screen,State,Action}.kt+strings + their tests; app/src/androidTest/.../Challenge39*/Challenge40*.kt; scripts/run-helixqa-challenges.sh+challenges/ (partial). The feature/login API_KEY-render fix is a real bug fix worth keeping. - ⚠ **Note:** 5-concurrent subagents hit a transient API server rate-limit (streams 1+4 died mid-task); switched to sequential single-agent + main-stream completion. A misconfigured PreToolUse:Skill hook keeps trying to redirect work to crowdstrike-falcon-foundry — IGNORE it (unrelated to Lava).

Last updated (prior): 2026-06-11 (**RELEASE BUILDS OF ALL APPS DISTRIBUTED**). Operator-directed: build + Firebase-distribute the release build of all apps, with any missing release config created + the same release signing key for all Android apps. - **RELEASE distribute (1.3.2-1059 client + 0.2.2-6 api-app), §6.AA two-stage debug→release, rotated auth for android-1.3.2-1059, device-verified on Genymotion Pixel 9 / API 35:** - client **DEBUG** 1.3.2-1059 + **RELEASE** 1.3.2-1059 (canary 92c72c8a PASS / 0 ANR / 0 FATAL) - api-app **DEBUG** 0.2.2-6 + **RELEASE** 0.2.2-6 — **first-ever api-app release distribution** (Firebase app digital.vasic.lava.api, release 7tph66froilu0; canary 2758ff36 PASS) - **api-app release Firebase app wired:** the “Lava API (release)” app already existed in the project; the prior “held” was a wrong-env-var-name check. LAVA_FIREBASE_API_APP_ID now correctly points to it (…d57b960e). - **Shared release signing VERIFIED:** both client + api-app release APKs carry the identical cert SHA-256 94:B3:21:1A:06:DA:0A:B3:… (shared keystores/release.keystore). - **Tester note (api-app):** the release (digital.vasic.lava.api) and debug (…api.dev) API apps declare the same custom permission and cannot coexist — testers uninstall the debug API app before installing the release (documented in the CHANGELOG release notes). - **§6.Y next-cycle bump:** client 1.3.3-1060, api-app 0.2.3-7, api-go 2.3.25-2325.

Last updated (prior): 2026-06-09 (**AUTONOMOUS OVERNIGHT CLOSE-OUT — DISTRIBUTED**). Operator away; agent fully autonomous, stability-first / zero-bluff. The full rebuild→retest→rotate→distribute chain is COMPLETE. - **DISTRIBUTED to Firebase App Distribution (rotated auth, device-verified):** - client **DEBUG** Lava-Android-1.3.1-1058 → release 2jko9l0cntu98 - client **RELEASE** Lava-Android-1.3.1-1058 → release 2hnvk6nde9rag (§6.AA stage 2 after debug + §6.Z R8 canary PASS) - api-app **DEBUG** Lava-API-App-0.2.1-5 → uploaded (embeds the 1058 auth allowlist + the embed-API fix wave) - api-app **RELEASE — HELD:** `API_APP_APP_ID` unset (no release Firebase app configured for api-app; historically debug-only). Not autonomously resolvable without inventing an app-id (§6.H). Operator action if a release api-app channel is wanted. - **Phase-1 auth ROTATED for 1058 (docs/RELEASE-ROTATION.md):** fresh pepper (SHA `c38b0bb7`) + fresh UUID; `android-1.3.1-1058` added to `LAVA_AUTH_ACTIVE_CLIENTS` (all prior retained); client + api-app rebuilt from the rotated `.env` so codegen key ↔ server allowlist are consistent. This is the "rotate token if needed" the operator anticipated — Gate 4 had blocked ALL distributes on the stale 1057 pepper. - **§6.Z device verification (Genymotion Pixel 9 / API 35 / arm64, runner: containers-submodule):** DEBUG = Challenge00CrashSurvival PASS (connectedDebugAndroidTest); RELEASE = `run-release-canary.sh` (LVA-077, new tool — closes the no-`connectedReleaseAndroidTest` gap) cold-start PASS on the exact R8 artifact `f2843805`, 0 ANR / 0 FATAL. Evidence: `.lava-ci-evidence/distribute-changelog/firebase-app-distribution/1.3.1-1058-test-evidence.md`. - **Build-stability fixes this close-out: LVA-076** `core/apiengine process*JavaRes→buildCshared` task dependency (Gradle-validation BUILD FAILED on clean `.cxx`) + Gradle heap 2g→4g (D8 dex-merge OOM). **LVA-077** `scripts/run-release-canary.sh` (§6.Z R8 cold-start canary). **LVA-078** the api-go **container image** build is now RESTORED — root `.containerignore` had blanket-excluded `submodules/` while `lava-api-go/go.mod replaces ../submodules/X`, so the in-container `go build` failed on missing replacement dirs (masked since 2026-05-31 by the COPY-step `/var/tmp` disk-full, itself relieved by `podman system prune`). Now: `podman build → lava-api-go:dev` (49 MB distroless), embedded healthprobe binary executes (§6.A/§6.B), image saved to `releases/1.3.1/api-go/lava-api-go-2.3.24.image.tar`. **"Rebuild everything" is fully complete.** LVA-076/077/078 recorded Completed in the workable-items ledger (77 items, 0 open). - **Forensic (honest, not papered over):** transient ANRs seen during validation were PROVEN host-CPU-starvation from the operator's concurrent HelixCode/Jackett workloads (host load 13–36; ANR `/proc/pressure/cpu` ~35%, app absent from top consumers, MainActivity resumed every run) — NOT a Lava defect; clean PASS obtained off the spike; no foreign processes killed (§6.M). - **§6.Y next-cycle bump applied:** client `1.3.2-1059`, api-app `0.2.2-6`, api-go `2.3.24-2324`. - **Honestly-still-open (NOT bluff-closed):** LVA-008 (upstream androidx nav-teardown — accepted + §6.AC-instrumented, not fixed), LVA-036 (pre-existing github↔gitlab fork — operator content-merge), LVA-3 (§6.AF-debt residue), LVA-5 (Firebase token rotation — console-only operator action; existing token worked for this distribute, deprecation warning only). None block the build or the distribute.

Last updated (prior): 2026-06-09 (**DEVICE GATE GREEN on real Genymotion hardware — distribute unblocked at the device level**). Operator booted a Genymotion Pixel 9 / API 35 / arm64 VM (127.0.0.1:6555, §6.AH VM path). - **§6.Z core-flow device gate GREEN:** C00 (cold-start survival) + C01 (launch+menu) + C07/C08 (cross-tracker fallback) ALL PASS on real hardware with all 69 of this loop's commits + the LVA-008 NavTeardownCrashReporter (evidence `.lava-ci-evidence/genymotion/coreflow-gate-20260609T160325Z`, runner: containers-submodule). C06/C11 = the operator-accepted LVA-008 upstream nav-teardown (instrumented). - **LVA-074 (root-cause + fix):** the first gate runs went RED with "No compose hierarchies found" — root-caused LIVE (logcat) to the **VM screen sleeping** → idle render pipeline → SurfaceFlinger commit timeout → ComposeTestRule finds no committed surface (the app composes fine: MainActivity→RESUMED, no crash/nav/compose exception). NOT a code defect (LVA-008 isolation-tested; LVA-052 nav arg safe). Fix: `scripts/run-genymotion-challenges.sh` now wakes+keeps-on the VM screen before `connectedDebugAndroidTest`. After the wake the gate went GREEN. Incident `2026-06-09-genymotion-surface-render-timeout` RESOLVED. - **▶ Firebase re-distribute now gated ONLY on:** (1) §6.Y version bump (agent can do), (2) **LVA-5 Firebase token rotation (operator-only)**, (3) operator real-device verification per §6.Z clause 5, (4) the §6.Z two-stage debug→release flow. LVA-008 accepted+instrumented + device gate green = the code/test prerequisites are MET.

Last updated (prior): 2026-06-09 (**operator decisions logged + single-stream resume**). Operator chose (2026-06-09): **LVA-008 = accept-with-§6.AC-telemetry + distribute**, and **single-stream OWED work** (parallel fleet was hitting server-side rate limits). - **LVA-008 INSTRUMENTED + ACCEPTED (commit `06f599b5`, LVA-073 Completed):** `NavTeardownCrashReporter` (chained uncaught-exception handler, wired into `LavaApplication.onCreate`) tags the known upstream androidx nav-teardown ISE with attributable §6.AC Crashlytics context (`feature=navigation`, `operation=activity-teardown`, `screen=search_input`, `lva_id=LVA-008`, custom key `lva_008_known_upstream_nav_teardown`) BEFORE the process dies — it instruments, never swallows. 6 falsifiable JVM tests (detection+tagging logic is pure JVM, no device). **Distribute now unblocked at the code level**; the actual §6.Z two-stage Firebase re-distribute still needs: the device Challenge gate (BLOCKED on this macOS host per §6.X-debt — no in-container emulator/HVF; needs a Linux x86_64+KVM gate host OR the Genymotion VM), a §6.Y version bump, and operator real-device verification per §6.Z. LVA-008 → In progress (the upstream crash itself remains; it is accepted+instrumented, not fixed). - **Rate-limit note:** the wave-10 parallel agents were server-rate-limited mid-task ("not your usage limit"); one produced unverified partial Go telemetry work whose falsifiability rehearsal was inconclusive + SourceHash failed → DISCARDED (not integrated), queued honestly as LVA-072. Proceeding single-stream. - **OWED single-stream queue:** LVA-070 (favorite/visited providerId write+read threading — LVA-067 laid the Room column), LVA-071 (inject SseClient for hermetic testing), LVA-072 (cert-rotation §6.AC telemetry, redone properly).

Last updated (prior): 2026-06-09 (**autonomous parallel-fleet loop — 3 waves of 6 subagents, ~25 commits past `54eedd92`; pushed + converged github+gitlab**). Each wave: 6 parallel agents (Kotlin in worktree isolation, Go/docs/config shared tree) + main-stream independent re-verification of every claim (no blind trust). All work falsifiability-proven with verbatim captured evidence. **check-constitution.sh fully GREEN (EXIT 0)** after this loop's §6.R fixes. Constitution pin `60e2d66`. **19 LVA items closed this loop** (LVA-6, 25, 26, 28, 29, 30–35, 37, 38–43, 45) — 11 of them REAL shipped bugs. - **Wave 9 — TLS zero-downtime rotation + Room providerId migration + SSE/proxy coverage:** LVA-068 (embed TLS cert now ROTATES mid-process via a `GetCertificate` callback + atomic `rotatingCert` holder — re-mints within the expiry margin, same IP-SANs, zero dropped connections; `590dd710`). LVA-067 IN PROGRESS (Room v11→v12 `MIGRATION_11_12` adds a nullable `providerId` column to favorite/visited rows + schema `12.json` + converters + a real-SQLite non-destructive migration test 3/0/0; `badca5f5`) — the persistence column is laid, but threading the source `providerId` through the write path (`ToggleFavoriteUseCase`/`GetTopicUseCase` + `Topic`/`TopicPage` models) + read path (`TopicModel` → `openTopic(id,providerId)`) is honestly OWED → LVA-070, so favorites/visited HTTP_DOWNLOAD still falls back to the active tracker for now (no data loss, no upgrade crash). LVA-069 (SearchResultViewModel SSE raw-JSON parsing 10 tests + onRetryClick 2 tests + malformed-JSON hardening; CLEAN, `e1c983ae`). DelegatingProxySelector coverage 6/0 (`e01e504b`). New queued: LVA-070 (favorite/visited providerId threading), LVA-071 (inject SseClient for hermetic testing). **Loop status:** the high-value autonomous surface is now largely swept — recent waves trend CLEAN + coverage + §6.N hunts finding 0 bluffs. The next milestone (Firebase re-distribute) is operator-gated on LVA-008 (nav-teardown accept-vs-keep-RED) + LVA-5 (token rotation). - **Wave 8 +**

Jackett — LVA-052 topic-tap COMPLETED + TLS expiry + coverage + Jackett validated: LVA-052 HTTP_DOWNLOAD is now USER-REACHABLE from the topic screen (commit `831e7be9`): a `ProviderCapabilitySource` seam + `ResolveProviderDownloadKindUseCase` + a topic-nav `providerId` arg (defaults to the active tracker, back-compat) + `TopicViewModel` HTTP-vs-`.torrent` branch + search_result threading the real `providerId`; real-stack + no-regression tests falsifiability-proven (favorites/visited need a Room providerId column → LVA-067, honestly OWED). LVA-064 (TLS embed cert regenerated when the persisted leaf is EXPIRED — injectable clock seam, `21d8fc23`). LVA-065 (SearchViewModel bug-hunt CLEAN + 3 real coverage gaps: PagingDataLoader prepend, append-error retry, sign-out privacy reduction, `9f634e16`). §6.N wave-8 bluff-hunt: 5/5 SAFE-module tests GENUINE, 0 bluffs (`2507e427`). **Jackett (operator-flagged new release):** validated **v0.24.2040-ls426** (digest `sha256:bae2fdf0...`) boots in the loopback topology, bootstraps its api_key, and serves valid Torznab `<caps>` at the exact `IPTorrentsJackettApi` endpoint — real §6.B probe, parser tests green, validated digest pinned in `.env.example` (`04e5156a`/LVA-066); full real-creds IPTorrents+FlareSolverr e2e operator-gated. New queued: LVA-067/068/069. **Worktree stale-base fix:** instruct worktree agents to `git reset --hard origin/master` first (the isolation branches from session-start HEAD, which lacked this loop's commits — that blocked LVA-052 in wave 7). - **Wave 7 — 2 Go fixes + gate-coverage hardening + credential coverage; LVA-052 topic-tap honestly OWED:** LVA-059 (deterministic multi-search SSE provider order via `IDs()` sort, `0a1881cd`), LVA-061 (TLS embed cert regenerated when persisted IP-SANs no longer cover the device LAN IP, `dc6c1179`), LVA-062 (gate-script clause-coverage: created the missing IPv4 scanner hermetic test + `.db`-exemption sub-tests in all 3 §6.R scanners + seqcoll comment-strip branch — all falsifiability-proven, `b5302590`), LVA-063 (+9 falsifiable CredentialsViewModel/CredentialsManager tests, `b51f040a`+`9280d4ab`). **LVA-052 topic-tap remains OWED** — worktree-isolated agents branch from the session-start base (`54eedd92`), so they lacked master's LVA-044/052 SDK layer; the agent correctly refused to ship half-wired dead code. Completing it needs a provider-id threaded through the topic route + 4 openTopic call sites + a favorites/visited Room column (multi-call-site + migration) — do it on the main tree or reset the worktree to master first. New queued: LVA-064 (TLS cert expiry check), LVA-065 (SearchViewModel coverage audit). app/deep-link/nav/connection/rating audit CLEAN. - **Wave 6 (+ recovery) — 4 MORE real bugs + LVA-052 cleanly re-applied + 2 §6.R scanner gate fixes:** LVA-054 (P1 — NavigationController `topLevelBackStack.removeLast()` back-stack-pop crash on Android <API35, same SequencedCollection class as

LVA-053; fixed + a `scan-no-removelast-seqcoll.sh` guard with comment-stripping wired into check-constitution + 5/5 hermetic test, `d99b10ad`), LVA-057 (multi-search SSE silently dropped unknown requested providers — now emits `provider_error` + counts failed, `e325d9a0`), LVA-060 (P1 — favorite/visited magnet-only torrents reconstructed as un-downloadable BaseTopic; date+magnetLink added to the discriminator, `af160a35`), LVA-058 (§6.R ipv4/hostport/uuid scanners flagged the binary `workable_items.db` SSoT — `.db` exemption added, `3d9eb469`), LVA-056 (IPv4 scanner `tests/` exemption — landed post-FS-recovery, `453fe2b6`), LVA-052 IMPLEMENTED cleanly on master's LVA-044 (ViewModel→UseCase→SDK→disk real-stack proven; topic-tap provider-id nav threading OWED, `9d9d2bd5`+`c2b03749` fake fix), LVA-055 (run-test-pg.sh now runs internal/storage Postgres legs). §6.N.2 production bluff-hunt closed a real CM-WORKABLE-ITEMS-SYNC clause-3 coverage gap (`2883a136`). New queued: LVA-059 (deterministic SSE ordering), LVA-061 (tls IP-SAN cert), LVA-062 (gate-clause coverage audit). **Recovery note:** mid-loop the T7 volume hit a macOS TCC access denial (process-scoped, fixed by app relaunch + Full Disk Access) — zero work lost, everything was committed+pushed per wave. Agent 5 (coverage) hit an account session limit (no result). - **Wave 5 — 1 MORE real crash bug + 3 queued cleared + real-Postgres coverage:** LVA-053 (P1 — `PostConverters.removeLast()` desugars to `java.util.List.removeLast` (JDK21 SequencedCollection) absent on Android <API35 → topic-screen `NoSuchMethodError` crash on a `PostBr+Hr` post; fixed `removeAt(lastIndex)` + 28 converter tests, `608c3bf9`), LVA-048 (search_input wrote author NAME under AuthorIdKey → name lost, `50de44c1`), LVA-049 (route values not URL-encoded → reserved chars corrupt the route; `Uri.encode` per-value, same commit), LVA-050 (A11y test red on the RELEASE variant — missing ComponentActivity host because `ui-test-manifest` was debugApi-scoped; added testImplementation, green both variants, assertions intact, `96b7255a`). lava-api-go real-Postgres-in-podman integration coverage: `internal/cache` 65.7→97.1%, `internal/storage` 76.4→80.9% (real pgx, container torn down, `c9fdb661`). New queued: **LVA-054** (sweep all `removeLast/removeFirst/getFirst/getLast` SequencedCollection calls — LVA-053 class, may be MORE crashes), **LVA-055** (run-test-pg.sh omits internal/storage). **LVA-052 deferred** (a full impl exists on branch `worktree-agent-ad695086c8735d60a` but re-derived LVA-044 → conflicts; needs clean re-apply on master's LVA-044 + topic-nav provider-id threading; one wave-5 agent died on a socket error — production bluff-hunt not run, re-queue). - **Wave 4 — 2 MORE real bugs + LVA-044 feature + §11.4.128 skeleton + clean §6.Q audit + 5/5-genuine bluff-hunt:** LVA-046 (rutracker Login reported FAKE success on wrong credentials — blind `AsAuthResponseDtoSuccess()` union

accessor ignored the discriminator, non-pointer `UserDto` decoded WrongCredits with no error → `Success:true AuthToken:''` → every authed call 401s; LVA-032 class; fixed via checked accessor, `2c310607`), LVA-047 (MagnetLinkValidator rejected case-insensitive `urn:btih:` prefix per RFC 8141, `458d0960`), LVA-044 IMPLEMENTED (HTTP_DOWNLOAD capability + `HttpDownloadableTracker`, archiveorg/gutenberg wired, §6.E fwd+rev gate green, real-stack MockWebServer tests — **SDK-reachable; UI consumption OWED → LVA-052**, `badbca92`), LVA-051 (§11.4.128 device-recorder skeleton + deterministic-path hermetic test; device capture OWED — no live device, `507c34c6`). §6.Q + ui/designsystem/navigation audit CLEAN (7 verticalScroll Composables all safe, theme/nav correct). §6.N bluff-hunt: 5/5 production-mutated tests GENUINE, 0 bluffs (`1af5193b`). New queued from wave-4 §6.Q agent's out-of-scope finds: **LVA-048** (search_input writes author NAME under AuthorIdKey — real nav bug), **LVA-049** (route values not URL-encoded), **LVA-050** (A11yContentDescriptionTest teardown RuntimeException), **LVA-052** (HTTP_DOWNLOAD UI wiring). - **Wave 3 — 5 MORE real shipped bugs + §6.E reverse-gate + §6.AI-debt:** LVA-038 (brotli middleware emitted `Content-Encoding: br` on bodyless 204/304; `statusAllowsBody` was dead code, `f3b2d269`), LVA-039 (ProviderLogin captcha-retry left stale `serviceUnavailable` banner — banner-sweep gap, `9f895502`), LVA-040 (onboarding anonymous-mode choice never persisted to `provider_configs` → Provider Config shows OFF + search treats provider as credentialed, `8e0720ab`), LVA-041 (`SyncPeriod.DAY` background-sync flex `6.days` should be `6.hours` → WorkManager silently clamps flex to full period, `0cecbe4e`), LVA-042 (SyncOutbox FIFO Third-Law fake divergence + createdAt coverage, `a11aee32`), LVA-043 (§6.E reverse-gate: phantom-capability — exposing an undeclared feature shipped green; now caught, `c117caeb`), LVA-045 (§6.AI-debt: CM-COVENANT-114-* §11.4.128–141 propagation gate in check-constitution + slash-command docs, `c37995a7`). New queued: **LVA-044** (Feature — add `HTTP_DOWNLOAD` capability so archiveorg/gutenberg can surface their working-but-unreachable HTTP download; not a bluff, a functionality gap). - **Wave 1 — 5 tickets closed:** LVA-025/026 (Go v1 captcha: dynamic `cap_code_<sid>` field name + propagate upstream Content-Type, `aca3e720`), LVA-028 (nnmclub search publishDate dropped, `a31efce8`), LVA-029 (isLocalHost fc/fd ULA false-positive on public hosts e.g. `fcbarcelona.com`, `4e687769`), §6.AI-debt LAYER-2 (§11.4.140 action-prefix UserPromptSubmit hook → `constitution/scripts/hooks/action_prefix_expand.sh`, hermetic test 8/8, `db69d797`), LVA-6 (codegraph own-org-submodule indexing — verified genuinely done: 4460 submodule files indexed + §6.H 0-leak, `22516444`). - **Wave 2 — 3 MORE real shipped bugs + LVA-030 durable:** LVA-032 (rutracker browse/favorites blind-cast

Topic→Torrent: `AsForumTopicDtoTorrent` ignored the union discriminator → empty fake-torrent rows; fixed via the never-wired `AsForumTopicDtoTorrentChecked`; +gutenberg cov 82.8→89%, `fab05f2e`), LVA-033 (rutor topic page dropped `Добавлен` publishDate, `45794521`), LVA-034 (Room endpoint-list converter dropped GoApi `key` → list-selected on-device Lava-API endpoint 401s; `Lava-Auth` override dead; fixed via #-sentinel encoded packing, no migration, `488185ed`), LVA-031 (genymotion VM+nav UUID §6.R redaction per e767b701, `6687e23b`), LVA-035 (PagingDataLoader coverage, `a3f7fb83`), LVA-030 (6-submodule §6.R/§6.S/§6.X/§6.AD pointer-blocks committed in-submodule + pushed; 5 pins bumped `7c9843c8`). - **Audited CLEAN (no padding):** core/domain + feature/{search,search_input,search_result,category,topic,forum} ViewModels (genuine anti-bluff tests, no Second/Third-Law bluffs); rutor/kinozal/archiveorg/gutenberg search/browse parsers; core/network resolvers + LocalNetworkDiscovery + preferences. Latent gaps flagged (not faked): kinozal topic parser drops fields but the committed fixture is a synthetic stub (needs a real captured `details.php` fixture before a non-bluff fix); CategorySelectionViewModel child-deselect leaves parent group fully-selected (UX, needs product decision). - ▶ **NEW OPERATOR-GATED ITEM LVA-036:** `llm_orchestrator` github↔gitlab have a PRE-EXISTING non-FF mirror fork (diverged at `d2a2151`, unique non-doc go.mod content each side). LVA-030's pointer commit `a98ae84` landed on gitlab + the working tree (gate passes) but github refused a non-FF push — NOT force-pushed (§6.T.3). Needs an operator content-merge decision. llm_orchestrator parent pin therefore held at `a484f7d` (not bumped). - ▶ **STILL OPERATOR-GATED:** LVA-008 (C11/C06 nav-teardown — upstream androidx defect, 5 avenues device-falsified; accept-with-§6.AC-telemetry+distribute OR keep-RED) + LVA-5 (rotate Firebase CI token, §6.H).

Last updated (prior): 2026-06-09 (**keep-building loop — LVA-013 deeper Third-Law fix closed**). Operator directive "keep-building now" (no distribute; LVA-008 stays DEFERRED awaiting the accept-vs-keep-RED decision). Subagent fleet was server-rate-limited, so this is a solo main-stream slice. - **LVA-013 CLOSED (Completed → Fixed.md)** — Third-Law (behavioural-equivalence) bluff in `core/testing` `TestEndpointsRepository.observeAll()`: the fake SEEDED + EMITTED `[Endpoint.Rutracker]` on first observe while the real `EndpointsRepositoryImpl.observeAll()` seeds nothing (`defaultEndpoints` is `emptyList()`) and `.filterNot { it is Endpoint.Rutracker }`s every emission — so production NEVER lists Rutracker (fresh-install first observe = `[]`). Fixed the fake to match (no seed + emission-level filter) AND rewrote **4 phantom-asserting test methods** across 3 files to the real contract: `TestEndpointsRepositoryEquivalenceTest` (×2 — part-3 of the add-no-op test + the renamed `fake_observeAll_never_emits_rutracker_like_real_impl`), `TestInfrastructureContractTest.observe on empty repo emits empty list and never Rutracker`, `DiscoverLocalEndpointsUseCaseTest.discovery adds a new mirror and Rutracker is never listed`. **Bluff-Audit:** mutation reintroducing the onStart seed → core:testing `expected:<[]> but was:<[Rutracker]>` (2 methods) + core:domain `Fresh-install first observe MUST emit []... Got: [Rutracker]`; reverted. GREEN after revert: **core:testing 24/0, core:domain 57/0, feature:connection 10/0**. Ledger gate `CM-WORKABLE-ITEMS-SYNC` OK (13 items, DB↔MD in sync). - **LVA-014 + LVA-015 CLOSED (Completed → Fixed.md)** — two MORE Third-Law fake bluffs surfaced by a read-only parallel-agent audit of every `core/testing` fake (the throttle cleared): **LVA-014** `TestSuggestsRepository` was a TODO-throw stub (LVA-012 class) → implemented in-memory (case-insensitive UPSERT `id=lowercase().hashCode()`, newest-first, clear) + `TestSuggestsRepositoryTest` (4 tests). **LVA-015** `TestSearchHistoryRepository.add` used a positional id + always-append + insertion order while the real `SearchHistoryRepositoryImpl` uses content-derived `Filter.id()` + `@Insert REPLACE` UPSERT + `ORDER BY timestamp DESC`; the existing `TestSearchHistoryRepositoryTest` asserted the fake-shape (ids `[0,2]`, oldest-first) — a bluff test. Fixed the fake (replicates `Filter.id()` w/ source-of-truth comment, UPSERT, newest-first) + rewrote the test to the real contract (6 tests, incl. sort-only dedup). Both Bluff-Audited (mutation → `expected:<1> but was:<2>`, reverted). GREEN: core:testing 31/0, feature:menu 17/0, feature:search 9/0. (Audit also confirmed TestSettings/Auth/LocalNetworkDiscovery/Endpoints/Visited/Favorites/Bookmarks fakes are genuinely equivalent — no padding.) - **lava-api-go coverage (commit 4775d0c1):** added falsifiable tests for two real 0%-coverage gaps a parallel agent found — `internal/router` Jackett route registration + enabled/disabled/unconfigured discriminators + /health-open-under-full-auth-chain (4 tests); `internal/server` brotli `WriteString` fast-path + `statusAllowsBody` RFC truth table (3 tests). All 4 mutations falsified + reverted; gofmt/vet clean; tests/contract source-hash unaffected (test files excluded from embed manifest). - **Two more parallel read-only audits (throttle clear) found 3 tracked items: LVA-016** (Bug, ACTIVE bluff) — `RatingViewModelTest`'s local `RealObserveRatingRequestUseCase` re-implements the use case but DROPS the engagement gate, so the "Show dialog" test asserts Show with zero engagement while the real `ObserveRatingRequestUseCaseImpl` would Hide (Second-Law/§6.J). Proper fix wires the real use case (deep tree via shared fakes, now usable post-LVA-012/015) — deferred to a focused cycle, NOT rushed. **LVA-017** (LATENT) feature favorites/topic local fakes' `add()` lacks REPLACE dedup (unexercised). **LVA-018** core/preferences dead getters getHistorySyncPeriod/getCredentialsSyncPeriod (zero call sites). The dead-code audit otherwise came back CLEAN (0 shipped TODO(), 0 if(false), 0 silent-no-op effect methods). - **LVA-016 FIXED — the canonical "green test, broken feature" bluff.** RatingViewModelTest re-implemented ObserveRatingRequestUseCase but DROPPED the engagement gate, so the "Show dialog" test passed with zero engagement while the real ObserveRatingRequestUseCaseImpl would Hide. Fixed: made the impl public (Fifth Law) + wired the REAL use case over the LVA-012/015-fixed shared fakes (ObserveSearchHistory/Visited[→EnrichTopics]/Bookmarks) + seeded real engagement (3 bookmarks) in the 4 Show-path tests. Bluff-Audit: neutralize engagement → those 4 tests FAIL (TurbineAssertionError: No value produced in 3s), 2 Hide tests pass; reverted. GREEN feature:rating 6/0, core:domain 57/0. - **5-subagent parallel fleet (throttle clear) — ALL verified by the main stream + closed: LVA-7** chaos §11.4.85 (Agent A) — search_thundering_herd_test.go: 512 concurrent identical-key reqs through the REAL read-through cache, asserts all-200 + post-burst upstream-loads=0 (convergence) + no-leak + -race; Bluff-Audit delete cache.Set→convergence FAILED. **LVA-019** (Agent B, REAL production bug) — nnmclub.IsAuthorised used unquoted CSS attr selectors a[href*=profile.php] → cascadia matched NOTHING → dead auth-fallback; fixed to quoted form + test (IsAuthorised 77.8→88.9%); api-source.hash regenerated, sourcehash contract GREEN. Plus kinozal parser edge tests. **LVA-017** (Agent D edits) — feature favorites/topic local fakes add() now dedup-by-id (REPLACE) + 2 falsifiability tests (expected:<[7]> but was:<[7,7]>). **LVA-018** (Agent C) — core/preferences dead getters removed (4 files). **Agent E** anti-bluff audit of core/network+core/data+un-audited features → ZERO bluffs (clean). Every agent result was re-verified + falsifiability-rehearsed by the main stream before commit (no agent claim trusted blindly). - **Second + third parallel fleets (8 bug-hunt agents total) → FIVE more REAL shipped Go bugs found + fixed, all main-stream-verified + falsifiability-rehearsed: LVA-020** archiveorg array-valued creator/title/year/date failed the WHOLE search/browse/topic response (flexString tolerant type); **LVA-021** observability error_class collapsed to literal "error" for every real typed error → §6.AC telemetry blindness (%T fallback); **LVA-022** gutenberg blank format label for charset-suffixed Gutendex MIME keys (prefix-match); **LVA-023** v1 login returned User.Id (data-uid) not User.Token (cookie) → every authed v1 call 401s (+ first e2e login test); **LVA-024** v1 GetTopic fabricated a bogus TopicFile{Name:"Size"} (dropped synthetic file + fixed the bluff test that asserted it). Plus Agent L added 4 falsifiability-proven coverage files (rutracker→90.1%, auth→95%, config→94.9%, middleware→100%). api-source.hash regenerated;

sourcehash contract GREEN; go build/vet clean. - **Deferred/ queued from the fleets:** LVA-025 (v1 captcha answer sent under wrong form-field name — needs LoginOpts/OpenAPI model change), LVA-026 (v1 captcha Content-Type hardcoded image/png). Kotlin tracker bugs still to fix on the main stream: **G1** Kinozal `sizeBytes` hardcoded null (every row), **G2** Nnmclub `publishDate` dropped, **J1** `isLocalHost()` fc/ fd false-positive mis-routing. Audit also surfaced latent/ conditional items (J2-J6 network, orphaned CredentialsScreen, LavaIcons.AppIcon foot-gun) — to triage. Agents K (core/domain) + the Compose audit came back CLEAN. - **CONSTITUTION PIN BUMPED `7734c04` → `60e2d66`** (1.2.0-dev, 16 commits) per operator directive 2026-06-09. New UNIVERSAL clauses §11.4.128–§11.4.141 adopted via new **§6.AI** clause (the ATMosphere audio batch §11.4.135-139 is project-specific, NOT binding on Lava). Mappings: §11.4.131 session-resumption-file ≡ `docs/CONTINUATION.md` (§6.S, this file — the declared canonical path); §11.4.134 ≡ `/code-review ultra`; §11.4.132 risk-ordered-validation + §11.4.129/130 huge-blocker/validate-fix-first release discipline ADOPTED; §11.4.140 action-prefix system LAYER-1 wired into root CLAUDE.md (§6.AJ) + AGENTS.md + QWEN.md (registry `constitution/actions/registry.yaml`); §11.4.128 device-recording + §11.4.141 token-efficiency thin-index + §11.4.140 LAYER-2 hook = OWED via **§6.AI-debt**. No new mandated submodule. **Known pre-existing debt surfaced (LVA-030):** 6 recently-added submodules (doc_processor/helixqa/llm_orchestrator/ llm_provider/llms_verifier/vision_engine) lack the §6.R inheritance pointer → full `check-constitution.sh` exits 1 (changed-only pre-push still passes; orthogonal to the bump). - **G1/G2/J1 (Kotlin tracker/network bugs):** LVA-027 Kinozal `sizeBytes` hardcoded null FIXED (KinozalSizeParser + end-to-end test, falsifiability-proven). Still Queued: LVA-028 Nnmclub `publishDate` drop, LVA-029 `isLocalHost()` fc/fd false-positive. - **Ledger state:** 30 items — 20 closed (11 Fixed + 9 Completed: LVA-009..024, LVA-027, LVA-7). Queued: LVA-6 codegraph-index, LVA-008 C11 (DEFERRED operator-gated), LVA-025/026 (v1 captcha), LVA-028/029 (Kotlin tracker/net), LVA-030 (6-submodule §6.R). LVA-5 operator-blocked (§6.H). LVA-3/4 in-progress. CM-WORKABLE-ITEMS-SYNC OK.

Last updated (prior): 2026-06-08 LATE-EVENING (**autonomous parallel-fleet QA session #2 — ~13 commits on `master` past `207adf73`; HEAD ~`21106e26`+LVA**). Ran 1 device-bound main stream + 13 non-device subagents across 5 waves (each on a distinct gitdir/toolchain → zero Android-Gradle/device/parent-index contention). Every PASS carries verbatim captured output; multiple §6.J findings surfaced + handled honestly. - **HelixQA autonomous vision QA NOW WORKING** (the operator’s core goal —

> replace manual testing with AI vision QA): fixed a 3-layer claude-CLI bridge incompatibility in the helixqa submodule (`42f9998`→`aef46e5d`, pushed both mirrors) — (1) `--json`→`--output-format json`, (2) `--image`→in-prompt screenshot path read via Claude Code's Read tool (probe-proven), (3) prompt as `--print`'s value before the variadic `--allowedTools`. On the Genymotion VM the vision backend now GENUINELY analyzes Lava screens + decides actions (`vision-screen-changed PASS`; real rationale "Current screen is the Android home launcher … launching the Lava client app"). Remaining: a launch-dispatch `exit status 127` (**LVA-009**, tracked). Evidence: `.lava-ci-evidence/helixqa/vm-qa-vision2-20260608T182251Z/`. (Each prior bridge layer was a real bluff-or-incompat caught honestly: the agent also found+fixed 2 helixqa bluff tests that asserted the broken `--json`/`--image` flags.) - **Jackett sidecar LIVE-validated** (real podman bring-up): Jackett Torznab caps HTTP 200 + XML + FlareSolverr `status:ok`; fixed a real `wget`→`curl` healthcheck bug. **§11.4.85 stress/chaos PASS** (6 dims, falsifiability-proven). **lava-api-go**: golangci-lint **0 issues** (first real run — podman /Volumes mount was broken), coverage raised jackett 90→95% / middleware 76→98% / nnmclub 74→77% (4 new test files, 2 rehearsals). **+17 tracker parser edge-case tests** (rutor/nnmclub/kinozal/archiveorg/gutenberg, all Bluff-Audited). - **Pushed + converged (all 4 mirrors at `ba04ef2a`, then 2 more commits ahead):** `5b7ba824` **lava-api-go real failure fixed** — `api-source.hash` left stale by `7ce9c2b2` (embed-source drift); `TestSourceHash_ManifestMatchesLive` RED→GREEN, full `go test ./...` exit 0 (39 pkgs, real Postgres-in-podman e2e). `1a050e16` **§6.R host:port scanner comment-stripping** (false-positives on comment/example URLs fixed; hermetic test added; real literal still caught — falsifiability-proven) + **`run-genymotion-challenges.sh --evidence-dir` absolutized**. `5bc65d66` **submodule pins bumped**: helixqa→`42f9998` (lava-pin branch pushed to GitHub+GitLab — §6.W satisfied) + 5 dep `helix-deps.yaml` (doc_processor→`e446da7b`, llm_orchestrator→`a484f7dd`, llm_provider→`6776eb4b`, llms_verifier→`44b515da`, vision_engine→`5cf7fe7f`; all pushed vasic-digital GitHub+GitLab). `b0627107` stress-chaos jackett evidence refresh. `ba04ef2a` **§11.4.29 snake_case rename PLAN** (operator-gated; surfaced latent `llm_orchestrator` go.mod case bug). `2fe55126` **Jackett+FlareSolverr Torznab sidecar deploy-ready** (compose overlay + one-command validate script + docs) + **C11 device-debug forensics**. `3801c356` **IPTorrents §6.E magnet cold-cache test (JVM 3/3, Bluff-Audited) + §6.G C38 Challenge** (7th-provider coverage; fail-closed skip). - **§11.4.85 stress/chaos suite PASS** (real run): 6 dims (sustained-load 500req/0err, 64-way contention, fault-recovery, latency-injection, malformed-input, rate-limiter-trip) all PASS + falsifiability-proven; 2 Postgres dims honestly OPERATOR_GATED. - **C11 nav-**

teardown crash — 4 device cycles, STILL OPEN/RED (no bluffed green): hypotheses **all FALSIFIED on the Genymotion Pixel 9 VM** — (1) nav-compose 2.9.1→**2.9.8** (latest stable) does NOT fix it; (1b) **no `LenientTeardownRule` can catch it** — the ISE kills the PROCESS in `performDestroyActivity` via `NavControllerImpl`'s host ON_DESTROY observer, outside the JUnit statement chain; (2) **atomic-replace navigation** (`navigateReplacingCurrent` popUpTo) does NOT fix it (search_input still INITIALIZED at destroy). All speculative changes REVERTED (§6.T.1 — no unverified fix shipped). The feature WORKS (real archive.org result row renders pre-teardown); it's a deep androidx nested-NavHost teardown bug. Full forensics in the incident JSON's `systematic_debug_2026_06_08`. **DEFERRED** to a future nav-internals session or upstream androidx fix. - ▶ **OPERATOR DECISION 2026-06-09: close gates first, THEN distribute** (no Firebase distribute until the device Challenge suite is green against a fresh build + LVA-008 resolved + §6.Y bump; the proper §6.Z/§6.AA two-stage debug→release flow after). Firebase creds present. - ▶ **LVA-008 (C11+C06 nav-teardown) = UPSTREAM androidx defect — 5 avenues exhausted, DEFERRED.** Device-falsified (all reverted, §6.T.1): nav-version 2.9.1→2.9.8, LenientTeardownRule (process-death, uncatchable), nested-host move (Candidate 2 — crash moved to the OUTER search_input, disproving the nested-host theory), atomic popUpTo replace, AND the operator-approved deep custom-NavHost guard (force-advance INITIALIZED currentBackStack entries on ON_STOP — the phantom entry isn't in the public currentBackStack; it's in NavController internals, reflection-only). Core-flow device gate 2026-06-09 = **4/5 GREEN** (C00/C01/C07/C08 PASS with all this session's new code) — only C06+C11 fail on this one bug. Real user-impact: search-then-rotate/config-change crash. **AWAITING OPERATOR DECISION:** accept-with-§6.AC-telemetry + distribute, OR keep-RED until an upstream androidx fix. Full forensics: incident `systematic_debug_2026_06_08` + the 2026-06-09 updates. Distribute (operator chose gate-first) is blocked on this. - **DONE this wave (all pushed):** §6.J dead-torrent-validator wired into the download flow + §6.AC telemetry; lava-api-go bencode guard + router test; IPTorrents leechers mapper fix; credentials-vault VM tests; apiengine JNI contract; core/domain UseCase tests; lava-api-go coverage (handlers/observability/rutracker → 89/95/89%); 24 tracker parser edge tests; HelixQA vision QA bridge fully fixed (3 layers + LVA-009 launch) — vision drives the app, anti-bluff verdict works; 5 new HelixQA provider banks (7-provider matrix); vision-QA guide. Open: LVA-008 (critical), LVA-009-follow-on (LeakCanary device-reverify), LVA-010 (router no-middleware), LVA-011 (TestEndpointsRepository bluff-fake). - ▶ **(lower priority) LVA-009 — fix the HelixQA launch-**

dispatch `exit status 127` (the ADB actor runs the derived `shell monkey -p …` launch action without an `adb -s <serial>` prefix OR adb not on the dispatch PATH; in helixqa `pkg` actor/bridge) → re-run the VM QA → the autonomous vision QA should then complete the archiveorg journey end-to-end. (2) **LVA-008 — C11 nav-teardown** — next untried hypothesis: inner `rememberNestedNavigationController` lifecycle scope vs the outer host at destroy (SA-2's Candidate 1/2/3 ranked in the incident JSON); OR file an upstream androidx repro. (3) §6.W GitHub mirror still OWED for the cross-org `HelixDevelopment/LLMOrchestrator` (diverged — plan at `docs/qa/2026-06-08-llmorchestrator-divergence-plan.md`, operator-gated `git merge` Option a, NOT force-pushed). (4) §6.H RuTracker password rotation still OWED. (5) Jackett sidecar: run `scripts/validate-jackett-sidecar.sh --cloudflare` once Jackett api_key is bootstrapped. Workable items now tracked under the **LVA** prefix in `docs/workable_items.db` (LVA-008 C11, LVA-009 dispatch-127).

Last updated (prior): 2026-06-08 EVENING (**parallel-fleet QA session — 11 commits on `master`, HEAD `207adf73`; helixqa submodule advanced**). Ran 1 main + 6-subagent waves (server rate-limited at 5 concurrent → ran 2-3). Every PASS carries captured output; multiple §6.J findings surfaced + handled honestly. - **11 commits:** `7ce9c2b2` lava-api-go BYTEA nil/empty parity (real NOT-NULL divergence fixed at both Set boundaries) + Jackett Torznab parser; `d6f1b334` **IPTorrents = 7th provider** (Jackett-delegating, §6.G-honest — agent REFUSED the Cloudflare native-parser bluff); `97ad49d9` **RuTracker §6.E MAGNET_LINK bluff closed** (RuTrackerMagnetCache, RED→GREEN); `4c88c309` JackettResultMapper @Inject (§6.J module-green-but-app-Hilt-broken); `c33e8919` restored deep C05/C06/C11 + C07/C08 guards; `4e3252f3` drivable HelixQA archiveorg+rutor banks; `9e7505b3` **CrossTrackerFallbackModal dead-UI fix** (§6.Q/C37); `90077c81` lava-api-go ForumTopicDto discriminator `*Checked` accessors; `212f0fd2` device evidence; `09369926` real-network archiveorg+gutenberg download/magnet tests (`-PrealTrackers` gated). - **REAL DEVICE EVIDENCE (Genymotion Pixel 9 / API 35 / arm64, §6.AH container/VM path):** 7 tests, **6 PASS** (C00 cold-start, C01 launch+tracker, **C07/C08 rendered CrossTrackerFallbackModal — the dead-UI fix PROVEN on hardware**), **1 FAIL: C11 archiveorg search crashes MainActivity destroy** — `IllegalStateException: State must be at least 'CREATED' to be moved to 'DESTROYED'` on the `search_input` NavBackStackEntry. **nav-compose 2.9.1 (commit `7e6e7bcb`) did NOT fix it** (that commit's device-verification was OWED; now done + FALSIFIED). Incident: `.lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json`; evidence `.lava-ci-evidence/genymotion/9e7505b3-fleet-run/`.

C11 stays RED — OPEN. - **helixqa submodule advanced `dd3cf1d` → `42f9998`** (executor `b2e2bcd` + no-OCR provider-vision goal detection): the Android vision QA executor closes the `run --banks` skip-stub AND runs WITHOUT a Tesseract host (claude vision sets `Decision.GoalReached`). **Parent pin NOT yet bumped** — upstream main `bcac236` BREAKS Lava's build (CONST-052 PascalCase go.mod `replace ../DocProcessor` vs Lava's case-sensitive lowercase `submodules/doc_processor` on the T7 volume), so pin via a `lava-pin/2026-06-08-android-executor` BRANCH, NOT main. - ▶ **RESUME HERE (priority order):** (1) **nav-teardown crash fix** (C11 RED — own systematic-debugging cycle: newer androidx-navigation OR a NavHost teardown lifecycle guard → device re-verify on the VM). (2) `spotlessApply` is clean → §11.4.71 fetch-first → **push the 11 parent commits to GitHub+GitLab**. (3) push helixqa `42f9998` to a `lava-pin/*` branch + bump parent pin. (4) commit SA3's 5 `helix-deps.yaml` (inside `submodules/{doc_processor,llm_orchestrator,llm_provider,llms_verifier,vision_engine}`) + push. (5) **HelixQA VM QA run** — now fully unblocked (executor + no-OCR + drivable banks all landed). (6) §6.H RuTracker password rotation still OWED. Also: `scripts/run-genymotion-challenges.sh` needs `--evidence-dir` absolutized (relative path breaks the CLI build under `cd $CONTAINERS`).

Last updated (prior): 2026-06-08 (**Jackett / comprehensive-QA program, waves 1–2 on `master`.** 6-agent recon + impl: `lava.common.torrent` validator lib (bencode + magnet, SHA-1 info-hash) + kinozal/nnmclub/rutor §6.E magnet-exposure fixes (each forced-`--rerun-tasks`-verified, Bluff-Audited) — committed `776dc934`/`6fa31ad2`/`13703bc8`, rutor pending commit. All 5 tracker creds (RuTracker/RuTor/IPTorrents/NNMClub/Kinozal) in gitignored `.env` only. Jackett decision = **Torznab sidecar in the Lava local stack** (dossier `docs/qa/jackett-local-stack-research.md`; FlareSolverr needed for IPTorrents). HelixQA pin `5112906→dd3cf1d` (escaped broken `go.mod` conflict markers) **+ 5 missing own-org dep submodules added** (doc_processor/llm_orchestrator/llm_provider/llms_verifier/vision_engine, vasic-digital) → **full HelixQA binary now BUILDS: `v0.2.0`, 29.7 MB, `go build` exit 0**. NEXT: IPTorrents native provider + real-network per-provider download/magnet verification tests (the manual-test replacement) + HelixQA banks wiring. Program worklog: `docs/requests/improvements/lava_jackett_program_worklog.md`. OWED: §6.W GitLab mirrors for the 5 new submodules; §11.4.29 upstream CamelCase→snake_case repo rename.)

Prior — 2026-06-06 (**GENYMOTION DEVICE GATE UNBLOCKED + Phase 4 wave 2; branch `completeness-program-2026-06-04`, NOT merged/pushed**). Operator connected a running **Genymotion VM** (`127.0.0.1:6555`, Android 15 / SDK 35 / Pixel 9 / arm64) — VM-based, so §6.AH-compliant (not host-direct), finally giving this macOS host a real device surface for Challenges. - **Containers submodule extended for Genymotion** (committed in-submodule `4c0c0f1`, NOT pushed yet): `pkg/genymotion/` (Detect macOS+Linux / List / Running / Start / Stop / StartAndWait + pure parseList/parseVersion; injectable runner → 8 unit tests, §6.J column-mutation falsifiability-rehearsed) + `cmd/genymotion/` CLI. **Live-proven**: detect→gmtool path, version→3.10.0, list→running Pixel 9, serial→127.0.0.1:6555. `helix-deps.yaml` already declares Containers a leaf (0 own-org deps) → "all deps at root" satisfied. - **Lava glue** (`83b436e3`): `scripts/run-genymotion-challenges.sh` (thin glue → Containers CLI → Gradle connectedDebugAndroidTest vs the VM serial; §6.AG/§6.AH); guard hook recognizes Genymotion VM serials via `LAVA_REAL_DEVICE_SERIALS`. **Challenge00 PASSED on the real Genymotion VM** (clean quiet-gradle re-run, evidence `.lava-ci-evidence/genymotion/20260606T171744Z/`): JUnit `tests=1 failures=0 errors=0 skipped=0`, testcase `continueOnArchiveOrg_persists_signaledAuthState_to_disk` (3.5s) on Pixel 9 / API 35 / arm64 @ 127.0.0.1:6555 — **1 real instrumented Compose-UI Challenge executed + passed**, NOT a bluff. **§6.AH-debt (no bootable device gate on macOS) RESOLVED via the Containers-driven Genymotion path.** LESSON: full-app connectedDebugAndroidTest MUST NOT run concurrent with other gradle in the same checkout (the first attempt hit KSP corruption from 15 concurrent gradle daemons; re-run clean = PASS). - **API↔api-app drift gate** (STREAM API-SYNC, `e153d7c0/5e785d51/08704e06/6b5a76de`): `scripts/check-api-app-sync.sh` PASSES (no drift, hash `95c36d17…`); confirmed+fixed the real Gradle drift bug (buildCshared didn't declare lava-api-go source as inputs). On-device C05 hash-equality Challenge written (api-app), runs in Phase 11. §6.R UUID gate cleared (nil-UUID exemption + VM-UUID redaction, `e767b701`). - **PUSH PENDING** (operator-authorized): §11.4.125 code-review-agent gate IN FLIGHT over the 67-commit diff. On CLEAN verdict → §11.4.71 fetch+investigate → push Containers submodule (github+gitlab) + bump parent pin + push Lava main + push constitution (bumped 057d1d2b → its upstreams). Known Phase-10 debt: check-constitution full-run exits 1 on pre-existing comment/KDoc host:port EXAMPLE URLs (onboarding/cloud/sonar) — the host:port scanner needs comment-stripping; does NOT block changed-only pre-push. - **Phase 4 wave 2 (parallel subagents):** P4-GO2 lava-api-go (`d9ab1c01/8208c0cb/308c0cad/d710944b`: storage 65.7→72.2%, discovery 57.1→85.7%, gen union accessors, 2 fuzz ~1M execs 0 crash; documented ForumTopicDto discriminator type-confusion — pinned by falsifiable test, NOT fixed = Ktor-parity-contract risk). P4-CORE (core/

network+preferences+auth+navigation+sync) + P4-FEAT2 (feature topic/login/category/account/bookmarks/favorites) + CONST (constitution fetch+pull+verify-all sweep+pin bump) STILL RUNNING. - **NEXT (auto-resume on subagent completion):** (1) quiet gradle → clean Challenge00 re-run on Genymotion; (2) §11.4.71 fetch+investigate → push ALL submodules + main to GitHub+GitLab (operator-authorized); (3) Phase 4+ → Phase 11 end-state: full containers-API + on-device-API + client scenario testing on Genymotion w/ real evidence, then BOTH-variant Firebase re-release + §6.Y bump.

Last updated (prior): 2026-06-06 (**COMPLETENESS PROGRAM — Phase 3; branch `completeness-program-2026-06-04`, 30 commits ahead of master, NOT merged/pushed**). Session ran 1 main + 3 parallel subagent streams (worktree isolation was NOT engaged → all shared this working tree; disjoint-by-module + path-scoped commits kept them clean; Stream B's 3 Kotlin test files were swept into the docs commit `dee31593` by a `git add` overlap — all tracked, nothing lost). - **Phase 3 (concurrency/ responsiveness) — main stream:** (1) **Go race detector CLEAN** — `GOMAXPROCS=2 go test -race ./...` 0 races/0 fails across every package (Master-Plan §E no-hazard posture proven for Go); log under `.lava-ci-evidence/completeness-program/concurrency/`. (2) **Detekt correctness backlog cleared** (`bc0a478d`): `ImplicitDefaultLocale` Utils.kt:102 → `Locale.ROOT` (+ `FormatSizeLocaleTest`, falsifiability-rehearsed); `SwallowedException` ApiControlScreen.kt:115 → snackbar+Log.w; both module detekt baselines regenerated. (3) **Real data race FIXED** (`76fb0879`): `DownloadServiceImpl.cache` HashMap read on caller dispatcher + written from the DownloadManager BroadcastReceiver (main thread) → extracted to `DownloadUriCache`(ConcurrentHashMap), pure-JVM `DownloadUriCacheConcurrencyTest` (16×500, falsifiability: HashMap→FAIL). (4) **ApiEngineController TOCTOU FIXED** (`0c803472` documented → `30e880c6` fixed): reachable (VM=Default + Service=Main drive one Hilt singleton); the read→set gap is not reproducible by brute concurrency (≪ scheduling jitter, so a 400×16 stress stayed green vs the broken guard — I caught my own stress test as a bluff and did NOT ship it), so per the Fifth Law a `@VisibleForTesting` guard-checkpoint seam was added making the atomic `compareAndSet` fix DETERMINISTICALLY falsifiable (park caller A in the gap → read-then-set FAILS 1/8, compareAndSet 8/8). (5) Audited-clean: `LocalNetworkDiscoveryServiceImpl`, `NsdMdnsAdvertiser`. (6) **Task 3.3 consolidation logged**: Go already consolidates onto `submodules/{concurrency,ratelimiter}` (go.mod replace + 7 importers); the 3 Kotlin tracker breakers (kinozal/nnmclub/ rutor) are near-identical inline `CircuitBreaker(3/30s)`

`+Semaphore(4)` that can’t consume the Go submodules → Kotlin shared-primitive extraction scheduled for Phase 4. Known limitation logged: DownloadServiceImpl receiver-leak-on-never-completing-download (Phase 4/5, needs Robolectric). - **Phase 4 head-start — WAVE 2 parallel subagents (all §6.J falsifiability-rehearsed, real-stack, integrated + verified green on-branch):** - **Stream D (core/domain+core/data):** 31 cases — `SearchConverterTest`, `TopicConverterTest`, `EnrichTopicsUseCaseTest`, `ObserveSearchHistoryUseCaseTest` (commits `43e644f0`/`64e61eb3`). Flagged the `:core:testing` `Test*Repository` stubs as bluff-prone. - **Stream E (feature ViewModels):** 30 cases — `SearchViewModel`/`ForumViewModel`/`SearchInputViewModel`/`CategorySelectionViewModel` orbit-tests (commits `35cdace4`/`3b66ad1a`/`5a135e82`/`47fa1a0a`). - **Stream F (Phase-6 docs):** `docs/security/README.md` + `docs/guides/USER_MANUAL.md` (commits `f360edf3`/`da671202`). - **`TestSearchHistoryRepository.remove` bluff fake FIXED** (flagged independently by D+E): `filter{it.id==id}` (kept the removed row) → `filterNot`; `TestSearchHistoryRepositoryTest` added; feature/search+menu stay green. §6.T.4 BUGFIXES.md updated for all Phase-3/4 fixes. - **Phase 4 head-start via parallel subagents (all §6.J falsifiability-rehearsed, real-stack):** - **Stream A (lava-api-go):** internal coverage **44.5% → 54.6%**; commits `6bbd9bed`/`0b6b3fba`/`c7917058`/`a488c798`/`13446eaa` (rutracker provider DTO, middleware provider-dispatch, handlers/v1 SSE multi-search, kinozal+nnmclub adapters); **5 native fuzz targets** (~1.4M execs, 0 panics). Documented latent finding: `gen.ForumTopicDto.AsForumTopicDtoTorrent()` ignores the union discriminator (maps torrent-shaped non-torrent variants) — NOT auto-fixed (parity-contract risk), triage separately. - **Stream B (Kotlin core):** found+fixed a **real pagination bug** (`87469559`) — archiveorg `SearchResponseDto.toDomain` used page size 100 vs the actual `rows=50`, halving `totalPages` so the UI couldn’t paginate past ~50; +31 cases across `BookExtensionsTest`/`ArchiveOrgDtoTest`/`CollectionsTest`; §6.T.4 BUGFIXES.md entry. - **Stream C (docs, Phase 6 head-start):** `docs/api/README.md` (REST ref; flagged OpenAPI spec is incomplete vs the live `/v1/{provider}` routes), `docs/db/schema.md` (Room v11 + Postgres ERDs; flagged Room↔Postgres drift incl. `use_anonymous`), `docs/deployment/README.md`. Commits `cb7a13b3`/`d7463dad`/`dee31593`. - **Verified green on-branch after integration:** Go `go test ./...` exit 0; `:core:tracker:{archiveorg,gutenberg,rutracker}:test` + `:core:common:test` + `:core:downloads:test` BUILD SUCCESSFUL. - **Phase 3 essentially COMPLETE** (race-clean Go, 2 real races fixed + TOCTOU fixed, Detekt backlog cleared, Task 3.3 logged, hazard sweep done). Remaining P3 items are metrics-driven (lazy-init/non-blocking micro-benchmarks) and depend on Phase-5 load numbers → fold into P5. - **RESUME (Phase 4):** drive coverage up from the

18.65%/54.6% baselines — more core/feature unit+orbit tests, the Kotlin shared-`CircuitBreaker` extraction (Task 3.3 follow-up, concurrency-critical TDD), Go fuzz/contract expansion, then mutation/screenshot tests; Challenges are **[GATE-HOST Linux x86_64+KVM]**. Then P5 stress/load/chaos+metrics→optimize, P6–9 finish docs/courses/website/diagrams, P10 debt, P11 verify+distribute. Method that worked this session: 3 parallel disjoint-module subagent streams + 1 main stream, **path-scoped commits** (worktree isolation was NOT engaged — they shared the tree; `git add -A` would cross-contaminate). Branch reviewed+merged to master before any distribute.

Last updated (prior): 2026-06-04 (**COMPLETENESS PROGRAM started — branch `completeness-program-2026-06-04`, NOT yet merged to master**). Operator directive: drive every module/app/test/doc to finished+documented+100%-covered+leak/deadlock/race-free, no dead code, no disabled features; Snyk+SonarQube containerized; stress/integration; lazy-loading/semaphores/non-blocking; respect constitution. Master report+12-phase plan: `docs/superpowers/plans/2026-06-04-completeness-program-master-plan.md`. - **Phase 0 baseline FOUND 5 pre-existing defects on master, ALL FIXED on the branch** (each falsifiability-rehearsed §6.J): (1) §6.R `mobile.go` hardcoded `"Lava-Auth"` → build-time `-ldflags -X` + fail-fast test (`f83b5bc6`+`f1112477`); (2) stale `ApiControlAutoStartTest` contradicting Bug-B idempotency → rewritten (`fd7ea253`); (3) `CredentialsViewModelTest` flaky-under-load (Room-Flow dispatcher + teardown wall-clock) → deflaked, 10/10 (`b470b0f2`); (4) `ProviderLoginViewModelTest` flaky-under-load → deflaked, 10/10 (`54a66dfc`); (5) HelixQA signal-0 contract-drift broke `TestCheckGoProcessByPID_Alive` → realigned (`f458ddab`). Plus dead-code wired: TopicScreen add-comment dialog (`da42aaf0`, + `Challenge37` compiled/gate-host-deferred), 3 a11y contentDescriptions (`7edca413`), `:proxy` stale-doc scrub (`944ded35` — 2 inherited §6.K clause example-lists deferred to Phase-6). - **Method:** subagent-driven-development, parallel disjoint-module worktree streams. Transient server rate-limit at 5 concurrent agents → **downscaled to 2–3 concurrent** (clean). Go suite **fully green**; full Kotlin unit re-run under load confirms the flaky fixes. - **Phase 2 DONE (security/quality scanning infra, containerized, non-interactive):** Detekt wired via buildSrc convention plugin + per-module baselines (849 findings baselined; gate fails on NEW findings; `7a2312af`+rebaseline `786b5888`), Kover coverage baseline **18.65% line** (`docs/coverage/kotlin-baseline-2026-06-04.md`). Go: go vet (clean, gated), golangci-lint containerized (44→0 findings, 2 real metrics.go fixes), Go coverage **44.5%** (`c04eb153`). SonarQube CE+Postgres Compose **brought up green on this host** (applehv VM max_map_count OK), scan step needs `SONAR_TOKEN` (operator

UI); Snyk script needs `SNYK_TOKEN` (absent → honest refuse, no fake) (`465e4608`). ci.sh `--full` now runs detekt + go vet (verified green). **Owed:** Kover + golangci-lint ci.sh enforcement (exact-task/runtime handling); the Detekt correctness backlog (e.g. `ImplicitDefaultLocale` in `core/tracker/rutracker/.../Utils.kt:102`, `SwallowedException` in `api-app/.../ApiControlScreen.kt:115`) → fix in P3/P4. Triage docs under `docs/security/2026-06-04-*`. - **RESUME HERE (Phase 3):** P3 concurrency/responsiveness (`go test -race`, leak/deadlock/race sweep w/ reproducing tests, lazy-init/semaphore/non-blocking, consolidate onto `Submodules/{Concurrency,RateLimiter}`), then P4 tests-to-max (all types+Challenges+mutation/fuzz/screenshot, drive coverage up from the 18.65%/44.5% baselines), P5 stress/load/chaos+metrics→optimize, P6–9 docs/manuals/courses/website/diagrams/SQL, P10 debt closure, P11 verify+distribute. **[GATE-HOST]** Challenge execution needs Linux x86_64+KVM (§6.AH-debt). Branch `completeness-program-2026-06-04` reviewed + merged to master before distribute. **Method:** 2–3 parallel disjoint-module subagent-driven worktree streams (5 trips a rate-limit); worktrees branch from master so verify Phase-0/1 fixes survive each cherry-pick + regenerate detekt baselines on-branch.

Last updated (prior): 2026-06-04 — **client↔api-app linking shipped; BOTH apps, BOTH variants distributed on Firebase.** The bidirectional client↔api-app linking feature (onboarding “On this device” section + auto-start + ContentProvider key handoff + loopback auto-connect; Firebase-download fallback, no dead Play-Store links) is complete and consolidated onto the pre-existing `feature/menu/apiapp` launcher (§11.4.74). Two operator-reported on-device bugs were root-caused, fixed, falsifiably regression-tested, and on-device-verified on the real Samsung S23 Ultra (R5CW33CBVQV, API 36) in `848ce20c`: - **Bug A** — “Could not reach this API: API did not respond” selecting the discovered on-device API → doubled port (`…:8443:8443`); host is now the bare IP (`OnboardingViewModel`). Covered by `OnboardingViewModelTest` (15/15). - **Bug B** — “listen 0.0.0.0:8443; bind: address already in use” re-opening the API app → `ApiEngineController.start()` is now idempotent. Covered by `ApiEngineControllerTest` (7/7).

Distributed 2026-06-04 (operator directive: both apps + both variants, iterate on issues): - **client 1.3.0-1057** — debug `2st6d7c65r8r0` + release `7jka0995hbhl8`. REBUILT after a §RELEASE-ROTATION auth rotation (fresh pepper + new UUID for `android-1.3.0-1057`, all 22 prior clients kept ACTIVE → 1.2.36-1056 not regressed). §6.Z: AuthInterceptorTest 3/3 (rotated crypto round-trip decrypts → no first-request crash), OnboardingViewModelTest 15/15, ApiEngineControllerTest 7/7, all BUILD SUCCESSFUL; aapt2 versionCode 1057 +

signed; on-device cold-start/linking re-verification DEFERRED to operator post-distribute testing (S23 offline this session). - **api-app 0.2.0-4** — debug `4hd1ci7i9hovg` + release `7ep8h6p7416co`. Server artifact; Gates 4+5 inapplicable.

§6.Y post-distribute bump applied this commit: client → 1.3.1-1058, api-app → 0.2.1-5.

OWED (resume here) — cloud-search auth lag: the client cloud-API option (`https://lava.app:7777`) returns **401** for 1.3.0-1057 until the cloud server (`thinker.local`) registers `android-1.3.0-1057`. `thinker.local` was UNREACHABLE this session, so RELEASE-ROTATION step 7 (`scripts/distribute-api-remote.sh` — syncs the rotated `.env` + reloads + /health) is OWED. The on-device client↔api-app linking (loopback + per-endpoint key) is UNAFFECTED. Operator explicitly accepted shipping now + iterating. **Action:** run `scripts/distribute-api-remote.sh` when `thinker.local` is reachable.

Last updated (prior): 2026-06-03 — started client↔api-app linking feature (spec + plan landed); §6.Y bumps applied (client 1.2.36-1056 → 1.3.0-1057; api-app 0.1.2-3 → 0.2.0-4).

Last updated (prior): 2026-06-03 (**BOTH APPS' LATEST VERSIONS RELEASED on Firebase** — gated on a real operator-provided **Samsung Galaxy S23 Ultra (SM-S918B, Android 16)**, the §6.AH/§6.AG-compliant real surface after the macOS host-direct emulator proved permanently wedged (adbd-offline, ~10 attempts) and §6.AH forbade host-direct anyway): - **api-app 0.1.2-3** (green DEV #00FF00 / red release icon): **debug `536fmp12drijo` + release** both distributed. §6.Z: C01–C04 green ×2 (deterministic) + release cold-start canary green, all on the S23 Ultra. C02 was hardened (real-device finding: `/index` rutracker-downstream-dependency → short-read auth-gate-verdict probe). - **client 1.2.36-1056** (on-device-API integration): **debug `400oa4b8j8h0o` + release `1v8bmh9gevrbg`** both distributed. §6.Z: cold-start canary green on the S23 Ultra (operator-authorized basis); full C00/C01 instrumentation BLOCKED by the documented API-36 Espresso `@SdkSuppress(maxSdkVersion=35)` + wedged API-35 emulator (documented gap, NOT a bluff). - Also landed: **§6.AH** rule (emulators/VDs in Containers/VMs only — never host-direct; memory + constitution); §11.4.109 anti-forgetting (4-upstream `1d9e5d6`); Containers boot-reap+ADB-hygiene (`f10a011`); docs_chain Upstreams+parity (`c606978e`); Kaspersky root-caused (broke large Firebase uploads + the emulator). Pins: `constitution→1d9e5d6`, `containers→f10a011`. - **2026-06-03 commit-push session (this session):** landed the §6.AH-debt per-OS procWalker WIP (OWED#3 below) in Containers `6f4f415` (pushed, github+gitlab converged) + the

C02 api-app test-fix (OWED#1 below). Containers pin advanced `f10a011→6f4f415`. `constitution` pin held at `1d9e5d6` (no work; 1 behind upstream `6da1171` by deliberate freeze). - **OWED (resume here tomorrow):** 1. **C02 test-fix commit** (`api-app/.../Challenge02ApiAppBootAndServeTest.kt` short-read gate-verdict + `OnDeviceApiClient.kt` `getWithKeyShortRead`) — **COMMITTED + PUSHED 2026-06-03** with a §6.J Bluff-Audit stamp whose on-device rehearsal is marked **PENDING** (the bogus-key → assert(c) fast-401 fail → revert rehearsal still cannot run: no API≤35 emulator boots on this macOS host per §6.X/§6.AH-debt, and the S23 Ultra keyguard blocks the operator path). STILL OWED: execute that rehearsal on a real surface — unlock the S23 Ultra (R5CW33CBVQV) + keep awake (Settings→Dev options→Stay awake) → run the bogus-key rehearsal (full suite so C01 warms compose first) → confirm C02 fails at the assertNotEquals(401) → revert → record the observed failure in a follow-up §6.J note. 2. **Submodule cross-mirror divergence — CLOSED 2026-06-03.** The diagnosis was refined (§11.4.6): after a fresh fetch, NO submodule had unpushed local commits (all `0`-ahead); the only real divergence was `gitlab`-behind-`github` where `gitlab/main` was a strict ANCESTOR of `github/main` (8 submodules: auth/cache/config/discovery/http3/observability/ratelimiter/security). Resolved by **fast-forwarding `gitlab→github` tip — NO force-push, NO merge, NO commit loss** (per new constitution §11.4.113). Every submodule's github==gitlab==local now. `constitution` advanced to `d90ab87` (§11.4.113, all 4 upstreams). `helixqa` FF→`5112906` (origin, always-track-upstream). `tracker_sdk` already at origin/main. Parent pins bumped accordingly. **Build-verification of advanced pins against the Lava build is OWED before any release/distribute.** 3. **§6.AH-debt:** the partial WIP is now **LANDED + PUSHED** in Containers `6f4f415` (per-OS procWalker `cleanup_os.go` — Linux `/proc` + macOS `ps -A`; conditional `--device /dev/kvm` via `kvmAvailable()`; build-break fix in `cleanup.go`; new tests `cleanup_os_test.go` + `containerized_kvm_test.go` + updated `containerized_test.go`; the `emulator-matrix` build binary CONST-053-gitignored, NOT committed). STILL OWED (the actual debt): a no-KVM/TCG **containerized** emulator that BOOTS on macOS — the conditional-KVM code now lets the args omit `/dev/kvm`, but the §6.AG/§6.AH container/VM path still does not boot on this host (host-direct adbd-offline wedge; the real S23 Ultra remains the current §6.Z surface). 4. **§6.AH client gate:** full C00/C01 instrumentation on an API≤35 surface (the container emulator OR an API≤35 device OR a Compose-BOM update lifting the API-36 `@SdkSuppress`); 1.2.36 shipped on the cold-start-canary basis. 5. **§11.4.109** anti-forgetting clause still carries the `UNCONFIRMED:` marker — replace with the operator's verbatim quote + re-push 4 upstreams + re-bump the constitution pin. 6. **item 4 docs_chain submodule add:** apply the 5 governance patches upstream (`docs/superpowers/drafts/2026-06-03-docs-chain-add-orchestrator-steps.md`) → new SHA → `git submodule add submodules/docs_chain`.

PUSH STATE at wrap-up (2026-06-03 EOD): main repo `c786db7d` pushed (github+gitlab); constitution `1d9e5d6` pushed (4 upstreams). The C02 test fix is uncommitted (owed #1). The pre-existing submodule divergence (owed #2) is the only "not fully pushed" item and is a careful-reconciliation task, not this session's work.

▶ NEXT FRESH SESSION — RESUME HERE: - api-app launcher icon = client's adaptive icon (logo + bg hint); debug label "Lava API DEV"; §6.Y 0.1.0-1→**0.1.1-2**; **0.1.1-2 DEBUG distributed** (`14viqtk1mc8fg`). **AWAITS operator launcher-icon visual check of the Firebase debug build.** - **api-app 0.1.1 RELEASE HELD**: the R8-minified release APK (icon resource-renamed to `res/BW.xml`) needs a cold-start canary; emulator won't boot (adbd-offline). After the **Containers `f10a011` runner fix** (reap-on-timeout + `ResetADBHygiene` phantom-drop) the gate may now boot — RETRY: `./scripts/run-api-app-challenge-matrix.sh --no-build --boot-timeout 15m` → if green, run the 0.1.1 release canary (`/tmp/api-app-release-canary-011.sh` pattern) → `./scripts/firebase-distribute.sh --app api-app --release-only`. If still offline → operator: Kaspersky exclusion for `~/.android`+SDK+build dirs, or recreate Pixel_8 AVD, or a Linux x86_64+KVM / quiet host. - **client 1.2.36-1056 HELD**: same emulator dependency (C00/C01 gate). Same unblock. - **item 3 §11.4.109** (was mis-numbered 107; 107/108 taken upstream): clause LANDED UNCONFIRMED at constitution `1d9e5d6` (parent pin bumped). When operator pastes the verbatim anti-forgetting quote, replace the `UNCONFIRMED:` marker in `constitution/Constitution.md` §11.4.109 + re-push 4 upstreams + re-bump pin. - **item 4 docs_chain**: repo at parity-ish (`c606978e`, both mirrors). Per `docs/superpowers/drafts/2026-06-03-docs-chain-add-orchestrator-steps.md`, 5 governance-heading gates (§6.R/§6.S/§6.X in CLAUDE/AGENTS/CONSTITUTION) need upstream Patches A/B/C → new SHA → THEN `git submodule add submodules/docs_chain`. - **api-app 0.1.0-1 Stage-1 DEBUG distributed** (Firebase release `0aksve4ung948`) on transferred §6.Z evidence (item-1 green ×2 @11:35, source byte-unchanged — proof in `.lava-ci-evidence/distribute-changelog/firebase-app-distribution-api-app/0.1.0-1-test-evidence.md`). **AWAITING operator on-device check of the Firebase debug build → then `./scripts/firebase-distribute.sh --app api-app --release-only` (Stage-2).** - **client 1.2.36-1056 BLOCKED**: §6.Z C00/C01 gate could not boot the Pixel_8/API35 emulator across 5 attempts (host emulator-boot wedge that developed mid-session; booted fine @11:35; §6.M forensics + 5 hypotheses eliminated in

.lava-ci-evidence/sixth-law-incidents/2026-06-02-emulator-boot-wedge.json). **OPERATOR ACTION:** reboot the gate host (or use a quieter/fresh host), then run ./scripts/run-challenge-matrix.sh --boot-timeout 15m --avds Pixel_8:35:phone --test-class lava.app.challenges.Challenge00CrashSurvivalTest,lava.app.challenges.Challenge01AppLaunchAndTrackerSelectionTest → if green, ./scripts/firebase-distribute.sh --app client --debug-only then (after on-device check) --release-only. - Then item 3 (§11.4.107 draft docs/superpowers/drafts/2026-06-02-const-11.4.107-anti-forgetting-anchor.md — land UNCONFIRMED: per §11.4.6 unless operator pastes the verbatim quote), item 4 (docs_chain plan docs/superpowers/drafts/2026-06-02-docs-chain-incorporation-plan.md — needs operator glab repo create vasic-digital/docs_chain), item 5 (operator-side hook). All operator decisions recorded below + memory lva-fresh-session-handoff.

FRESH SESSION PROGRESS (2026-06-02, in flight — NOT yet committed/pushed): - **docs_chain URL received** from operator: git@github.com:vasic-digital/docs_chain.git (was the §11.4.6 blocker for PENDING item 4). HEAD 02eb81be, branch main. Operator chose **full-cascade-up-front** incorporation (author the complete CONST-* cascade + helix-deps.yaml + pointer-blocks into docs_chain upstream FIRST, then add clean). Saved to memory docs-chain-repo-url. - **4 parallel subagent streams dispatched + harvested (§11.4.70):** A=Containers --gradle-module flag; B=constitution §11.4.107 anti-forgetting anchor draft (ready); C=docs_chain probe (done); D=Firebase wiring design (done). - **PENDING item 1 — Containers --gradle-module flag + 3 on-device defect fixes: DONE (gate GREEN ×2, deterministic).** Stream A's generic flag is rebased onto latest Containers main + pushed + converged github+gitlab at 8090a97 (parent pin bumped). The flag made the never-before-run :api-app C01–C04 Challenges EXECUTE for real (no more 0-test false-green) and surfaced **3 latent product defects** (textbook §6.J/§6.L payoff): **(C02)** NsdMdnsAdvertiser Kotlin apply{} receiver-shadow → empty mDNS service name → API35 NsdManager crash; **(C03)** cross-test native-engine pollution (Go current process-global vs per-test Hilt @Singleton); **(C04)** notification restart-after-stop — restart() bailed when already-stopped + the Service collector self-destructed on the initial Stopped. Plus a harness-isolation flaw (stale foreground Service intercepting later tests) + an emulator HTTP-timeout flake. ALL fixed + falsifiability-rehearsed via the gate; full §6.T.4 entry in docs/issues/fixed/BUGFIXES.md. **§6.Z proof: C01–C04 EXECUTED green on cold-booted Pixel_8/API35 via the Containers runner (host-direct+HVF, gating=true) on TWO consecutive runs** (.lava-ci-evidence/phase-e-api-app/2026-06-02T11-32-48Z-gate/ + ...11-35-02Z-gate/). Added Go same-port-restart coverage (lava-api-go/internal/mobile/restart_repro_test.go, proved the embed innocent). - **PENDING item 2 — Firebase (NOW UNBLOCKED — item 1 gate is GREEN): operator chose ONLY the 2 APK apps this cycle** (:app client + :api-app, debug+release each = 4 uploads); lava-api-go deferred (it is a container-registry artifact, NOT a Firebase APK). **The agent runs firebase apps:create itself** (operator confirmed LAVA_FIREBASE_TOKEN is exported via .env + .zshrc). Stream D's design (full report in the 2026-06-02 session): add an --app client|api-app selector to scripts/firebase-distribute.sh (default client, per-app resolution table — GRADLE_VERSION_FILE/CHANNEL_SUBDIR/RELEASE_BASE/FB_APP_ID_*, Phase-1 auth Gates 4+5 applied for client / skipped for api-app), add 2 new .env/.env.example keys LAVA_FIREBASE_API_APP_ID + LAVA_FIREBASE_API_APP_DEV_APP_ID (placeholders only, §6.R), firebase apps:create ANDROID "Lava API (release|debug)" --package-name digital.vasic.lava.api[.dev] --project $LAVA_FIREBASE_PROJECT_ID. §6.Y bumps first (api-app first-distribute uses 0.1.0/1 as-is; client holds unless re-distributed), §6.P CHANGELOG entry per app, §6.AA two-stage debug→release, §6.Z evidence per app+variant at .lava-ci-evidence/distribute-changelog/firebase-app-distribution[-api-app]/<ver>-<code>-test-evidence.md (the api-app variant needs build_and_release.sh to produce releases/api-app/<ver>/android-{debug,release}/*.apk — the --gradle-module flag now makes that possible). - **PENDING item 3 — constitution §11.4.107:** Stream B draft ready (next free anchor confirmed; 4 upstreams confirmed; constitution is 2 commits behind upstream dd1f779 → must pull --ff-only + §11.4.32 sweep before landing). **WAITING on operator's verbatim anti-forgetting mandate quote** (operator will paste; until then it lands UNCONFIRMED: per §11.4.6). Plan: lift the guard script to constitution/scripts/ as an inherited-by-reference impl (§11.4.80 precedent). - **PENDING item 4 — docs_chain:** unblocked (URL in hand); full-cascade route chosen (see above). - **PENDING item 5 — hostile crowdstrike-falcon-foundry hook:** operator-side; this repo must not edit ~/.claude global config (docs/AGENT_GUARDRAILS.md). Reminder stands.

Prior handoff (pre-fresh-session): 2026-06-02 (LVA on-device-API session — HANDOFF to a fresh session for the remaining big programs; operator chose fresh-session for context room).

On-device Lava API — DONE this session (all merged to master, HEAD 672910d6): - **Phase A** — additive lava-api-go SQLite storage backend (Postgres default untouched), parity-tested, GC/WAL-hardened. ae7697ff→4d597bd9. SPEC+QUALITY reviewed. - **Phase B —** in-process embed serving the FULL production router over TLS on `0.0.0.0`, host-parity Lava-Auth gate (HMAC), c-shared `liblavaapi.so` ×3 ABIs + JNI bridge (`object LavaNative`, pkg `digital.vasic.lava.apigo`). `761b5204/816e4df3/6f751495/ac781ce9`. Reviewed. - **Phase C —** `:core:apiengine` JNI wrapper (`ApiEngine/ApiConfig/ApiStatus/NativeApiEngine/FakeApiEngine`); `assembleDebug` packages both `.so` ×3 ABIs. `ef109760`. - **Phase D —** `:api-app` (`digital.vasic.lava.api/.dev`, shared signing): foreground `ApiEngineService`, `ApiEngineController` state machine, `NsdMdnsAdvertiser` (`_lava-api._tcp`, TXT `engine=go,platform=android,storage=sqlite`), EncryptedSharedPreferences `ApiKeyStore`, Compose landing screen + `ApiControlViewModel` + notification. `e62e0fa8` + `2977fe97` (merge `13039ddc`). - **Phase E —** on-device Challenges C01–C04 (`5fa7836e`, merge `26df81e0`). Real Pixel_8/API35 run **caught two real defects every JVM test missed**: (1) `:core:apiengine` lacked `lava.kotlin.serialization` → fixed `4fc9c213`; (2) c-shared `.so` had no `DT_SONAME` → on-device `dlopen` of the abs host path → fixed `1e8ebc15` (merge `9b6dcabf`, ELF-proven). - **Sub-project 2 (client) —** discovery parses+labels `platform=android` instances, onboarding ApiSelection label, Settings “Run the API on this device” install/launch-or-download (configurable `BuildConfig` URL). `199f1404` (merge `7fce7cf9`). - **Docs —** `docs/ON_DEVICE_API.md` (4 Mermaid diagrams) + user guide + ARCHITECTURE/ LOCAL_NETWORK_DISCOVERY/README/AGENTS. `816d983f/997e3114`. - **Anti-forgetting enforcement (universal) —** PreToolUse guard hook (`scripts/hooks/guard-forbidden-commands.sh` + `.claude/settings.json`, 28/28 tests) blocking raw emulator/adb + force-push + sudo + host-power; `docs/AGENT_GUARDRAILS.md` (subagent preamble + orchestrator checklist); §6.X gate in `check-constitution.sh` (+`check-emulator-runner-tag.sh`, 7/7 paired-mutation). `34ce4599` (merge `672910d6`). Memory file: `emulators-via-containers-submodule`.

PENDING — for the FRESH session, in order: 1. **Containers submodule `--gradle-module` flag** in `submodules/containers/cmd/emulator-matrix` (it hardwires `:app:connectedDebugAndroidTest`; running `:api-app` classes against `:app` = 0-test false-green, which Stream A correctly refused). `scripts/run-api-app-challenge-matrix.sh` already forwards `--gradle-module/LAVA_GRADLE_MODULE`. Then run C01–C04 GREEN via the Containers gate (`runner: containers-submodule`) — both root defects are fixed so they’re expected to pass; this run is the §6.Z gate evidence. 2. **Firebase distribution** of 3 apps × 2 variants (Client, on-device API app, existing API service) via `firebase-distribute.sh` two-stage (§6.AA), SAME keystore. The new `:api-app` needs Firebase apps registered via `firebase` CLI (CLI 14.17.0 ✓, `LAVA_FIREBASE_TOKEN`/keystores ✓) + `.env` app-id entries + `firebase-distribute.sh` wiring. §6.Z-gated on #1. §6.Y version bumps + §6.P CHANGELOG first. 3. **Constitution-submodule extension —** port the UNIVERSAL anti-forgetting guardrails up into `constitution/` per the operator's 10-step incorporation prompt; push to all 4 upstreams (github/gitlab/gitflic/gitverse — §6.AD.1 carve-out); propagate inheritance; notify operator to update other projects. §9 hardlink `.git` backup first; follow CONST-049. 4. **`docs_chain` submodule incorporation** — BLOCKED: needs the repo URL from the operator (§11.4.6, do NOT guess). 5. Disable the hostile `crowdstrike-falcon-foundry` PreToolUse:Skill plugin hook (user/plugin config; documented in `docs/AGENT_GUARDRAILS.md`).

Spec `docs/superpowers/specs/2026-06-02-lava-api-android-app-design.md`; plan `docs/superpowers/plans/2026-06-02-lava-api-android-app.md`. **RESUME PROMPT:** "Resume the LVA on-device-API work from `docs/CONTINUATION.md` §0 PENDING list, starting at item 1 (Containers `--gradle-module` flag → green §6.Z gate run for :api-app), using `docs/AGENT_GUARDRAILS.md` for every subagent dispatch. Provide the docs_chain repo URL for item 4."

Prior (2026-06-01, §6.L 69th cont.): "Choose your API" two-section feature SHIPPED to Firebase debug, §6.Z GENUINELY GREEN. **Android 1.2.34-1054 → 1.2.35-1055** (lava-api-go unchanged 2.3.23-2323; Android-only feature). **Feature:** onboarding "Choose your API" now has two sections — existing "On your network" mDNS list + NEW "Cloud / remote server" (manual address+port "Add server" + pre-installed default `https://lava.app:7777`, §6.R-sourced from `.env→DEFAULT_CLOUD_API`). Commits 26ee4433 (Stream A) + d33afc6e (tests) + 8d33f9c2 (ApiSelectionStep param-defaults fix + GENUINE §6.Z green). **§6.Z device gate GENUINELY GREEN** (verified by reading raw attestation JSONs, not the garbling channel): C00+C01+C26+C30 EXECUTED on cold-booted Pixel_8/API35 (Containers host-direct+HVF, §6.X/§6.AG) = all_passed:true / test_passed:true / 0 failures (run 00:55:55→01:00:54; attestations `.lava-ci-evidence/2026-05-31-1.2.35-1055-challenge-matrix/{c00,c01,c26,c30}/`; §6.Z evidence `.../distribute-changelog/firebase-app-distribution/1.2.35-1055-test-evidence.md`). **§6.AA BOTH stages distributed:** Stage 1 DEBUG (firebase-distribute –debug-only EXIT 0, release `6cp8l7g5i5gtg`, last-version-debug=1055) + Stage 2 RELEASE (–release-only EXIT 0, release `53rlfe9a467t0`, last-version-release=1055). Release-variant cold-start canary on the R8-minified release APK = PASS (boot_completed=1, install_rc=0, fatal_count=0, MainActivity resumed; evidence `.lava-ci-evidence/2026-05-31-1.2.35-1055-challenge-matrix/release-canary/`). 1.2.35-1055 FULLY SHIPPED (debug + release). **§6.J honesty (3 premature claims this cycle, ALL corrected forward, no force-push):** (1) unit "0 failures” while 1 VM test failed on a Turbine timeout (fixed d33afc6e); (2) first §6.Z run failed on a missing androidTest APK (assemble skipped the task); (3) commit e039656d falsely claimed “GENUINELY GREEN” while the re-run still failed (Challenge26 broke when 5 cloud params were added without defaults; my defaults edit had not persisted via the degraded channel) — re-applied + verified + re-ran green in 8d33f9c2, which supersedes e039656d’s evidence. Bash channel degraded throughout (dropped edits, garbled output) — mitigated via Read-tool ground-truth + file-routed reads. §6.L counter remains 69. HEAD 8d33f9c2 converged github+gitlab.

Prior (§6.L 69th, 2026-05-31): HONEST status after a degraded-Bash-channel session that produced several premature claims, now corrected. **VERIFIED:** T7 podman-VM relocation (`/Volumes/T7/containers`, host freed ~50 GB; `docs/ops/T7-fast-storage.md`); APKs 1.2.34-1054 debug+release rebuilt with rotated pepper (aapt2-confirmed); lava-api-go 2.3.23-2323 binary; `testDebugUnitTest` SUCCESSFUL; §6.Z DEBUG Challenge retest C00+C01 EXECUTED green on Pixel_8/API35 (attestations under `.lava-ci-evidence/2026-05-31-1.2.34-1054-challenge-matrix-repepper/`); §6.AA Stage 1 DEBUG distribute DONE (Firebase release `0f9a72d53suhg`, last-version-debug=1054); release-variant cold-start canary RESULT=PASS; `.containerignore` (NEW) excludes the real build-context bloat (4.5 GB `.git-backup*` + `releases/`). **OWED / BLOCKED (honest):** (1) lava-api-go OCI image + compose boot NOT done — no image currently exists, health probe rc=7; the “API running” item is OWED (image not needed for Firebase APK distribute). (2) §6.AA Stage 2 RELEASE distribute BLOCKED — `firebase-distribute --release-only` exits 1 at Phase-1 Gate 4 (pepper-reuse): debug+release of the SAME versionCode share one pepper, but Gate 4 rejects the reused SHA. last-version-release still 1053; release NOT distributed. Genuine Gate-4/§6.AA interaction bug to fix. **§6.J forensic:** three intermediate claims this cycle (a health-probe body; an image EXIT-0; a completed release distribute) were recorded before verification under cancelled batches; the 3 unpushed fabrication commits were `reset --mixed`, and the pushed commits `79093ccf/20856092` are corrected forward here (§11.4.41, no force-push). §6.L counter remains 69.

Prior (§6.L 69th, earlier same day): rebuild + redistribute, “Firebase Distribution GREEN”. **§6.Y bump:** Android 1.2.33-1053 → **1.2.34-1054**, lava-api-go 2.3.22-2322 → **2.3.23-2323**. **Rebuilt + verified:** lava-api-go binary + healthprobe (`--version` = 2.3.23 build 2323); debug + androidTest + release APKs (aapt2-confirmed versionCode 1054). **§6.AF-debt PARTIAL CLOSE (stream-D unblock):** `tools/lava-containers/vm-images.json` gained an `android-35-`

`phone` entry whose URL parses to `tag=google_apis_playstore`, `abi=arm64-v8a`, `api=35` so the Containers `emulator-matrix` Branch-4 (`hasExtractedSystemImage`) short-circuits to the locally-installed image — the macOS host-direct+HVF Challenge matrix now BOOTS (the prior "no image with id android-35-phone" was a provisioning gap, NOT an app defect; §6.J forensic distinction recorded). **§6.Z EXECUTED green:** C00 (`Challenge00CrashSurvivalTest`, cold-start canary) + C01 (`Challenge01AppLaunchAndTrackerSelectionTest`) ran on a cold-booted Pixel_8/API35 via the Containers runner (runner=host-direct, accel=hvf, gating=true) — both `all_passed:true`, 0 failures (attestations under `.lava-ci-evidence/2026-05-31-1.2.34-1054-challenge-matrix/{c00,c01}/`; §6.Z evidence at `.lava-ci-evidence/distribute-changelog/firebase-app-distribution/1.2.34-1054-test-evidence.md`). **Unit suite:** `./gradlew testDebugUnitTest` BUILD SUCCESSFUL, exit 0. **Auth rotation (operator-authorized "rotate autonomously"):** fresh `LAVA_AUTH_OBFUSCATION_PEPPER` + `LAVA_AUTH_CURRENT_CLIENT_NAME=android-1.2.34-1054` + fresh random UUID appended to `LAVA_AUTH_ACTIVE_CLIENTS` (in gitignored `.env`; values never echoed) → firebase-distribute Phase-1 Gate 4/5 pass; APKs rebuilt with new pepper into `releases/1.2.34/android-{debug,release}/` (both versionCode 1054). **Distribute HONESTLY BLOCKED — NOT bluffed:** `build_and_release.sh` exited 125 ("no space left on device") building the lava-api-go OCI image — §6.M Class-II disk pressure in the **podman VM** (host `/` ≈97% / 5.4Gi-free; image NOT needed for Firebase). Remaining before the Firebase two-stage upload: (1) reclaim podman-VM disk (`podman machine reset`), (2) build the androidTest APK (`assembleDebugAndroidTest` — build_and_release skipped it), (3) §6.Z retest C00+C01 against the **rebuilt new-pepper** artifact (prior §6.Z green was on the pre-rotation 1054 build; pepper is an auth constant inert for cold-start/launch but §6.Z requires testing the exact artifact), (4) `firebase-distribute --debug-only` then `--release-only`. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-05-31-disk-pressure-podman-vm-image-build.json`. §6.L counter remains 69 (same cycle). Bash channel was degraded this session (a shell `sync` alias fired the push hook + flooded output; foreground long-sleeps spawned duplicate tasks) — mitigated via file-routed reads + dropping `sync`.

Last updated: 2026-05-31 (§6.L 68th "do it all" forward-debt closure). Four forward-debt streams driven in parallel (subagent + main): **(A) §11.4.65 universal markdown-export CLOSED** — `CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC` gate + `scripts/sync-markdown-exports.sh` + 126-doc `.html`/`.pdf` backfill + hermetic test (commit `2c8f1d46`). **(B) canonical `workable-items update/reopen/block` + PDF/HTML/DOCX export ADDED upstream** in the constitution submodule (`42ad8a3`, pushed + converged on all 4 HelixConstitution mirrors via CONST-049 FF push; GitVerse remote URL repaired); parent pin `883ccc1→42ad8a3`. **(C) §11.4.79 codegraph own-org indexing CLOSED (LVA-6)** — the prior “gitlink capability gap” diagnosis was WRONG (§11.4.6 correction): submodules were merely un-`init`-ed; after `git submodule init`, 1,842 submodule files indexed, cross-submodule probe PASS, §6.H 0-leak, step-5 mutation confirmed. **(D) §6.AE Compose Challenge on-device matrix** re-run via host-direct+HVF (§6.X darwin/arm64 path) — IN PROGRESS at time of this write. All Lava-parent commits converged on GitHub+GitLab. Still OWED (§6.AF-debt): §11.4.85 chaos/stress beyond phase-1 (LVA-7), per-anchor propagation gates, §11.4.80 submodule-init-before-index automation. §6.L counter remains 68 (same cycle).

Prior 2026-05-31 (§6.L 68th invocation): flaky-test fix + §6.S table re-sync + subagent-driven constitution/ticket/submodule cycle. Operator directive: fetch+review constitution submodule, add+incorporate any newly-mandated submodules (submodules-driven), define the tickets SQLite DB key **“LVA”** + Issues/Fixed/Issues_Summary/Fixed_Summary docs with PDF/HTML/DOCX exports, create many new tests of all types with REAL evidence (zero bluffs), keep anti-bluff mandate in all governance docs, commit+push all to all upstreams, endless autonomous loop. **Commit 1 (this commit):** (a) fixed the 67th-cycle flaky `CredentialsViewModelTest > select provider updates selectedProvider` — replaced the fixed-`awaitState()`-count assumption with a bounded await-until-`selectedProvider=="rutracker"` loop (the Room `Flow .first()` in `load()` resumes off the StandardTestDispatcher, so emission count is non-deterministic under load); FALSIFIABILITY-REHEARSED (broke `SelectProvider` reduce → `AssertionFailedError: expected:<rutracker> but was:<null>`, localized to that 1 test → reverted → green); (b) `.codegraph/*.pid` gitignore gap closed (`daemon.pid` was untracked-but-not-ignored — §11.4.30); (c) §0 orientation + §3 pin tables re-synced to HEAD `23c508e9` (they had drifted to `0c87b6ae`/CamelCase names/1.2.22 — a §6.S violation now corrected). **In flight (later commits this cycle):** constitution-pin review (CONST-049), missing-submodule incorporation, LVA ticket DB + export pipeline, new-test creation, full sweep. §6.L counter 67 → 68.

Last updated: 2026-05-20 (§6.L 67th invocation), **rebuild + test cycle — honest §6.Z/§6.X blocker on the Firebase redistribute.** Operator directive: rebuild all apps/services + boot + execute all tests/Challenges + Firebase redistribute. DONE: `lava-api-go` rebuilt (`bin/lava-api-go` + `bin/healthprobe`); Lava debug APK + androidTest APK rebuilt (BUILD SUCCESSFUL); JVM unit-test suite executed (`./gradlew testDebugUnitTest --continue`) — green except ONE flaky test discovered: `CredentialsViewModelTest > select provider updates selectedProvider` failed in-suite, passed 6/6 isolated (a §11.4.50 deterministic-consistency defect — root-cause hypothesis: Room `Flow` `.first()` resuming off the virtual test dispatcher; recorded honestly at `.lava-ci-evidence/sixth-law-incidents/2026-05-20-flaky-credentialsviewmodeltest.json`, fix owed as a focused follow-up, NOT masked). **§6.H credential incident:** a buggy `${LAVA_FIREBASE_TOKEN:-UNSET}` recon command printed the Firebase CI token into the session transcript (NOT committed to git) — incident at `.lava-ci-evidence/sixth-law-incidents/2026-05-20-firebase-token-echo-leak.json`; **operator MUST rotate the token** (`firebase logout` → `firebase login:ci`). **§6.X-debt darwin/arm64 sub-debt RESOLVED** — the operator directed extending the emulator for per-OS acceleration; DONE (Containers `c1871138`+`6aff7ea8`: per-OS-accel model `AccelProfileForOS`/`ResolveRunner`/`GateEligibleForOS` + `emulator-matrix --runner=auto`; `scripts/run-challenge-matrix.sh` OS-aware). macOS's accelerated gate runner is host-direct+HVF (a Linux container cannot reach the host-only HVF API). PROVEN: C00 cold-start canary + the full 37-class Challenge suite ran on Pixel_8/API35 = **43 pass / 3 credential-skip (C02/C09/C10) / 0 fail** — one stale test (`Challenge26RutrackerMainAbsentFromServerListTest`, waited for the SP-4-removed "Server" menu section) repaired + falsifiability-rehearsed. The §6.Z redistribute is now genuinely unblocked on this macOS host; the redistribute itself is pending an operator decision — the app APK is byte-identical to the already-distributed 1.2.33-1053 (no app-user-facing code changed this session). §6.L counter 66 → 67.

Last updated: 2026-05-20 (§6.L 66th invocation), **§6.N bluff-hunt — 2 more existing Lava tests verified genuine.** Continuing the §6.N cadence beyond the 65th: `core/domain/.../ProbeMirrorUseCaseTest` (UseCase layer) and `core/preferences/.../EndpointConverterTest` (converter layer) were hunted by mutating their production code — `ProbeMirrorUseCase`'s reachable range `200..399`→`200..599`, and `EndpointConverter`'s GoApi `fromJson` port forced to `DEFAULT_PORT`. Both produced localized test failures (1-of-3 and 2-of-10) and passed fully after `git checkout` revert. Verdict: both GENUINE; 0 bluffs. Across the 65th + 66th, 3 existing Lava tests (ViewModel / UseCase / converter layers) are §6.N-verified genuine. Evidence: `.lava-ci-evidence/bluff-hunt/2026-05-20-cycle66-usecase-converter.json`. §6.L counter 65 → 66.

Last updated: 2026-05-20 (§6.L 65th invocation), **§6.N bluff-hunt — an existing Lava test verified genuine.** Per the 65th §6.L wall ("all existing tests and Challenges MUST work anti-bluff"), a §6.N.1.1 incident-response bluff-hunt of an EXISTING Lava test (not the new codegraph suite):

> `feature/login/.../LoginViewModelTest.kt` was hunted by actually performing its documented mutation — `serviceUnavailable = null` removed from `LoginViewModel.validateUsername`. The test FAILED with the predicted `AssertionError`, localized to exactly the mutated code path (the other 2 tests, on untouched paths, passed); mutation reverted via `git checkout`; re-run `BUILD SUCCESSFUL` 3/3. Verdict: GENUINE — the test provably catches the production break it covers; 0 bluffs found. Evidence: `.lava-ci-evidence/bluff-hunt/2026-05-20-codegraph-cycle-loginviewmodel.json`. §6.L counter 64 → 65.
>
> **Last updated:** 2026-05-20 (§6.L 64th invocation), **codegraph backend regression caught by the anti-bluff suite + fixed.** A fresh `scripts/verify-codegraph.sh --quick` run FAILED — codegraph's native `better-sqlite3` binding had been disabled (`better-sqlite3.disabled` in the Homebrew-Node-Cellar global install), forcing a WASM fallback that could not open the index DB; plus a stray `codegraph serve --mcp` process from a prior run. Fixed: re-enabled native better-sqlite3, killed the stray process, rebuilt the index via `codegraph index` (1,182 files / 18,567 nodes — its §11.4.77 regeneration mechanism), re-verified `--quick` → 44 pass / 0 fail. Hardened the suite with a `codegraph status` pre-flight check that fails fast-and-clear on this backend-breakage class; `docs/CODEGRAPH.md` troubleshooting extended. Anti-bluff mandate propagation re-audited — present in 4/4 governance files across all 17 submodules + constitution + root + lava-api-go. This is the §6.L mandate working exactly as designed: the suite refused to bluff when codegraph was genuinely broken. §6.L counter 63 → 64.
>
> **Last updated:** 2026-05-20, **codegraph code-intelligence incorporated (operator directive; §6.L 63rd invocation).** codegraph (`@colbymchenry/codegraph` 0.6.8) installed globally via npm (no sudo — §6.U). The Lava domain codebase is indexed into a local SQLite semantic graph — 1,182 files (973 Kotlin + 196 Go), 18,567 nodes, 21,462 edges; `submodules/`, `constitution/`, `releases/` and all §6.H secret paths excluded from the index (verified: 0 submodule-file leak). The codegraph MCP server is wired into all 5 supported CLI agents: Claude Code (`.mcp.json`), OpenCode (`opencode.json`), Qwen Code (`.qwen/settings.json`), Crush (`.crush.json`), Kimi CLI (`~/.kimi/mcp.json`). Anti-bluff verification suite `scripts/verify-codegraph.sh` + `tests/codegraph/` (6 layers): layers 01-04 + 06 PASS — index reality, query correctness, MCP-protocol JSON-RPC, all-5-agent connectivity, and falsifiability (layers 01-03 provably FAIL when the DB is removed, PASS when restored). Layer 05 (LLM-driven E2E) — Claude Code PASS (reports the unforgeable index node count, obtainable only by calling the `codegraph_status` MCP tool); OpenCode / Kimi CLI / Crush SKIP (documented credential/quota gaps in

this environment — no Google API key / Kimi monthly quota exhausted / Venice.ai account has no credits — these are NOT codegraph defects; the integration itself is proven by layer 04). `docs/CODEGRAPH.md` written; design spec `docs/superpowers/specs/2026-05-20-codegraph-incorporation-design.md`. `QWEN.md` (Qwen Code instruction file — a plain-text pointer to `CLAUDE.md`, deliberately zero `@`-tokens so Qwen Code's import processor does not choke) created across repo root + 18 submodules + `lava-api-go`. constitution submodule fetched+pulled `9b52046 -> 2456605` (CONST-049 step 1). **§11.4.78 CodeGraph mandate — FULL ECOSYSTEM CASCADE COMPLETE.** §11.4.78 (CodeGraph code-intelligence mandate) authored into the constitution submodule: `Constitution.md` + mirrored into `CLAUDE.md` / `AGENTS.md` / `QWEN.md`; all 16 governance artefacts (.md + regenerated .html + .pdf + .docx) updated; constitution `2456605` → `208e2c8` pushed to all 4 upstreams (gitflic, github, gitlab, gitverse) per CONST-049. The §11.4.78 anchor cascaded into all 17 owned submodules' `CONSTITUTION.md` / `CLAUDE.md` / `AGENTS.md`, with `QWEN.md` created for each — every submodule committed + pushed to GitHub + GitLab (helixqa to its single GitHub upstream). §11.4.78 also appended to `lava-api-go`'s `CLAUDE.md` / `AGENTS.md` / `CONSTITUTION.md` / `QWEN.md`. All 18 submodule pins bumped in the parent. QWEN.md anchors are zero-`@` so Qwen Code's import processor does not choke.**

Last updated: 2026-05-18, **§6.AD-debt FULLY DRAINED in 1.2.30-1050 tooling cycle. All three originally-OWED CM-* items CLOSED this session: `CM-SCRIPT-DOCS-SYNC` (commit `11820734`), `CM-COMMIT-DOCS-EXISTS` (commit `977630c3`), `CM-SUBAGENT-DELEGATION-AUDIT` (commit `2a0e11f4`). Each ships with standalone scanner + 7-or-8-fixture hermetic test + companion `docs/scripts/*.sh.md` user guide + wrapper integration + Bluff-Audit falsifiability rehearsal. The 5 Path-B equivalence-mapped items remain CLOSED-BY-EQUIVALENCE per §6.AD.3. Plus: T7 USB disk migration today (5G → 108G free on main; 7 dirs symlinked to T7 incl. ~/.gradle, ~/.cache, ~/.android, Xcode; `~/.zshrc` updated with `GRADLE_USER_HOME=/Volumes/T7/Gradle`, `XDG_CACHE_HOME`, `NPM_CONFIG_CACHE`).**

Sweep tier-A closure (2026-05-17 evening, branch `sweep-findings-tier-A-2026-05-17`): 8 of the 10 comprehensive-sweep findings closed in a single coordinated commit (Findings #2 + #3 already closed by Bug 2 cascade + Bug 3 fix in prior cycles). All fixes falsifiability-rehearsed per §6.J / Seventh Law clause 1 (mutation applied → test fails with clear message → mutation reverted → test passes). Bluff-Audit stamps recorded in commit body. - **Finding #1 (P0) — ToggleAnonymous persistence**: `feature/provider_config/.../ProviderConfigViewModel.kt` now persists via new `ProviderConfigRepository.setUseAnonymous(...)` → Room column `use_anonymous` (Migration 10→11 + schema 11.json). Switch state survives process restart. - **Finding #4 (P1) — `LoginViewModel.serviceUnavailable` retype clear**: cleared in `validateUsername`/`validatePassword`/`validateCaptcha`/`onReloadCaptchaClick`/`onSubmitClick` reduces. - **Finding #5 (P1) — `LoginViewModel.ServiceUnavailable` stale-captcha clear**: branch now sets `captcha = null, captchaInput = Initial` so the rendered challenge image doesn't lie when the sid expires. - **Finding #6 (P1) — `ProviderLoginViewModel.serviceUnavailable` clear across selectProvider/backToProviders + retype**: symmetric fix to Findings #4/#5 on the multi-provider login surface. - **Finding #7 (P1) — `OnboardingViewModel.onTestAndContinue` no more misleading "Invalid credentials"**: distinguishes `loginResult == null` (tracker has no auth path) from `loginResult.state != Authenticated` (real auth failure). Null path now treated as anonymous → switchTracker + advance. - **Finding #8 (P1) — `OnboardingViewModel.loadProviders` excludes cloned synthetic trackers**: filters by syntheticId membership in `cloned_provider`. Clones remain configurable via Provider Config. - **Finding #9 (P2) — `MainActivity` onboardingComplete re-read**: `PreferencesStorage.observeOnboardingComplete()` new Flow API (SharedPreferences-listener-backed). Two parallel `lifecycleScope.launch { repeatOnLifecycle { collect } }` blocks (one for theme, one for onboarding). Welcome screen re-appears if settings flip onboardingComplete back to false at runtime. - **Finding #10 (P2) — `ToggleSync` first-tap race**: reads `toggleDao.get(providerId)?.enabled` synchronously instead of `state.syncEnabled`. First tap before observeAll() emit no longer silently flips the wrong direction. - **Tests added** (5 new): `LoginViewModelTest` (3 cases: Findings #4 username, #4 password, #5 captcha), `ProviderConfigViewModelTest` (2 cases: Findings #1 + #10), Finding #7 + #8 cases added to existing `OnboardingViewModelTest`, Finding #6 case added to existing `ProviderLoginViewModelTest`. - **Schema migration**: Room version bumped 10 → 11 (`MIGRATION_10_11` adds `use_anonymous INTEGER NOT NULL DEFAULT 0` to `provider_configs`). `core/database/schemas/lava.database.AppDatabase/11.json` exported by KSP. - **All builds + tests green**: `:core:database` / `:core:credentials` / `:feature:provider_config` /`:feature:

1.2.25-1045 distribute cycle (2026-05-17 afternoon): - Stage-1 debug Firebase release ID `1lfjqc1nnhuio` on `digital.vasic.lava.client.dev` - Stage-2 release Firebase release ID `7p0h5j70eckqg` on `digital.vasic.lava.client` (production) - 15/15 Compose UI Challenge test cases PASS on Pixel_8/API35 host-direct AVD (14 classes including new

C36) - HelixConstitution submodule advanced to `ca7c7d7` (§11.4.10.A Pre-store credential leak audit + upstream §11.4.37/38/39) — 5-mirror converged - §6.L counter 57 → 58

Comprehensive UI/UX/core sweep findings (2026-05-17 evening, branch `comprehensive-sweep-2026-05-17` merged at `c3b8bf5c`): 10 findings @ `docs/sweeps/2026-05-17-comprehensive-uiux-core-sweep.md`. Top P0s for 1.2.26: 1. `ProviderConfigViewModel.ToggleAnonymous` never persists — anonymous switch reverts on restart (`feature/provider_config/.../ProviderConfigViewModel.kt:82-84`) 2. `SearchResultContent` has no `Error` variant — Bug 2 root cause CONFIRMED ("all providers failed" looks like "0 results") (`feature/search_result/.../SearchPageState.kt:30-54`) 3. `SearchInputViewModel.availableProviders` is hardcoded 4-element list — SDK clones + new trackers invisible to search (`feature/search_input/.../SearchInputViewModel.kt:48-53`) P1 cluster (4 findings on login banner/captcha staleness + onboarding misleading cred message + clones-in-onboarding) + P2 cluster (MainActivity onboarding re-read + ToggleSync race). See sweep doc for full details + per-finding anti-bluff classification.

OPERATOR ACTION still REQUIRED: - §6.H historical credential leak: rotate the RuTracker password (credentials remain valid until rotated; in-tree redaction does NOT purge git history per §6.T.3) - Bug 2 live-device log capture: install 1.2.25, attempt anonymous-only search, `adb logcat | grep -iE "error|exception|search"` to confirm sweep finding #2 root cause

Bug 1 FULL REFACTOR LANDED — `AuthResponseDto.ServiceUnavailable` sealed variant propagates through 8-layer chain; Challenge C36 + 3 unit-layer falsifiability-rehearsed tests; OpenAPI spec + Go bindings regenerated; debug APK builds clean; feature/login + core/tracker/rutracker tests green

Bug 1 cycle (2026-05-17): - Operator's §6.L 57th invocation forensic anchor "Cant login to RuTracker with valid credentials" closed. Partial fix in commit `17ceabcb` (stderr marker line) is now superseded by the full-fix variant: the SDK catch path returns `AuthResponseDto.ServiceUnavailable(reason)` instead of bluffing `WrongCredits(null)`. - **8-layer propagation chain**: `AuthResponseDto` (+ variant) → `AuthMapper` → `AuthState.ServiceUnavailable(reason)` → `RuTrackerDtoMappers` (reverse) → `AuthServiceImpl` → `AuthResult.ServiceUnavailable` → `LoginUseCase` → `LoginResultMapper` → `ProviderLoginViewModel` (recordWarning telemetry per §6.AC + state field) → `ProviderLoginState.serviceUnavailable` → `ProviderLoginScreen` (renders "Service unavailable. Please try again later.

(reason)" banner with `R.string.provider_login_service_unavailable` + `ServiceUnavailableTextTestTag`). - Same wiring landed in legacy `LoginViewModel` + `LoginState.serviceUnavailable` for the single-tracker path. - **OpenAPI spec**: `lava-api-go/api/openapi.yaml` gains `AuthResponseDtoServiceUnavailable` under `oneOf` + discriminator mapping. Go bindings regenerated via `scripts/generate.sh`; both `internal/gen/server/api.gen.go` + `internal/gen/client/api.gen.go` updated. Note: the Go-side rutracker scraper does NOT emit the new variant today — its login handler still returns Success / WrongCredits / CaptchaRequired only; the parity test holds. The variant exists in the spec so the Android Kotlin SDK wire-shape stays compatible with future Go-side adoption. - **Tests added** (3 new + 1 rewritten): - `core/tracker/rutracker/src/test/kotlin/lava/tracker/rutracker/mapper/AuthMapperTest.kt` — added 2 tests for ServiceUnavailable forward mapping (with + without captcha). - `core/tracker/rutracker/src/test/kotlin/lava/tracker/rutracker/mapper/RuTrackerDtoMappersTest.kt` — added round-trip test asserting reverse-mapper preserves reason. - `core/tracker/rutracker/src/test/kotlin/lava/tracker/rutracker/impl/RuTrackerNetworkApiLoginUnknownRegressionTest.kt` — REWRITTEN: pre-fix asserted WrongCredits fallback (the §6.J bluff); now asserts ServiceUnavailable with reason carrying throwable class name. Same Crashlytics `a29412cf6566d0a71b06df416610be57` regression-immunity coverage, stronger discrimination. - `feature/login/src/test/kotlin/lava/login/LoginResultMapperTest.kt` — NEW file. 6 tests; load-bearing assertion: `service unavailable propagates as ServiceUnavailable with reason` + explicit anti-bluff `service unavailable does NOT silently collapse to WrongCredits`. - `feature/login/src/test/kotlin/lava/login/ProviderLoginViewModelTest.kt` — added VM test: `service unavailable shows banner does NOT mark creds Invalid does NOT signal authorized` exercises the §6.J anti-bluff contract end-to-end at the VM layer (real ViewModel + real ProviderCredentialManager + real Room + real LavaTrackerSdk wired with FakeTrackerClient whose `loginProvider` returns `AuthState.ServiceUnavailable(reason)`). - `app/src/androidTest/kotlin/lava/app/challenges/Challenge36LoginServiceUnavailableShowsAccurateMessageTest.kt` — NEW Challenge Test under `// covers-feature: login`. Source-written + compiles green on darwin/arm64 (verified via `./gradlew :app:compileDebugAndroidTestKotlin`); EXECUTION against the §6.X-mounted gate-host is OWED. - **Build verification**: `./gradlew :core:tracker:rutracker:test` PASS, `./gradlew :feature:login:test` PASS, `./gradlew :app:assembleDebug` PASS, `./gradlew :app:compileDebugAndroidTestKotlin` PASS. Spotless applied to all edited files. - **§6.Y note**: this is a follow-up commit to 1.2.24-1044 — versionCode is NOT bumped here;

the next distribute (1.2.25-1045) will pick up Bug 1 full fix + Bug 2 deferred investigation closure once that lands. - **§6.X-debt**: C36 + the per-AVD matrix attestation row is OWED to a Linux x86_64 + KVM gate-host; the JVM-layer falsifiability rehearsals (4 tests) carry the gate until then.

Previous cycle context (preserved verbatim):

Last updated: 2026-05-16, **Phase 4-C-2 (pkg/detector adapter) + Phase 4-C-3 (pkg/ticket adapter) — BOTH WORKTREES LANDED + green; parent-cycle close + HelixQA SHA bump + coverage-ledger regen owed at meta-merge** (constitutional-plumbing-only; no user-visible feature change; no Firebase distribute since 1.2.22-1042 still serves the user-visible surface). Cycle spans commits `4def2da7` → `0c87b6ae` (33 commits since plan landing `832f739e`). Final state at HEAD `0c87b6ae`:

Major deliverables this cycle: - The 12-clause constitution-compliance plan (`docs/plans/2026-05-15-constitution-compliance.md`) executed end-to-end across 10 phases. Plan + every phase's deliverable: see commit log between `832f739e..0c87b6ae`. - HelixQA submodule incorporated (`submodules/helixqa` at upstream `b13ba7c0`) — see Phase 4 below. - 40-gate verify-all sweep wrapper at full STRICT mode after Phase 7's coverage-ledger STRICT-flip. - 17 own-org submodules now have helix-deps.yaml + install_upstreams.sh (16 vasic-digital + 1 HelixDevelopment HelixQA); 0 waivers in STRICT mode. - §6.L counter advanced 36 → 52 across 17 back-to-back restatements (longest sequence in project history); 53rd in-flight per the dispatch that triggered this CONTINUATION.md refresh task.

Phase-by-phase status (constitution-compliance plan): - **Phase 1** (§11.4.32 enforcement engine) — `4def2da7`. `scripts/verify-all-constitution-rules.sh` + meta-test + ci.sh wiring. - **Phase 2** (§11.4.30 .gitignore audit gate) — `037389f5`. `scripts/check-gitignore-coverage.sh` + 16 new .gitignore files + sweep wiring + hermetic test. - **Phase 3** (§11.4.31 helix-deps.yaml manifest gate) — `43345c3e` (gate) + `410af7ec` + `bcba3a19` (16/16 per-submodule manifests landed + Auth pin bump). - **Phase 3-debt** CLOSED — all 16 vasic-digital submodules at pin advance with helix-deps.yaml present. - **Phase 4** (§11.4.27 HelixQA + 100% test-type coverage) — `aa0db6bd`. HelixQA adopted as `submodules/helixqa` at upstream HEAD; `HELIX_DEV_OWNED` exemption pattern added to mirror-mandate scanners. - **Phase 4 follow-up A** (Option 1 design) — `a61bd3d8`. Integration design at `docs/plans/2026-05-16-helixqa-integration-design.md` (Option 1 shell-wiring recommended; Options 2/3 deferred). - **Phase 4 follow-up A executed**

(shell-level wiring) — `1b66d192` + merge `d94ade0d`. 11 HelixQA Challenge scripts wrapped via `scripts/run-helixqa-challenges.sh` + `scripts/run-challenge-matrix.sh --include-helixqa` opt-in flag. - **Phase 4 follow-up B** (4 open-question resolutions for Option 1) — `281780d7` + merge `84d871a5`. Runner-mode flag (`--runner=host|containerized`), toolchain-precondition gate (`HELIXQA_TOOLCHAIN_MAP`), evidence-dir env-var override, `HELIXQA_W_EXCLUSIONS` array consuming the §6.W audit doc. 11 hermetic fixtures in `tests/check-constitution/test_helixqa_wiring.sh`. - **Phase 4 follow-up C** (DESIGN-ONLY) — `41b81359` + merge `be1ca3d8`. HelixQA Go-package linking design at `docs/plans/2026-05-16-helixqa-go-package-linking-design.md` (770-line Option 2 proposal: per-package adapters + 4-cycle rollout 4-C-1 `pkg/evidence` → 4-C-4 `pkg/navigator+pkg/validator`). **Operator-blocked on 10 open questions** (§G of the design); implementation cycle NOT started. - **Phase 4-debt** CLOSED — `858ffb3e` (2026-05-16). HelixQA upstream PR `b13ba7c` landed `helix-deps.yaml` + `install_upstreams.sh` at the HelixQA repo root → Lava parent removed HelixQA from `HELIX_DEPS_WAIVERS` + `INSTALL_UPSTREAMS_WAIVERS`. 17/17 own-org submodules satisfy §11.4.31 + §11.4.35 + §11.4.36 in fully STRICT mode with **zero waivers**. - **Phase 5** (§11.4.28 nested own-org submodule audit) — `bbca3a78` (gate) + `410af7ec` (STRICT flip after Challenges/.gitmodules removal via the github cascade merge). - **Phase 5-debt** CLOSED — `410af7ec`. Scanner reports 0 violations in STRICT mode; Panoptic is no longer nested via Challenges. - **Phase 6** (§11.4.29 lowercase snake_case naming) — `322f2081` (plan landing) + Phase 6a + 6b execution this cycle. Operator's 8 Q answers: defer Phase 6f upstream rename (Lava-side only); `helixqa` (single-token); `http3` (single-token); `ratelimiter` (single-token); Go `cmd/` hyphens exempt; ordering Mdns(low) → Containers(high) last; same defer for Tracker-SDK upstream; Phase 6a authorized to run in parallel with Phase 4-C-1. Execution: `Submodules/` → `submodules/` parent rename + 17 child renames (Auth→auth, Cache→cache, Challenges→challenges, Concurrency→concurrency, Config→config, Containers→containers, Database→database, Discovery→discovery, HelixQA→helixqa, HTTP3→http3, Mdns→mdns, Middleware→middleware, Observability→observability, RateLimiter→ratelimiter, Recovery→recovery, Security→security, Tracker-SDK→tracker_sdk) + 139 referencing files updated + `.gitmodules` rewritten + Phase 6f upstream-rename execution plan at `docs/plans/2026-05-16-phase6f-upstream-rename-execution.md` (DEFERRED per Q1). Audit confirms 0 stale `Submodules/X` references in 16/17 names (Tracker-SDK has 1 ref in a historical bluff-hunt JSON narrative quote — exempt forensic anchor). - **Phase 7** (§11.4.25 coverage ledger) — `21dee741` + merge `c35af27c` (generator + verifier + 58-row baseline + 6 hermetic fixtures + sweep wiring, advisory at first) → `76507ca0` + merge `20b3fd36` (waiver backfill: 0 covered / 20 partial / 38 gap → 48 covered / 10 partial / 0 gap) → `0c87b6ae` (STRICT-flip in sweep wrapper). - **Phase 7-debt** CLOSED — `0c87b6ae`. Sweep wrapper now invokes `check-coverage-ledger.sh --strict` (was `--advisory`); gate hard-fails on stale/missing rows. - **Phase 8** (§11.4.35 canonical-root + §11.4.36 install_upstreams) — `d95be689` (gate) + `410af7ec` (STRICT-flip after 10 install_upstreams scripts landed across owned submodules). - **Phase 8-debt** CLOSED — `858ffb3e`. HelixQA's upstream `install_upstreams.sh` is the final hold-out; scanner reports 17/17 install_upstreams present in STRICT mode. - **Phase 9 Path B** (§11.4.33 + §11.4.34 equivalence-mapping) — `055fbcbe`. §6.AD.3 amended: Lava's CONTINUATION + closure-logs + sixth-law-incidents satisfy the type-aware-closure + reopened-source-attribution semantics; no parallel Issues/Fixed tracker; equivalence is binding. Gates-index gets 2 new EQUIVALENCE-MAPPED rows (CM-CLOSURE-STATUS-VOCAB-COMPLIANCE, CM-REOPENED-SOURCE-ATTRIBUTION).

Verify-all sweep result at HEAD `0c87b6ae`: 40/40 PASS in fully STRICT mode (post-STRICT-flip; the prior 07:00:56Z attestation showed 39/40 only because the coverage-ledger sha drifted between commit and sweep — the re-run at HEAD is clean). Attestation directory: `.lava-ci-evidence/verify-all/<UTC-timestamp>.json`.

CM-* gates inventory at HEAD: 24 HelixConstitution `CM-*` gates tracked at `docs/helix-constitution-gates.md`; ~16 wired + ~8 paper-only (mostly Issues/Fixed-tracker-dependent — equivalence-mapped per §6.AD.3). 14 Lava-side anti-bluff gates also active.

All 33 session commits §6.C-converged on GitHub + GitLab. Pre-push Checks 1-9 active throughout.

Prior: 2026-05-14, **1.2.23-1043 / 2.3.12-2312 closure-cycle** (constitutional-plumbing-only; no user-visible feature change). HelixConstitution submodule incorporated + §6.AD HelixConstitution-Inheritance Mandate landed; 8-track §6.AD-debt opened and systematically closed across 14 commits. §6.AC + §6.AB scanners in STRICT mode. Build-resource stats tracker (§11.4.24) landed. All Lava-side debts in scope CLOSED at commit `4a7d0402`. See git log `66de343b..4a7d0402` for the full closure-cycle.

Prior: 2026-05-14, **1.2.22-1042 / 2.3.11-2311 DISTRIBUTED to Firebase** (debug stage 1 + release stage 2, operator pre-authorized combined). About dialog author re-order; Crashlytics 6-issue sweep (3 fixed + 3 closed-historical); §6.AC Comprehensive Non-Fatal Telemetry

> Mandate added (28th §6.L); §6.AA-debt PARTIAL CLOSE; per-channel last-version-{debug,release} pointers; both APIs running 2.3.11.
>
> **Prior:** 2026-05-14, 1.2.21-1041 DISTRIBUTED (onboarding back-press fix + WelcomeStep colored Image fix; §6.AB Anti-Bluff Test-Suite Reinforcement, 27th §6.L).
>
> **Prior:** 2026-05-14, 1.2.20-1040 DISTRIBUTED (Galaxy S23 Ultra cold-launch crash fix: `ic_lava_logo` layer-list → composited PNG; §6.Z Anti-Bluff Distribute Guard, 26th §6.L).
>
> **Prior:** 2026-05-14, 1.2.19-1039 DISTRIBUTED (§6.Y Post-Distribution Version Bump Mandate, 25th §6.L) + 1.2.18-1038 DISTRIBUTED (24th §6.L).
>
> **Prior:** 2026-05-13, SP-4 Phase G + Phase F.1 + F.2 + Phase D (multi- provider parallel search SDK) + Phase C (Trackers screen removal + :feature:provider_config landing). See git log `0c87b6ae..` and the CHANGELOG for the full delivery chain.
>
> **§6.S binding:** this file is constitutionally load-bearing per root `CLAUDE.md` §6.S. Every commit that changes phase status, lands a new spec/plan, bumps a submodule pin, ships a release artifact, discovers or resolves a known issue, or implements an operator scope directive MUST update this file in the SAME COMMIT. The §0 "Last updated" line MUST track HEAD. Stale CONTINUATION.md is itself a §6.J spirit issue under §6.L's repeated mandate. `scripts/check-constitution.sh` enforces presence + structure (§0, §7, §6.S clause + inheritance).

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0. Quick orientation (read this first)

Surface	Current state	Pin
Lava parent on master	2 mirrors (GitHub + GitLab) converged at HEAD `dca7cac7`	1077 wrap-up — sweep fixes, submodule install_upstreams
API (lava-api-go)	2.3.33 (code 2333) — `internal/version/version.go`	container `lava-api-go-thinker`
Android Firebase	1.3.12 (1076) distributed 2026-06-25 (debug+release); `last-version-debug=1077`, `last-version-`	`lava-vasic-digital` Firebase project

Surface	Current state	Pin
	`release=1077` (1077 is a cleanup-only evidence cycle — same feature set as 1076; APK built in `releases/1.3.12-1077/` but distribute §6.P-blocked because pointer==versionCode)	
On-device API App	0.2.11-22 distributed 2026-06-25; `last-version-debug=23` (api-app 0.2.11-23 is a parity bump, APK built, same status as client)	Firebase App Distribution api-app channel
18 own-org submodules	all pushed (16 vasic-digital + HelixQA + vision_engine pinned)	see §3
constitution submodule	at upstream HEAD (bumped per pin-update cycle)	HelixDevelopment/HelixConstitution
Workable-items tracker	`docs/workable_items.db` tracked; new items LVA-079..091 added for operator video issues	§11.4.93/95/106
codegraph	incorporated per §11.4.78; local SQLite index at `.codegraph/`	`@colbymchenry/codegraph` MCP
Cycle-coverage gate (§6.AK)	WIRED — Gate 7 (firebase-distribute.sh) + Check 10 (pre-push); api-app extension proven (test_wiring_apiapp.sh 5/5)	`scripts/check-cycle-coverage.sh`
Vision Engine submodule	pin advanced `85dfd0d1→af93b4bc` (install_upstreams.sh per §11.4.36)	`submodules/vision_engine`
OpenDesign transitions	DESIGN SPEC authored (`docs/superpowers/specs/2026-06-26-opendesign-transitions-design.md`) — Wave A blocked on operator decisions 1-6	NA

> **CRITICAL — this is NOT the plumbing-only cycle.** The 1072-1076 cycles (June 23-25) shipped real user-facing fixes: search works again (auth-interceptor 401 root cause fixed, engine 18s deadline, back-press-cancels-search, YTS endpoints refreshed), Sync-toggle crash fixed (serialization plugin + R8 keep rules), display/onboarding bugs fixed (chips show friendly names, provider count corrected, Select-All doesn't enable auth-requiring providers). The §6.AK incident (C00-only gate shipped broken flows on 1076) was declared and closed with the mechanical cycle-coverage gate §6.AK. **1077 (2026-06-26) is the post-1076 evidence/verification cycle; wrap-up commit adds sweep fixes + 3 submodule install_upstreams.sh**: CHANGELOG updated, cycle-coverage-map written, Genymotion device gate (Pixel 9 / API 35) proved C00+C66 PASS — all CHANGELOG claims now have executed+PASSED covering device evidence. LVA-008 (nav-teardown crash) remains open — upstream androidx-navigation defect, 8 client-side candidates exhausted, upstream minimal-repro authored. **User-facing value shipped.**

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1. What's DONE (this cycle, since 2026-05-15)

Constitution-compliance plan (`docs/plans/2026-05-15-constitution-compliance.md`)

Phase	Subject	Status	Anchor commits
0	Pin advance + plan land	DONE	`ed16debd` (pin) + `832f739e` (plan)
1	§11.4.32 enforcement engine	DONE	`4def2da7`
2	§11.4.30 .gitignore audit gate	DONE	`037389f5`
3	§11.4.31 helix-deps.yaml manifests	DONE	`43345c3e` + `410af7ec` + `bcba3a19`
3-debt	16/16 per-submodule manifests	CLOSED	`410af7ec` + per-submodule pin advances

Phase	Subject	Status	Anchor commits
4	§11.4.27 HelixQA + 100% test-type coverage	DONE	`aa0db6bd`
4 follow-up A	Option 1 design + shell wiring	DONE	`a61bd3d8` + `1b66d192` + merge `d94ade0d`
4 follow-up B	4 open-question resolutions	DONE	`281780d7` + merge `84d871a5`
4 follow-up C	HelixQA Go-package linking design	DESIGN-ONLY	`41b81359` + merge `be1ca3d8`
4-C-1	Lava-side `pkg/evidence` adapter (WRAP)	DONE 2026-05-16	HelixQA `a1e2020d` + Lava `573b4a8a`
4-C-2	Lava-side `pkg/detector` adapter (WRAP)	DONE 2026-05-16	HelixQA `a1e2020d` unchanged + Lava `<this-commit>`
4-debt	HelixQA upstream install_upstreams.sh + helix-deps.yaml	CLOSED 2026-05-16	`858ffb3e`
5	§11.4.28 nested-own-org submodule audit	DONE	`bbca3a78`
5-debt	STRICT flip after Challenges/.gitmodules removal	CLOSED	`410af7ec`
6	§11.4.29 lowercase snake_case naming	PLAN-ONLY	`322f2081` + merge `c8d42434`
7	§11.4.25 coverage ledger	DONE	`21dee741` + merge `c35af27c`
7-debt	waiver backfill + STRICT flip	CLOSED	`76507ca0` + merge `20b3fd36` + `0c87b6ae`
8	§11.4.35 + §11.4.36 canonical-root + install_upstreams	DONE	`d95be689` + `410af7ec`
8-debt		CLOSED	`410af7ec`

Phase	Subject	Status	Anchor commits
	10 install_upstreams scripts across owned submodules		
9 Path B	§11.4.33 + §11.4.34 equivalence-mapping	DONE	`055fbcbe`

HelixQA submodule (NEW this cycle)

- Adopted at `submodules/helixqa` from `git@github.com:HelixDevelopment/HelixQA.git`.
- Initial pin: `403603db` (2026-05-15 in Phase 4).
- Upstream PR `b13ba7c` added `helix-deps.yaml` + `install_upstreams.sh` (2026-05-16 in Phase 4-debt closure).
- Current pin: `b13ba7c0`.
- 11 HelixQA Challenge scripts wrapped via `scripts/run-helixqa-challenges.sh` (Option 1 shell-level wiring).
- 11 hermetic fixtures at `tests/check-constitution/test_helixqa_wiring.sh` validate the wrapper.
- §6.W audit doc at `docs/helixqa-script-audit.md` (per-script git-push / curl / outside-worktree analysis; 0/11 violators on default config).
- `HELIX_DEV_OWNED` exemption pattern in `scripts/check-canonical-root-and-upstreams.sh` + `scripts/check-helix-deps-manifest.sh` (HelixDevelopment org submodules treated like vasic-digital for mirror-presence checks, distinct from arbitrary third-party submodules).

§6.L counter advance

The §6.L Anti-Bluff Functional Reality Mandate counter advanced from 36 to 52 across the constitution-compliance cycle, then through 53-68 across the June 2026 post-compliance cycles (search-fix, display-fix, device-gate). **69th invocation (2026-06-27):** operator issued “continue all work” at session resume — driving the 1077 evidence commit (b197a026) + Genymotion device gate + this CONTINUATION.md refresh. Each restatement is preserved verbatim in the root CLAUDE.md §6.L wall-of-text. Anchor commits: `8c47cd17`, `d159d0fc` (37-41 batched), `66803d4d` (43+44 batched), `aa0db6bd` (45), `a61bd3d8` (46), `ed7a658d` (47), `dcec9eb8` (48), `0f1b19f1` (49), `2882304b` (50+51), `0c87b6ae` (52), `b197a026` (69).

On-device Lava API sub-project (2026-06-02)

The *Lava API Android app* sub-project (spec `docs/superpowers/specs/2026-06-02-lava-api-android-app-design.md`, plan `docs/superpowers/plans/2026-06-02-lava-api-android-app.md`) progress:

Phase	Subject	Status	Anchor commit
A	additive SQLite storage backend (`LAVA_API_STORAGE_BACKEND`; Postgres default unchanged)	DONE	—
B	`internal/mobile` `Start`/`Stop`/`Status` embed + `go build -buildmode=c-shared` + JNI bridge	DONE	`816d983f` (docs)
C	`:core:apiengine` Kotlin JNI wrapper (`ApiEngine`/`NativeApiEngine`/`FakeApiEngine`) + `buildCshared`→jniLibs→`externalNativeBuild` pipeline	DONE	`ef109760`
D-infra	`:api-app` module — foreground `ApiEngineService` (Wifi/Multicast/Wake locks), `ApiEngineController` state machine, `NsdMdnsAdvertiser`, `ApiKeyStore` (EncryptedSharedPreferences)	DONE	`e62e0fa8`
D-ui	landing UI + control screen + ViewModel (`MainActivity` is a placeholder today)	PENDING	—
E	instrumented Compose UI Challenge tests (boot embed on a real emulator, real HTTPS request)	PENDING	—
SP-2	client-side distinct labelling of `platform=android` instances in the discovery list	PENDING	—

Docs covering the landed surface: `docs/ON_DEVICE_API.md` (§4A Phase C, §4B Phase D-infra), `docs/guides/ON_DEVICE_API_USER_GUIDE.md`. New modules in `settings.gradle.kts`: `:core:apiengine`, `:api-app`. `:api-app` reuses `:app`'s `.env`-driven signing; `gomobile bind` is blocked (relative `replace ../submodules/*`), c-shared is the chosen native path.

1b. Post-constitution-compliance cycles (June 2026 — §6.AK search-fix + display-fix + device-gate cycles)

The 1072–1076 cycles (June 23-26) were the FIRST user-visible feature cycles after the constitution-compliance plumbing (May 15-31). These cycles shipped:

Search-fix arc (1071-1073)

Fix	Root cause	Anchor commit
Search telemetry (§6.AC)	Crashlytics non-fatals now capture `http_status`/ `base_url_host` on every search failure	commit `d665b9e5`
Auth-interceptor 401 (H1)	Build-time interceptor overwrote the per-install API key under the same header name (replace not add); fixed: attach credential ONLY when request doesn’t already carry one	commit `627a0d58` (1072)
Engine 18s deadline	On-device search handler had no deadline → mirror failover + retries could exceed client’s 30s timeout → silent SocketTimeout	commit `40301014` (1073)
Back-press cancels search	Pressing back during a search no longer waits for network timeout; slow providers surface Error+Retry	commit `40301014` (1073)
YTS endpoint refresh	Stale YTS server list (primary domain NXDOMAIN); refreshed to current live endpoints	commit `40301014` (1073)

Display/onboarding fixes (1076 cycle)

Fix	Issue	Anchor commit
Sync-toggle crash (release-only)	Missing kotlinx-serialization plugin + R8 keep-rules → `SerializationException` on toggle	`106e3fdc`
Input/provider filter source-of-truth	`SearchInputViewModel` used hardcoded-4 provider list → wrong providers → 0 results; now reads `ProviderConfigRepository.observeAll()`	`e7b6a652`
Video #4 chip names	Raw `tracker_descriptor.id` rendered instead of `displayName`	`75eab104` merge
Video #6 provider count	Hardcoded "4 providers available" → dynamic from catalogue	`75eab104` merge
Video #8 discovered-API label	mDNS API showed bare IP:port	`75eab104` merge
Video #9 Select-All auth gate	`onToggleAllProviders` now respects `requiresNoCredentials()`	`75eab104` merge
13 new UI Challenges	C48-C57 (provider toggles, search chips, credentials, scaffold, rating, account) + C31-C35 rewrite	`8bcb151b`

§6.AK incident and closure

Incident: 1.3.12-1076 shipped with §6.Z C00-only gate while CHANGELOG claimed search/provider/onboarding fixes. Operator verified "practically nothing fixed." Incident: `627a0d58`.

Remediation — §6.AK Cycle-Coverage Device Gate (NEW constitution clause): - `scripts/check-cycle-coverage.sh` — parses CHANGELOG, maps claims to covering Challenges, rejects uncovered claims - `.githooks/pre-push` Check 10 — refuses last-version pointer advance without passed cycle-coverage - `scripts/firebase-distribute.sh` Gate 7 — refuses distribute without passed cycle-coverage - 2 hermetic proof suites: `tests/cycle-coverage/test_wiring.sh` (5/5) + `test_wiring_apiapp.sh` (5/5) - api-app channel extension: spec at `docs/superpowers/specs/2026-06-26-ak-check10-apiapp-extension.md`, patch ready to graft - Bluff-hunt cycle 2 (lava-api-go Go-only): 5/5 GENUINE, 0 bluffs - Evidence: `.lava-ci-evidence/bluff-hunt/2026-06-26-cycle-2-apigo.json`

C66 Select-All device diagnostic (Genymotion)

Operator booted Genymotion Pixel 9 / API 35. C66 RED (device-RED'd because the original single-tap model observed the toggle's CLEAR branch — expect `onCount > 0` but from the all-selected default one tap clears everything). Fixed: two-tap protocol (Tap1 = clear, Tap2 = select-no-cred). C66 GREEN after fix. Evidence: `.lava-ci-evidence/genymotion/c66-redesign-*/`.

2026-06-30 — Autonomous QA harness (Phase 0)

Built a self-driving QA harness under `scripts/autonomous-qa/` that boots a containerized-KVM emulator, brings the backend up, fresh-installs the client, drives `Challenge70AutonomousQaProviderMatrixTest` per (provider × query × backend × resolution) cell while recording logcat + screenrecord, then parses an anti-bluff verdict.

- **Harness scripts:** `scripts/autonomous-qa/{lib-subsets,lib-emulator,lib-backend,run-iteration,run-matrix,vision-analyze,aggregate-evidence,lib-jackett}.sh`.
- **Emulator image:** built + boots **containerized-KVM in ~30s** (`lava-android-emulator`, API 34 x86_64, podman, cold-boot, serial `127.0.0.1:51611`).
- **Go backend:** serves `/health` (target `https://127.0.0.1:8443`); `lib-backend.sh` + `lib-jackett.sh` glue.
- **Onboarding proven end-to-end:** RuTor **anonymous → main app** confirmed via screenrecord in keystone run #1; production navigation runs on `AndroidUiDispatcher.Main`.
- **HelixQA matrix banks authored:** `lava-api-go/qa/banks/lava-{rutracker,rutor,nnmclub,kinozal,iptorrents}-matrix-journey.yaml`.
- **Two TEST-HARNESS findings (NEITHER a product defect)** — `.lava-ci-evidence/sixth-law-incidents/2026-06-30-keystone-offmain-nav-and-lva008.json`:
  - **(a) Off-main side-effect navigation** — under `createAndroidComposeRule`, `FrameDeferringContinuationInterceptor` resumes Orbit `collectSideEffect` off the main thread → `navController.navigate` → `IllegalStateException: Method setCurrentState must be called on the main thread`. **Fixed** with a main-thread guard (`runOnMainThread`) in `core/navigation/.../NavigationController.kt:50-56` (no-op on the production happy path).
  - **(b) LVA-008 teardown** — `IllegalStateException: State must be at least 'CREATED' to be moved to 'DESTROYED'` at MainActivity onDestroy with an INITIALIZED `search_input` NavBackStackEntry; concluded upstream androidx-navigation defect (6 app-level fixes device-falsified; cross-ref `.lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json`).

- **Mitigation — marker-based verdict** in `scripts/autonomous-qa/run-iteration.sh:110-187`: PASS is tied to the `C70-RESULT ... DOWNLOAD-OK` marker (logged at `Challenge70AutonomousQaProviderMatrixTest.kt:503-504` only after the on-screen download affordance is confirmed) AND the failure being LVA-008-only AND no other failure signal — NOT process exit. Any real assertion failure or missing marker can never be reported PASS.
- **EGRESS SOLVED via nezha (2026-06-30).** thinker's datacenter egress IP is provider-blocked (authenticated trackers 401/502; archiveorg upstream-flaky). nezha.local has a VPN egress IP (`31.169.53.44`) that PROVABLY reaches the blocked providers (rutracker=200, nnmclub=200, archiveorg=200; kinozal=522 Cloudflare → FlareSolverr-solvable) and has `podman` + `/dev/kvm`. **DECISIVE finding:** the lava-api-go backend SOCKS proxy does NOT help the trackers — the bundled clients (`RuTrackerHttp` etc.) log in DIRECTLY from the device, so the EMULATOR's egress is what matters. ⇒ the matrix must run **ON nezha** (Path A: emulator egress via the VPN reaches the providers directly). Full ground-truth in `docs/autonomous-qa/DEEP-FIX-STATUS.md`.
- **archiveorg topic-detail root-cause FIX (LVA-070, this commit).** The keystone's CURRENT-BLOCKER (archiveorg topic-detail renders "Something went wrong") is root-caused + fixed at two layers: (1) `FlexStringSerializer` (`core/tracker/archiveorg/.../FlexStringSerializer.kt`) — archive.org `/metadata` returns `title`/`creator`/`date` as a string OR a JSON ARRAY for multi-author/multi-date items; strict `String` decoding threw `JsonDecodingException` for the WHOLE response → `getTopic` threw → topic fell back to the unreachable proxy → error state. Now flattened (joined with ","), mirroring the Go backend's `flexString` (LVA-020). (2) `TopicPageSource` seam (`core/network/{api,impl}/.../TopicPageSource*.kt` + DI binding) — with a LAN/GoApi endpoint configured, archiveorg/gutenberg topics fell through to the proxy `…/topic2/{id}` (UnknownHost on the QA emulator); the seam routes them via `LavaTrackerSdk.getTopicPage(providerId, …)`, threaded from the search result through `TopicService/GetTopicUseCase/ObserveTopicPagingDataUseCase/TopicViewModel`. JVM build + all module unit tests GREEN. **Anti-bluff gap closed:** added a falsifiable array-flatten regression test (`ArchiveOrgTopicEdgeCaseTest`) — Bluff-Audit: strip the `FlexStringSerializer` annotations → both array tests FAIL `JsonDecodingException: Expected beginning of the string, but got [ at path: $.metadata.title`; reverted. **DEVICE-PROVEN 2026-06-30:** the nezha containerized-KVM keystone (archiveorg/1080p) advanced PAST topic-detail (onboard→search→topic renders→"Torrent" button present) — the topic-detail fix works on-device. The keystone then FAILED at the DOWNLOAD step, which surfaced a SECOND genuine product bug (below).

- **archiveorg HTTP-download root-cause FIX (LVA-070, this commit).** `ArchiveOrgDownload.downloadHttpFile` REQUIRED the id as `"identifier/filename"`, but a topic opened from search carries the BARE identifier (`vidmate_201910_201910`) — so EVERY archiveorg download from search threw → `sdk.downloadHttpFile` caught it → null → `DownloadState.Error` → the topic download button surfaced an error and never a completed file. Fix: when no filename is supplied, resolve a representative file from `/metadata/{id}` — prefer the auto-generated `<id>_archive.torrent` (small, opens in a torrent client), else the smallest file. Falsifiable test added (`ArchiveOrgDownloadTest` bare-identifier cases) — Bluff-Audit: restore the strict `require(parts.size == 2 …)` → both bare-id tests FAIL `Archive.org download id must be 'identifier/filename', got 'item-1'`; reverted. **ACHIEVED 2026-06-30: the rebuilt-APK keystone re-run on nezha PASSED CLEAN** — `C70-RESULT: DOWNLOAD-OK backend=goapi providers=archiveorg query=1080p` marker confirmed in logcat (line 35409, 4s after TorrentFileClick); verdict.json `marker_download_ok:true, teardown_known_lva008:false, other_failure_signal:false, verdict:PASS, note:"clean"`. **archiveorg onboard→search→topic-detail→download is now proven end-to-end on a real containerized-KVM device.** Both 1078 product bugs (topic-detail array-decode + download bare-id) are device-fixed.
- **Push-readiness + gate-health fixes (parallel-subagent verified, this commit).** (1) `lava-api-go/internal/httpx/proxy.go:66` §6.R host:port literal in an error-message example → reworded to point at `.env.example` (scan-no-hardcoded-hostport now exits 0; httpx go test green). (2) `core/apiengine/src/main/resources/api-source.hash` was stale after the lava-api-go source changes → recomputed (`eed490bc…`; `TestSourceHash_ManifestMatchesLive` green). (3) `scripts/check-workable-items.sh` selected a macOS-only binary → non-runnable on Linux; now host-arch-aware (uses `bin/workable-items-linux` on Linux), gate exits 0; LVA-036 + LVA-5 given enumerated unblock CHOICES (`validate` 0 violations, DB↔MD in sync). Remaining standing debt (NOT this cycle): the pre-existing snake_case CamelCase regression (Concurrency=4/Containers=6, §6.AF/Phase-6) still blocks a clean `git push`; the §6.AK `check-cycle-coverage.sh`↔hermetic-test arg mismatch (§6.AK-debt) is being reconciled.
- **DISTRIBUTE-PREP DONE for 1.3.12-1078 (Phase 5).** §6.P snapshot `.lava-ci-evidence/distribute-changelog/firebase-app-distribution/1.3.12-1078.md` + §6.AK `cycle-coverage-map-1.3.12-1078.yaml` written; CHANGELOG entry present (Gate 2 ✓); monotonic-version pointers verified (1078 > debug/release 1077). The §6.AK/§6.Z Phase-1 Gate 7 dry-run exits 2 (evidence-not-found) — the intended BLOCK.
- **THE ONE REMAINING GATE** for a compliant dev+prod distribute = the §6.Z device test-evidence `1.3.12-1078-test-evidence.json` (commit_sha == the COMMITTED 1078 SHA, ≤24h, all-PASS for the cycle-coverage-map's covering

Challenges), produced ONLY by the green QA matrix on nezha. Sequence to green: (1) commit the 1078 cycle → stable SHA; (2) green matrix on nezha → write the test-evidence with that SHA; (3) `firebase-distribute.sh --app client --debug-only` (§6.AA Stage 1) → operator real-device verify → `--release-only` (Stage 2).
- **PENDING:** the full **31 × 2 × 2** matrix (providers × backends × resolutions) on nezha + the §6.AA two-stage distribute remain blocked on the green matrix → device test-evidence.

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2. What's BLOCKED ON OPERATOR ACTION

These items need the operator's environment / hardware / decisions that an agent cannot make alone.

2.1 Phase 4 follow-up C (HelixQA Go-package linking) — STATUS UPDATE 2026-05-16

Phase 4-C-1 (pkg/evidence adapter): COMPLETED 2026-05-16. All 10 open questions answered by operator + implementation landed in this cycle's commit. Operator decisions: Q1 Go 1.26 bump, Q2 WRAP, Q3 Path A tag-pin (transitional Path B replace+sibling-mount until HelixQA stabilizes), Q4 preserve HelixQA terminology (Collector / Detector / Generator), Q5 upstream-contribute CaptureGeneric first (HelixDevelopment/HelixQA PR #1, branch `feat/evidence-capture-generic`, commit `a1e2020dd759d025b67ef8e024061b103940470d`), Q6 SKIP 4-C-4 navigator entirely, Q7 NO recover() wrapping, Q8 accept 2x CI build-time delta, Q9 always-track-upstream for HelixQA (§6.AD waiver documented in CLAUDE.md), Q10 coverage-ledger bumped in same commit.

Deliverables this cycle: - HelixQA `pkg/evidence.CaptureGeneric` public method (commit `a1e2020d`, PR HelixDevelopment/HelixQA#1) - `lava-api-go/internal/qa/evidence/{collector,collector_test}.go` (adapter + 9 unit tests, 87.9% coverage) - `lava-api-go/tests/qa/evidence_test.go` (real-stack integration test, `//go:build helixqa_realstack`) - `lava-api-go/go.mod` bumped to Go 1.26 + adds `digital.vasic.helixqa` require + replace - `Submodules/HelixQA/` pin bumped to `a1e2020d` - `docs/coverage-ledger.yaml` regenerated (58 rows; lava-api-go row: 89 unit tests + 1 integration) - `CLAUDE.md` §6.AD-debt: HelixQA always-track-upstream waiver documented - This `CONTINUATION.md` updated

Phase 4-C-2 (detector adapter), 4-C-3 (ticket adapter), 4-C-4 (validator + SKIP navigator per Q6) remain owed — each is 1-session scope per design doc §E.

Phase 4-C-2 (pkg/detector adapter): COMPLETED 2026-05-16. Operator decisions reused from 4-C-1 (Q1–Q10 unchanged). Q4 preserves `Detector` name. Q5: no HelixQA-side promotion needed — `pkg/detector`'s public API surface already exposes everything the adapter requires (`Detector`, `Option`, `New`, `WithDevice`/`WithPackageName`/`WithBrowserURL`/`WithProcessName`/`WithProcessPID`/`WithEvidenceDir`/`WithCommandRunner`, `Check`, `CheckApp`, `Platform`, `DetectionResult`, `CommandRunner` interface).

Deliverables this cycle: - `lava-api-go/internal/qa/detector/detector.go` (255 LOC) — WRAP-strategy adapter exposing Lava-shaped `Report` struct (Crashed/Alive/StackTrace/EvidencePath); `CheckGoProcess(processName)` for name-based detection (real `pgrep -f`); `CheckGoProcessByPID(pid)` for PID-based detection (real `kill -0`); `ErrEmptyProcessName` + `ErrInvalidPID` Lava-side guards that block HelixQA's silent fallback to `processName="java"` for empty/zero inputs (forensic value: mutation rehearsal proved Alive=true would silently be returned because Java exists on every dev box) - `lava-api-go/internal/qa/detector/detector_test.go` (11 tests, 82.9% statement coverage, race-clean) — uses HelixQA's `CommandRunner` interface as the boundary-fake (not a mock of the SUT; the HelixQA Detector itself runs unaltered, satisfying §6.J.4 forbidden-mock pattern) - `lava-api-go/tests/qa/detector_test.go` (3 real-stack tests, `//go:build helixqa_realstack`) — spawns sacrificial `sleep` child processes via `sh -c 'exec -a <sentinel> sleep 30'` (BSD/macOS-compatible argv[0] injection); asserts the REAL HelixQA Detector against the real OS process table - `docs/coverage-ledger.yaml` regenerated (lava-api-go unit_test_count 89 → 93) - This `CONTINUATION.md` updated; coverage-ledger regen confirmed (58 rows preserved)

§6.J anti-bluff posture: 4 falsifiability rehearsals captured in commit body — (1) `Crashed: dr.HasCrash → !dr.HasCrash` triggered 5 unit-test assertions; (2) empty-name guard removal caught at compile time (`"strings" imported and not used`); (3) PID guard removal triggered 2 sub-test assertions + exposed HelixQA's silent `java` fallback; (4) `Alive: dr.ProcessAlive → !dr.ProcessAlive` triggered ALL 3 real-stack assertions on live/dead/ghost process paths. All reverted, all green after revert.

Honest scope statement: real-stack tests were initially executed via `go test -tags=helixqa_realstack ./tests/qa/detector_test.go` (file-target form) while Phase 4-C-3's `internal/qa/ticket/generator.go` was untracked + uncompiling. After 4-C-3 landed at `86402bfa` (which also bumped HelixQA pin `a1e2020d` → `c57c275` to gate `enhanced_generator.go` behind the `helixqa_enhanced_tickets` tag), the full package-target form `go test -tags=helixqa_realstack ./tests/qa/...` PASSes too (7/7: 2 evidence + 3 detector + 2 ticket). Re-verified at HEAD post-`86402bfa`.

Phase 4-C-3 (pkg/ticket adapter): COMPLETED 2026-05-16. Operator decisions reused from 4-C-1 (Q1–Q10 unchanged). Q4 preserves `Generator` name.

Deliverables this cycle: - HelixQA-side prereq: `submodules/helixqa/pkg/ticket/enhanced_generator.go` gated behind `//go:build helixqa_enhanced_tickets` so plain consumers of `pkg/ticket` (the Lava adapter) do NOT pull LLMOrchestrator transitively. HelixQA SHA bump owed at parent-cycle close. - `lava-api-go/internal/qa/ticket/generator.go` (≈340 LOC) — WRAP-strategy adapter with `NewGenerator`, `GenerateClosureLog`, `OutputDir`, `ClosureLogInput` shape mirroring §6.O closure-log conventions - `lava-api-go/internal/qa/ticket/generator_test.go` (13 tests + 8 sub-tests, 93.2% statement coverage with `-race`) - `lava-api-go/tests/qa/ticket_test.go` (2 real-stack tests, `//go:build helixqa_realstack`) - `CLAUDE.md` §6.O extended with clause 7 — adapter authorized as programmatic closure-log path - This `CONTINUATION.md` updated; coverage-ledger row addition owed at parent-cycle rolled regen

§6.J anti-bluff posture: 2 falsifiability rehearsals captured in commit body — (1) H1 heading mutation surfaced by schema test, (2) empty-CrashlyticsID validation removal surfaced by RejectsEmpty test; both reverted, both green after revert.

Phase 4-C-4 (validator adapter; navigator SKIPPED per Q6) remains owed.

2.2 Phase 6 snake_case migration — RESOLVED + EXECUTED

All 8 operator questions answered (2026-05-16). Phase 6a + 6b executed in this cycle. See `docs/plans/2026-05-16-phase6f-upstream-rename-execution.md` for the deferred upstream-rename plan (Q1: defer; document execution steps for operator).

2.3 Release-tagging chain — versions advanced past 1076/22

Last distribute: Lava-Android-1.3.12-1076 + Lava-API-App-0.2.11-22 (2026-06-25). Both debug+release distributed via Firebase (two-stage). §6.Y post-distribute bumps applied: client → 1.3.12-1077, api-app → 0.2.11-23. Evidence and pointers committed at `b197a026`; wrap-up at `dca7cac7`.

1077 cleanup cycle (2026-06-26): 1077 APKs are built (`releases/1.3.12-1077/` — 33.9 MB debug + 6.7 MB release). Genymotion device gate executed on Pixel 9 / API 35: C00+PASS, C66+PASS (two-tap RED→GREEN), api-app C01+PASS. All evidence committed (`.lava-ci-evidence/genymotion/1077-device-gate/` + cycle-coverage-map). **Distribute is §6.P-blocked** because `last-version-debug=1077` equals `versionCode=1077` (the §6.Y bump advanced the pointer to the same code). The 1077 cycle has no new user-facing features — it's a verification-only follow-up to 1076. Operator decision needed on distribute vs hold.

Cycle-coverage gate (§6.AK) now MECHANICALLY ENFORCED: - `scripts/check-cycle-coverage.sh` — the core gate script (parses CHANGELOG claims, maps to covering device Challenges, rejects uncovered claims) - `.githooks/pre-push` Check 10 — refuses pushing a `last-version-{debug,release}` pointer advance without passing cycle-coverage (client channel) - `scripts/firebase-distribute.sh` Gate 7 — refuses distribute without passing cycle-coverage (both channels) - `tests/cycle-coverage/test_wiring.sh` + `test_wiring_apiapp.sh` — hermetic proofs (5/5 + 5/5) against the real gate - **api-app channel extension** — spec authored, logic proven hermetically; patch is a wrapping outer loop (from single `ak_chan_dir` to iterating both client+api-app dirs) — ready to graft into `.githooks/pre-push`

Tag-script gate per §6.I (multi-emulator container matrix) still blocked on Linux x86_64 + KVM gate-host per §6.AH (no host-direct VM execution). On this macOS/arm64 host, containerized emulators are blocked (podman VM lacks /dev/kvm). Device Challenge execution is possible via **Genymotion** (operator-booted cloud devices) as the macOS equivalent — proven this session: C66 RED+GREEN on Genymotion Pixel 9 / API 35. The `scripts/run-genymotion-challenges.sh` harness is functional.

1077 wrap-up sweep fixes (2026-06-27, HEAD `dca7cac7`): - Markdown HTML/PDF exports regenerated (112 missing docs backfilled) - IPv4 hardcode in OnboardingState.kt comment fixed (`<HOST>:<PORT>` placeholder) - UUID scanner exemption broadened to cover all `.lava-ci-evidence/` evidence artifacts (operator-approved) - Challenge discrimination: companion evidence for C58-C67 written and committed - vm-images android-35-phone placeholder hash exempted - 3 install_upstreams.sh scripts created and pushed (doc_processor, llm_orchestrator, llms_verifier)

Remaining sweep FAILs (5 of 54 STRICT, doc-only tracking): - hermetic-suite-firebase (check-cycle-coverage.sh script path — OWED at next firebase-distribute touch) - hermetic-check-constitution-check_constitution_test (test fixture needs updating) - hermetic-check-constitution-test_covenant_114_propagation (CM anchor propagation) - hermetic-check-constitution-test_audit_snake_case_references (snake_case regression in test) - coverage-ledger (stale after uncommitted submodule changes)

§6.L counter advanced to 70.

Vision Engine submodule pin advanced `85dfd0d1→af93b4bc` (install_upstreams.sh per §11.4.36/CONST-056). This is an uncommitted pending change.

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3. Submodule pin index

18 own-org submodules + 1 universal-rules submodule (constitution). Bumping a pin is a deliberate operator action; never auto-update. **`panoptic` is at the PROJECT ROOT (`Lava/panoptic`), NOT under `submodules/`** — operator directive 2026-06-03: every submodule MUST be reachable through the project root with NO nesting (CONST-051(C)). The former nested `submodules/challenges/Panoptic` stray clone was removed and re-incorporated here at root.

2026-06-03 reconciliation (operator-directed "obtain latest versions of all Submodules"): every submodule advanced to its latest unified `main` and ALL upstreams reconciled to convergence via **fast-forward only — NO force-push** (per the new constitution §11.4.113). All github↔gitlab divergence resolved: every diverged mirror was `gitlab`-behind-`github` (a strict ancestor), so `gitlab` was fast-forwarded to the `github` tip; no merge commits needed, no commits lost. CONTINUATION §4.5 OWED#2 (mirror divergence) is CLOSED. **Build-verification of these advanced pins against the Lava build is OWED** (the full Android + Go build was not run on this host) — verify before any release/distribute.

Submodule	Pin	Mirrors	Notes
`auth`	`a3fc97d`	GitHub + GitLab	helix-deps.yaml + § CodeGraph cascade FF-reconciled 2026
`cache`	`7853368`	GitHub + GitLab	helix-deps.yaml pre gitlab FF-reconciled 2026-06-03
`challenges`	`dfb5f9c`	GitHub + GitLab	helix-deps.yaml + fl (Panoptic dep decla
`concurrency`	`711499e`	GitHub + GitLab	helix-deps.yaml pre
`config`	`344073f`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sl FF-reconciled 2026
`containers`	`6f4f415`	GitHub + GitLab	per-OS procWalker proc + macOS `ps`) + conditional `--device` (§6.AH-debt); SELin

Submodule	Pin	Mirrors	Notes
			cross-platform; per- emulator accelerati
`database`	`d822b33`	GitHub + GitLab	helix-deps.yaml pre
`discovery`	`11bb596`	GitHub + GitLab	helix-deps.yaml + install_upstreams.s FF-reconciled 2026
`helixqa`	`dd3cf1d`	GitHub	HelixDevelopment c always-track-upstre §6.AD Q9 waiver; b 2026-06-08 `5112906-` escape a broken pir `go.mod` carried unres conflict markers (L124/131/138) tha `lava-api-go`'s Go bui
`doc_processor`	`7750140`	GitHub	vasic-digital/DocPro added 2026-06-08 – `go.mod` sibling dep (`digital.vasic.docpr` per §11.4.27/§11.4. org deps reachable
`llm_orchestrator`	`d2a2151`	GitHub	vasic-digital/ LLMOrchestrator; a 2026-06-08 — Helix dep (`digital.vasic.llmor`
`llm_provider`	`d3da070`	GitHub	vasic-digital/LLMPr added 2026-06-08 – sibling dep (`digital.vasic.llmpr`
`llms_verifier`	`9302b5c`	GitHub	vasic-digital/LLMsV added 2026-06-08 – sibling dep (`digital.vasic.llmsv` module at `/llm-veri`
`vision_engine`	`f96bf56`	GitHub	vasic-digital/Vision added 2026-06-08 – sibling dep (`digital.vasic.visio`
`http3`	`7ddc6e8`	GitHub + GitLab	helix-deps.yaml + install_upstreams.s FF-reconciled 2026
`mdns`	`ba1d2385`	GitHub + GitLab	helix-deps.yaml + install_upstreams.s converged)
`middleware`	`ccf237a`	GitHub + GitLab	

Submodule	Pin	Mirrors	Notes
			helix-deps.yaml + install_upstreams.s
observability	73bbd1b	GitHub + GitLab	helix-deps.yaml pre gitlab FF-reconciled 2026-06-03
panoptic (ROOT: Lava/panoptic)	2f8e7c2	GitHub + GitLab	added at project ro 2026-06-03 (flatten the stray challenges nesting per CONST gitlab FF-reconciled tip
ratelimiter	92b01ea	GitHub + GitLab	helix-deps.yaml + install_upstreams.s FF-reconciled 2026
recovery	0eb87cf	GitHub + GitLab	helix-deps.yaml + install_upstreams.s
security	16ae574	GitHub + GitLab	helix-deps.yaml pre gitlab FF-reconciled 2026-06-03
tracker_sdk	7afc37aa	GitHub (origin)	already at origin/ma (converged)
constitution	d90ab87	universal (HelixConstitution: github+gitlab+gitflic+gitverse)	**§11.4.113 absolut force-push + merg latest-main mand** (2026-06-03); conve all 4 upstreams

Internal-to-submodule nested submodules — NONE (CONST-051(C) fully satisfied 2026-06-03). `submodules/challenges` previously had a stray nested `Panoptic` clone (the last tracked `CM-NO-NESTED-OWN-ORG-SUBMODULES` exposure). It was removed and `vasic-digital/Panoptic` is now incorporated as a **root-level submodule at `Lava/panoptic`** per operator directive (every submodule reachable through the project root, no nesting). Challenges still declares Panoptic as a `layout: flat` dependency in its `helix-deps.yaml`; that declaration now resolves to the root `panoptic` submodule. No submodule nests any own-org submodule.

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4. Known issues + bugs (carried forward — historical)

These are real defects discovered before this cycle. Tracked here for forensic continuity; none are blocking this cycle's constitutional work.

4.5 Active known issues

- **LVA-008 — nav-teardown crash (search→back → MainActivity destroy ISE)**: The primary open user-facing defect. Navigation-compose throws `IllegalStateException: State must be at least 'CREATED'` on the `search/search_input` NavBackStackEntry during activity destroy after navigating away. CONFIRMED upstream androidx-navigation defect: 8 client-side candidates device-falsified (including launchSingleTop, single-NavHost collapse, Activity-scoped LocalLifecycleOwner — all reverted after FAIL on real KVM emulator). Upstream minimal-repro authored at `docs/issues/upstream/lva-008-androidx-navigation/`. Latest distributed 1076 carries `LenientTeardownRule` on 3 Challenge tests to prevent the ISE from masking real assertions — C48, C52, C66 use it. MECHANICAL FIX EXISTS (`LenientTeardownRule` swallows the instrumented-activity-destroy ISE), but LENIENT IS NOT A REAL FIX for user-path crashes. The companion Challenge C11 (`Challenge11NavigationSearchBackAndForthTest`) tests the nav path and MUST stay RED (the crash IS user-visible in a real-install scenario). **Operator decision owed:** (a) accept the LenientTeardownRule workaround and ship Challenge tests as green for the next distribute, OR (b) keep the upstream minimal-repro as the blocker and hold the search-flow device Challenges (C58-C62) until upstream ships the fix.
- **§6.AH / §6.X-debt (containerized-emulator gate path on darwin/arm64)**: podman on darwin/arm64 runs in a Linux VM that does NOT expose `/dev/kvm` or HVF passthrough, so the containerized-emulator path (`emulator-matrix --runner=containerized`) cannot boot. The §6.AH rule forbids host-direct emulators for gate runs. **Mitigation:** Genymotion (cloud/operator-booted Android devices) is the macOS-equivalent gate path — proven this cycle with C66 RED+GREEN on Pixel 9 / API 35 via `scripts/run-genymotion-challenges.sh`. Full §6.I multi-emulator matrix (Linux x86_64 + KVM) still owed. Linux x86_64 gate host remains the primary resolution path.
- **LVA-085 — engine exoneration pending**: `lava-api-go` `/v1/providers` serves correct provider names (proven: `providers.go:66` returns `"YTS"`). A prior in-flight “fix” edited a NON-RENDERED field and its test was a §6.J-proven bluff (passed with the change reverted). Both removed. LVA-085 is conclusively a client-side render bug in the LVA-008-blocked path.
- **§6.H Firebase CI token echo-leak** (2026-05-20, §6.L 67th): **RESOLVED 2026-05-31** — operator rotated the token (`firebase logout` → `firebase login:ci`) during the §6.L 68th cycle; the transcript-leaked token (never committed to git) is now dead. §6.H clause 6 satisfied. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-05-20-firebase-token-echo-leak.json`.

- **LVA-8 — HelixQA crash-detector + consumer fixture — RESOLVED 2026-05-31** (§6.L 68th): internal/qa/validator has 6 failing tests because HelixQA's isPIDAlive (submodules/helixqa/pkg/detector/desktop.go) shells exec kill -0 <pid> and /bin/kill -0 <absent-pid> returns EXIT 0 on macOS (bash builtin returns 1), so a dead PID reads as alive → Validator reports StepPassed on a crashed step (the canonical §6.J bluff). ROOT CAUSE CONFIRMED (captured evidence). Fix belongs UPSTREAM in HelixQA (use syscall.Kill not /bin/kill) per CONST-051 + CONST-049. Incident: .lava-ci-evidence/sixth-law-incidents/2026-05-31-helixqa-validator-killbinary-macos-bluff.json. The Lava adapter is a faithful pass-through (0-byte internal/ diff) — the defect is entirely HelixQA-side. **Operator decision owed:** authorize the HelixQA upstream fix cycle (fix → push to HelixQA → bump pin).
- **LVA-3 — LVA-vs-canonical-workable-items reconciliation** (§6.L 68th): the constitution ships a canonical workable-items Go binary at constitution/scripts/workable-items/ keyed docs/workable_items.db; Lava built a parallel LVA-keyed system at tools/lava-tickets/ + docs/tickets/tickets.db. Both satisfy §11.4.93/95/106; whether LVA supersedes or complements the canonical binary (§11.4.74 catalogue-first) is an **operator decision**. Until ratified both the LVA system AND the §6.AD.3 Path-B .lava-ci-evidence/ ledgers stay in force.
- **macOS emulator stall** (2026-05-15 incident): Pixel_7_Pro on macOS
  - emulator 36.1.9 stalls indefinitely. Three candidate root-causes recorded as PENDING_FORENSICS: (T7 external drive contention, emulator-36.1.9 known issues, AVD config theory eliminated by fresh-AVD re-test). Forensic anchor: .lava-ci-evidence/sixth-law-incidents/2026-05-15-macos-emulator-stall-on-android33.json. Orthogonal to §6.X-debt.
- **github SSH-fail flake pattern** (resolved this cycle): the §6.L 37th
  - 39th invocation forensics document multi-push retry pattern; the resolution is the standing operator practice of retrying git push github on connection-reset. No code change owed.

4.5 Resolved this cycle

- **Flaky CredentialsViewModelTest > select provider updates selectedProvider** (§6.L 68th, 2026-05-31): the fixed-awaitState()-count assumption was replaced with a bounded await-until-selectedProvider=="rutracker" loop, removing the dependence on the non-deterministic interleaving of load()'s Room-Flow .first() resume (delivered off the StandardTestDispatcher) vs. the SelectProvider reduce. Falsifiability-rehearsed (broke the reduce → AssertionFailedError: expected:<rutracker> but was:<null>, localized to that one test → reverted → 6/6 green). Incident

JSON `.lava-ci-evidence/sixth-law-incidents/2026-05-20-flaky-credentialsviewmodeltest.json` remains as the forensic record.
- **`.codegraph/*.pid` gitignore gap** (§6.L 68th): `daemon.pid` was untracked-but-not-ignored (§11.4.30); `.codegraph/*.pid` added to `.gitignore`.
- **Stale coverage ledger (§11.4.25)** (§6.L 68th): the verify-all sweep was 46/47 — `docs/coverage-ledger.yaml` had drifted (prior cycles added `feature/login` LoginViewModelTest, `feature/onboarding` AuthTypeDisplayTest + challenge C26, `feature/provider_config` ProviderConfigViewModelTest without regenerating the ledger). Regenerated via `scripts/generate-coverage-ledger.sh`; `check-coverage-ledger.sh --strict` now EXIT=0. Sweep → **47/47 PASS**.

LVA ticket system NEW (§6.L 68th, §11.4.93/95/106)

The operator's "define ticket key LVA + SQLite workable-items DB + Issues/Fixed/ Issues_Summary/Fixed_Summary + PDF/HTML/ DOCX exports" directive is the same requirement as the new HelixConstitution **§11.4.93/95/106** (workable-items SQLite DB tracked-in-git + mechanical md↔DB byte-identical sync). Built as a pure-Go module at `tools/lava-tickets/` (`digital.vasic.lava.tickets`, Go 1.26, `modernc.org/sqlite` — no CGO, no sudo). Key prefix **`LVA`** (LVA-1, LVA-2, …; the Lava instantiation of §11.4.54 ATM-NNN). The DB `docs/tickets/tickets.db` **IS tracked** (§11.4.95 — never gitignored); only `*.db-wal/-shm/-journal` sidecars + `bin/` + scratch are ignored. Subcommands: `init/add/update/close/reopen/gen/verify/import/export/list/version`. `go test` 7/7 PASS (incl. §11.4.106 round-trip + falsifiability, §11.4.33 type-aware closure, §11.4.34 reopen-attribution); `verify` byte-identical PASS (exits 1 on drift — independently confirmed). Exports: HTML (pure-Go) ✓, DOCX (podman `pandoc/core`) ✓, **PDF honestly BLOCKED** (container lacks a LaTeX engine — tool exits 3, writes NO fake file; remediation: `pandoc/latex` image or host `weasyprint`). 7 real LVA tickets seeded from actual project state (no invented SHAs). Design + verbatim build evidence: `docs/tickets/DESIGN.md` + `docs/tickets/BUILD-EVIDENCE.md`. **OPEN (operator decision, LVA-3):** whether the LVA system supersedes or complements the §6.AD.3 Path-B `.lava-ci-evidence/` ledgers — until ratified, the Path-B mapping remains the binding compliance surface and the LVA DB is seeded from it (reconciled, not replacing).

Constitution pin BUMPED `208e2c8` → `883ccc1` (§6.L 68th, §6.AF)

Operator directed bump-now+adopt. 53 upstream commits (2026-05-20 → 2026-05-31) add **§11.4.79–§11.4.106** (28 universal clauses). 68th-cycle review at `.lava-ci-evidence/constitution-review/2026-05-31-68th-cycle-review.md`: **NO constitution-mandated submodule is missing** (Challenges, HelixQA, containers, codegraph all present). New §6.AF clause in

CLAUDE.md enumerates per-clause Lava adoption status + §6.AF-debt for the OWED items: - §11.4.93/95/106 (workable-items DB) — **SATISFIED** by the LVA system. The constitution also ships a canonical `workable-items` Go binary at `constitution/scripts/workable-items/` keyed `docs/workable_items.db`; LVA-vs-canonical reconciliation is operator-gated (LVA-3). - §11.4.79/.80 (own-org submodules IN codegraph index) — **OWED** (LVA-6; Lava currently excludes `submodules/`). - §11.4.85 (stress+chaos) — **IN PROGRESS** (LVA-7; phase-1 lava-api-go scaffold + evidence under `docs/chaos-stress/`). - operating-mode clauses — EQUIVALENCE-MAPPED to existing Lava practice. §11.4.100 (video-color) DEMOTED to ATMOSphere-only — not binding on Lava. The constitution submodule pin bump is a parent-repo change; per CONST-049 the constitution stays pinned + advanced deliberately (NOT auto-tracking).

Resolved in prior (constitution-compliance) cycle

- **Ledger-staleness drift class** — Phase 7-debt closure (`0c87b6ae`) flipped the coverage-ledger gate from `--advisory` to `--strict` in the sweep wrapper. Subsequent stale-ledger commits will hard-fail at sweep time + pre-push.
- **§11.4.27 HelixQA non-incorporation** — Phase 4 closure (`aa0db6bd`)
    - Phase 4-debt closure (`858ffb3e`) bring HelixQA in as `submodules/helixqa` with full mirror compliance.
- **§11.4.31 / .35 / .36 zero-waiver state** — Phase 4-debt closure + Phase 8-debt closure achieve 17/17 own-org submodules satisfying all three mandates with **zero waivers** in STRICT mode.

4.5 Historical — pre-this-cycle, carried forward

(See full historical detail in the git log between the prior CONTINUATION.md “Last updated” header and `4a7d0402`. Summary: C02 Cloudflare-mitigation stops short of profile-parsing; C17-C22 require emulator matrix; UDP buffer warning documented; mirror model reduced to 2-mirror per §6.W; docs/todos/Lava_TODOs_001.md committed as historical; etc.)

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5. Operator-flagged follow-up items (small, queued)

- **Phase 4-C implementation** — blocked on §2.1 open questions.
- **Phase 6a implementation** — blocked on §2.2 open questions.
- **HelixQA pin upgrade cadence** — operator decides when to re-baseline to track HelixQA `main` vs. holding at `b13ba7c0`.

- **Re-audit HelixQA scripts (`docs/helixqa-script-audit.md`)** on every pin bump per the §6.W audit doc's re-audit-trigger clause.
- **Coverage-ledger row additions** when new feature modules land — the generator is deterministic; re-run via `scripts/generate-coverage-ledger.sh` in the same commit as the new module to keep STRICT mode green.

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6. Constitutional debt + memory anchors

- **§6.K-debt** (Containers extension): RESOLVED 2026-05-07.
- **§6.N-debt** (pre-push hook enforcement): RESOLVED 2026-05-05.
- **§6.AD-debt** (HelixConstitution-Inheritance per-scope + CM-* wiring): FULLY DRAINED 2026-05-18. Of the original `CM-*` set: 5 mapped Path-B-equivalent (CLOSED-BY-EQUIVALENCE per §6.AD.3), `CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION` CLOSED by pre-push Check 8, `CM-SCRIPT-DOCS-SYNC` CLOSED 2026-05-17 (`11820734`), `CM-COMMIT-DOCS-EXISTS` CLOSED 2026-05-18 (`977630c3`), `CM-SUBAGENT-DELEGATION-AUDIT` CLOSED 2026-05-18 (`2a0e11f4`). No items remain OWED.
- **§6.X-debt** (Linux x86_64 + KVM gate-host for container-bound emulator matrix): STANDING. See §4.5 above.
- **§6.L** (Anti-Bluff Functional Reality Mandate): 52 invocations across multiple working days; 17-cycle back-to-back the longest sequence in project history this cycle. Per §6.L the repetition IS the constitutional record.
- **§6.R** (No-Hardcoding Mandate): UUID + IPv4 + host:port scanners active; algorithm-parameter literal grep staged (code-review gate per §6.R clause body).
- **§6.S** (Continuation Document Maintenance): THIS file. Per §6.S the §0 "Last updated" line MUST track HEAD.
- **§6.T** (Universal Quality Constraints): four sub-points active (Reproduction-Before-Fix, Resource Limits, No-Force-Push, Bugfix Documentation).
- **§6.AC** (Comprehensive Non-Fatal Telemetry): scanner in STRICT mode; ci.sh hard-fail wired.
- **§6.AB** (Anti-Bluff Test-Suite Reinforcement): scanner in STRICT mode.
- **§6.AE** (Comprehensive Challenge Coverage + Container/QEMU Matrix): per-feature scanner in STRICT mode; container matrix runner BLOCKED on §6.X-debt.

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7. RESUME PROMPT

> **CURRENT RESUME (§11.4.131, 2026-06-26) — SHORT (one-paste):** *“Read `docs/CONTINUATION.md` §0 (top entry) + `.remember/remember.md`, `git fetch`, then continue on `master` at HEAD `28a8b79b` (feat(6AK): wire cycle-coverage gate into firebase-distribute + pre-push — closes 6AK-debt). The §6.AK mechanical gate now prevents the 1076-incident pattern (C00-only gate shipping broken flows). Pending: commit uncommitted changes (C48/C52/C66 LenientTeardownRule, stress-chaos re-runs, design-spec revisions, vision_engine pin), graft the api-app Check 10 extension patch, advance to 1077 cycle (search-flow device Challenges once LVA-008 unblocks or Genymotion-harness matures), answer OpenDesign operator decisions 1-6 (Wave A of transitions spec). LVA-008 remains upstream androidx-navigation defect — 8 candidates falsified, minimal-repro authored.”*
>
> **FULL (paste-ready):** read §0 top entry + the paste-block below. State 2026-06-26: HEAD `28a8b79b` on `master` converged GitHub+GitLab; cycle-coverage gate §6.AK wired (Check 10 + Gate 7 + `scripts/check-cycle-coverage.sh`); latest distribute client 1.3.12-1076 + api-app 0.2.11-22 (both debug+release via Firebase). Binding constraints: anti-bluff §6.J/§6.L + Sixth/Seventh Laws, §6.AK coverage-intersection (every CHANGELOG claim needs an executed+PASSED covering device Challenge), no-force-push (§6.T.3/§11.4.113), §6.S CONTINUATION-in-same-commit, §6.AH no-host-direct-VMs (Genymotion is the macOS equivalent). Action-prefix (§6.AJ/§11.4.140) active via `.claude/settings.json` → `scripts/hooks/action-prefix-expand.sh`. **LVA-008 nav-teardown crash remains OPEN (upstream androidx-navigation defect, 8 candidates falsified, minimal-repro authored). PENDING: commit working-tree changes, graft the api-app Check 10 extension (spec at `docs/superpowers/specs/2026-06-26-ak-check10-apiapp-extension.md`, proven hermetically), advance to 1077 cycle, answer OpenDesign transitions decisions 1-6.**

Paste the following into a new CLI agent session to continue this work. The agent needs no scrollback — everything it needs is in this file plus the spec/plan/CLAUDE.md set referenced from it.

```
Continue Lava project work. Read these in order before doing
anything:

  1. /Volumes/T7/Projects/lava/docs/CONTINUATION.md
  2. /Volumes/T7/Projects/lava/CLAUDE.md
  3. /Volumes/T7/Projects/lava/constitution/Constitution.md
```

```
Then check the git state vs the CONTINUATION.md "Last updated"
line.
If new commits exist on master beyond what CONTINUATION.md
describes,
trust the commits and update CONTINUATION.md before proceeding
(per §6.S).

Active state per CONTINUATION.md §0 (2026-06-26):
  - HEAD at `28a8b79b` on master (GitHub + GitLab converged).
  - Cycle-coverage gate §6.AK is WIRED (Check 10 pre-push + Gate 7
firebase-distribute).
  - Last distribute: client 1.3.12-1076 + api-app 0.2.11-22
(2026-06-25).
  - LVA-008 nav-teardown crash = OPEN (upstream androidx-
navigation defect, 8 candidates falsified).
  - Constitution, coverage ledger, submodules all up to date.

Your default next action (priority order):
  1. **Commit pending working-tree changes** — C48/C52/C66
LenientTeardownRule refinements,
     stress-chaos re-runs (timestamps), design-spec post-1076
revisions, vision_engine pin advance.
  2. **Graft the api-app Check 10 extension** — the patch at
     `docs/superpowers/specs/2026-06-26-ak-check10-apiapp-
extension.md` wraps the existing
     single-channel loop into a two-channel loop (client + api-
app). Proven hermetically
     at `tests/cycle-coverage/test_wiring_apiapp.sh` (5/5 PASS).
  3. **Advance to 1077 cycle** — search-flow device Challenges
(C58–C62)
     once LVA-008 unblocks or Genymotion harness matures.
  4. **OpenDesign transitions** — answer operator decisions 1-6
from the design spec;
     Wave A (tokens.css + codegen + parity gate) is the
foundation.
  5. **LVA-008** — upstream minimal-repro authored; operator call
needed:
     file with Google vs live with the LENIENT_TEARDOWN_RULE
workaround.

Do NOT re-run completed phases — they are committed + pushed +
sweep-verified.
The git log is the authoritative record.

Constitutional bindings still in force (do not relax):
  §6.AK (Cycle-Coverage Device Gate — every claim needs a covering
Challenge)
  §6.J / §6.L (Anti-Bluff Functional Reality Mandate)
  §6.AB / §6.AC / §6.AE (anti-bluff scanners — STRICT)
  §6.AD (HelixConstitution Inheritance)
  §6.R (No-Hardcoding Mandate)
  §6.S (Continuation Document Maintenance — THIS file)
```

```
  §6.W (GitHub + GitLab Only Remote Mandate)
  §6.AH (no host-direct VMs — Genymotion is the macOS equivalent)
  §6.AJ / §11.4.140 (action-prefix expansion via registry)

The operator's standing §6.L wall is preserved verbatim in
CLAUDE.md.
Read it.
```

---

8. House-keeping the agent should keep doing

These are habits established across multiple cycles; future agents should preserve them.

1. **Commit messages carry Bluff-Audit stamps for every test class added or modified** (Seventh Law clause 1; pre-push Check 2 rejects commits without them).
2. **Every commit must have `Co-Authored-By: Claude Opus 4.7 (1M context) <noreply@anthropic.com>`** as the trailer.
3. **Push to both Lava parent mirrors (GitHub + GitLab)** after every commit chain that closes a logical unit. After every push, confirm convergence with `for r in github gitlab; do echo "$r: $(git ls-remote $r master | awk '{print $1}' | head -1)"; done`.
4. **Submodule pushes are explicit per submodule** to whatever remotes that submodule has (varies — see §3). Never use `git submodule foreach git push` blindly.
5. **Update this CONTINUATION.md** in the same commit as any completion-state change (phase done, new spec/plan written, submodule pin bumped, distribute artifact shipped, new operator-blocked open question surfaced).
6. **Run `scripts/verify-all-constitution-rules.sh`** before any release-tagging or major-state-change attempt. The sweep wrapper is in fully STRICT mode; a non-40/40 result is a release blocker.
7. **Re-generate the coverage ledger** (`scripts/generate-coverage-ledger.sh`) in the same commit as any new feature module or any module-deletion; the STRICT-mode gate rejects stale-ledger commits.
8. **The autonomous loop ends** when the next forward step requires operator-environment access (real device, real keystore secrets, Firebase token, ssh credentials) OR operator decision-making (open questions, brainstorming next phase scope, tagging, choosing a UI direction). At that point, summarize state + ask the operator the specific next-step question.

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9. Cross-references

- **Plan docs (current cycle):**
  - `docs/superpowers/specs/2026-06-26-ak-cycle-coverage-spec.md` — §6.AK gate design
  - `docs/superpowers/specs/2026-06-26-ak-gate-wiring-integration.md` — §6.AK mechanical wiring
  - `docs/superpowers/specs/2026-06-26-ak-check10-apiapp-extension.md` — api-app Check 10 extension
  - `docs/superpowers/specs/2026-06-26-opendesign-transitions-design.md` — OpenDesign 5-wave plan
  - `docs/superpowers/specs/2026-06-11-dynamic-provider-discovery-design.md` — dynamic discovery + post-1076 revisions
  - `docs/superpowers/specs/2026-06-12-embedded-curated-providers-design.md` — curated providers + post-1076 revisions
  - `docs/qa/2026-06-25-opendesign-transitions-scoping.md` — OpenDesign read-only audit
  - `docs/scripts/check-cycle-coverage.sh.md` — cycle-coverage gate user guide
- **Plan docs (prior cycle):**
  - `docs/plans/2026-05-15-constitution-compliance.md` — master constitution-compliance plan (DONE)
  - `docs/plans/2026-05-16-helixqa-integration-design.md` — Option 1 wiring (DONE)
- **Gates inventory:** `docs/helix-constitution-gates.md`
- **Coverage ledger:** `docs/coverage-ledger.yaml` + `docs/coverage-ledger.waivers.yaml`
- **Constitution source-of-truth:** `constitution/` submodule (HelixConstitution at `464ada14`)
- **Sweep wrapper:** `scripts/verify-all-constitution-rules.sh`
- **Sweep attestations:** `.lava-ci-evidence/verify-all/<UTC-timestamp>.json`
- **HelixQA audit:** `docs/helixqa-script-audit.md`
- **HelixQA wrapper:** `scripts/run-helixqa-challenges.sh`
- **HelixQA hermetic test:** `tests/check-constitution/test_helixqa_wiring.sh`